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Experimental Analysis of Binary Search Models in Graphs

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Abstract

In this work, we conduct an experimental analysis of the generalized binary search problem in tree graphs. The study is structured into two main sections:

The first part focuses on the theoretical foundations of the problem. It introduces key definitions, concepts, and pseudocodes of the analyzed procedures, along with a formal analysis of their parameters, including approximation ratios and time complexities.

The second part presents experimental verification of the theoretical claims established in the previous chapter. It includes a practical comparison of the quality of the solutions produced by the algorithms, both against each other and relative to the optimum. We also measure the computation time required by each algorithm to obtain these results. We evaluate the performance of the algorithms on randomly generated graphs of various classes: general random trees, random trees with bounded degree, and random trees with bounded diameter. Additionally, we employ different methods for generating cost functions, including uniform, binomial, and exponential distributions.

Keywords and phrases Trees, Graph Searching, Binary Search, Decision Trees, Ranking Colorings, Graph Theory, Approximation Algorithm, Combinatorial Optimization, Experimental Analysis of Algorithms

Streszczenie

W niniejszej pracy przeprowadzamy eksperymentalną analizę uogólnionego problemu wyszukiwania binarnego w drzewiastych grafach. Badanie podzielone jest na dwie główne sekcje:

Pierwsza część koncentruje się na teoretycznych podstawach problemu. Wprowadza kluczowe definicje, koncepcje i pseudokody analizowanych procedur, wraz z formalną analizą ich parametrów, w tym współczynników aproksymacji i złożoności czasowych.

Druga część przedstawia eksperymentalną weryfikację twierdzeń teoretycznych ustalonych w poprzednim rozdziale. Obejmuje praktyczne porównanie jakości rozwiązań generowanych przez algorytmy, zarówno ze sobą nawzajem, jak i w stosunku do optimum. Mierzymy również czas obliczeń wymagany przez każdy algorytm do uzyskania tych wyników. Oceniamy wydajność algorytmów na losowo generowanych grafach różnych klas: ogólne drzewa losowe, drzewa losowe o ograniczonym stopniu, oraz drzewa losowe o ograniczonej średnicy. Dodatkowo stosujemy różne metody generowania funkcji kosztów, w tym rozkłady jednostajny, dwumianowy i wykładniczy.

Słowa kluczowe Drzewa, Przeszukiwanie Grafów, Wyszukiwanie Binarne, Drzewa Decyzyjne, Kolorowania Rankingowe, Teoria Grafów, Algorytm Aproksymacyjny, Optymalizacja Kombinatoryczna, Eksperymentalna Analiza Algorytmów

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Glossary

G A graph consisting of vertices and edges.

$V(G)$ The set of vertices in a graph.

$E(G)$ The set of edges in a graph.

n The number of vertices, $n = |V(G)|$.

m The number of edges, $m = |E(G)|$.

T A tree (connected acyclic graph).

$N_G(v)$ The set of neighbors of vertex v in graph G , $N_G(v) = \{u \in V(G) \mid uv \in E(G)\}$.

$\deg_G(v)$ The degree of vertex v in graph G (number of incident edges).

$\Delta(G)$ Maximum degree of vertices in graph G .

$d(u, v)$ The distance between vertices u and v (length of shortest path).

$\text{diam}(T)$ Diameter of tree T (maximum distance between any two vertices).

c Query cost function, $c : V \rightarrow \mathbb{R}^+$, assigning a cost to each vertex.

$c(v)$ The cost of querying vertex v .

D A decision tree used to identify target vertices through queries.

$Q_D(G, x)$ The sequence of queries made to locate target x in graph G using decision tree D .

$C(D)$ Worst-case search cost of decision tree D , $C(D) = \max_{v \in V} \sum_{q \in Q_D(G, v)} c(q)$.

T_R Random tree generated via Prüfer sequence.

T_Δ Bounded degree tree with maximum degree Δ .

T_{diam} Bounded diameter tree with diameter at most diam .

T_{bal} Balanced m-ary tree.

$\mathcal{U}(a, b)$ Uniform distribution: all values in $[a, b]$ equally likely.

$\mathcal{B}(n, p)$ Binomial distribution: number of successes in n trials with probability p .

$\mathcal{E}(\lambda)$ Exponential distribution with rate parameter λ .

$c \sim \mathcal{U}$ Query costs drawn from uniform distribution.

$c \sim \mathcal{B}$ Query costs drawn from binomial distribution.

$c \sim \mathcal{E}$ Query costs drawn from exponential distribution.

$\alpha || \beta || \gamma$ Three-field notation: α = tree type, β = query characteristics, γ = objective (e.g., C_{max} for worst-case).

Aligned Cost A modified cost measure for aligned decision trees that penalizes light queries with down responses.

Approximation Ratio The ratio $\frac{\text{ALG}}{\text{OPT}}$ between algorithm's solution and optimal solution.

Binary Search A search strategy that repeatedly divides the search space by querying intermediate vertices.

Connected Component A maximal connected subgraph where every pair of vertices has a path between them.

Cycle A closed path where the first and last vertices are the same.

Decision Tree A rooted binary tree where internal nodes represent queries and leaves represent target vertices.

Down Response A response to query v in rooted tree T where the returned component does not contain the root $r(T)$.

EPTAS Efficient Polynomial-Time Approximation Scheme: PTAS with running time $f(\epsilon) \cdot \text{poly}(n)$ where f depends only on ϵ .

Forest An acyclic graph; a disjoint union of trees.

FPTAS Fully Polynomial-Time Approximation Scheme: produces $(1+\epsilon)$ -approximation in $O(\text{poly}(n, 1/\epsilon))$ time.

Heavy Module A maximal connected subset $H \subseteq V(T)$ where every vertex in H is heavy.

Heavy Vertex A vertex v with query cost $c(v) > pk$ for some parameter p and box size k .

Induced Subgraph For $S \subseteq V(G)$, the subgraph $G(S)$ with vertex set S and all edges from $E(G)$ with both endpoints in S .

Light Vertex A vertex v with query cost $c(v) \leq pk$ for some parameter p and box size k .

Path A sequence of vertices where each adjacent pair is connected by an edge.

PTAS Polynomial-Time Approximation Scheme: produces $(1 + \epsilon)$ -approximation in polynomial time for fixed ϵ .

QPTAS Quasi-Polynomial-Time Approximation Scheme: produces $(1+\epsilon)$ -approximation in $n^{O(\text{polylog}(n))}$ time.

Query Cost The cost $c(v)$ of querying vertex v to determine the target's location relative to v .

Rooted Tree A tree with a designated root vertex r that establishes parent-child relationships.

Search Cost The total cost of all queries performed to identify a target vertex.

Subtree A connected subgraph of a tree.

Up Response A response to query v in rooted tree T where the returned component contains the root $r(T)$.

Vertex Ranking A labeling $l : V \rightarrow \{1, 2, \dots, k\}$ where for each pair u, v with $l(u) = l(v)$, there exists z on the path between u and v with $l(z) > l(v)$.

Worst Case The maximum search cost over all possible target vertices.

Chapter 1

Introduction

1.1 The search problem

The Binary Search is a classical algorithm used to efficiently locate a hidden target element in a linearly ordered set. To do so, the searcher repeatedly picks the median element of such set, performs a comparison operation and in constant time learns whether it is the target and if not, whether it lays above or below the median (For example see Figure 1.1).

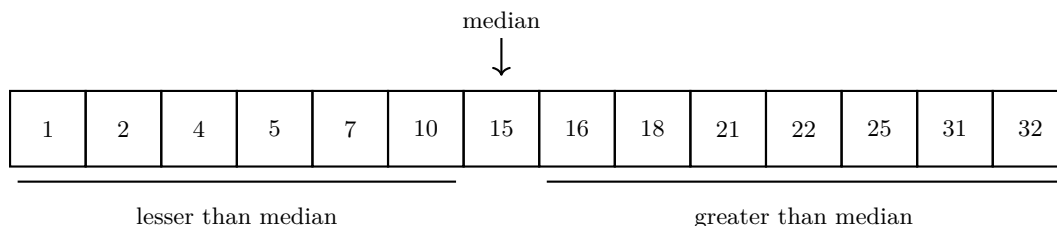


Figure 1.1: Example of a sorted array containing 14 elements. The subarrays with elements lesser than and greater than the median (15) are underlined. If the hidden element is for example 2, then the result of the comparison operation is "below" and the searcher can immediately discard all elements of value above 10. If the target were to be 15, then the comparison operation would yield "equal" meaning that the median element is in fact the target.

The study of searching was initiated by D. Knuth in his seminal book [Knu73] in which he discussed its various variants. However, the origins of the search problem reach the famous Rényi-Ulam game of twenty one questions in which a player is required to guess an unnamed object by asking yes-or-no questions¹. Throughout the years, the searching and its variants have been continuously rediscovered under various definitions and names. This hints that the intuitions behind this problem resurface among multiple use cases and research domains. In fact, the search problem in its many variants is deeply connected with many other algorithmic notions including: parallelization of the Cholesky factorization, scheduling join operations in database queries, VLSI-layouts, learning theory, data clustering, graph cuts and parallel assembly of multi-part products. This work aims to serve as a survey of the results obtained for the problem and an experimental analysis of algorithms aimed at solving it.

The importance of searching is also due to its various practical applications. For example consider the following scenario: a complex procedure contains a hidden bug required to be fixed. The procedure is composed of multiple (often nested) blocks of code. In order to find this hidden bug the searcher can perform tests which allow them to check whether a given block of code contains the bug. After performing each such test they learn whether the bug is in or outside of the tested block. This process then continues, until the bug is found. The problem is to find the best testing strategy for the tester in order to find the bug efficiently.

¹Note that in the twenty-one questions game one answer to a question may be a lie.

1.2 Problem statement

Tree Search More formally, we model the search space as a tree T . The *Vertex Tree Search Problem* is as follows: Among vertices of T there is a hidden target vertex x which is required to be located². During the search process, the searcher is allowed to perform queries, each about a chosen vertex $v \in V(T)$. In constant time the oracle responds whether the target is v and if not, it identifies which connected component of $T - v$ contains x . Upon learning this information the searcher then iteratively picks the next vertex to query until the target is found. The goal is to create a strategy of searching, which is an *adaptive* algorithm that provides the next query based on the previous responses, which optimizes a given criterion (e.g. the worst case or average case search time). One may also define an analogous process in which the queries concern edges. After a query to an edge e the searcher learns which connected component of $T - e$ contains the target. We will call this problem the *Edge Tree Search Problem*. For a visual example for both query models see Fig 1.5.

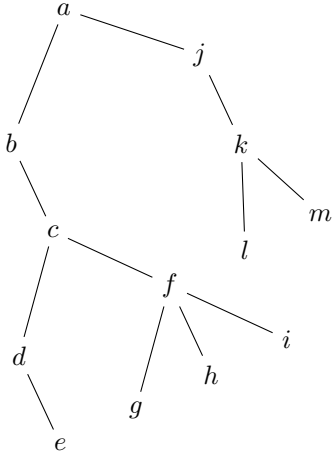


Figure 1.2

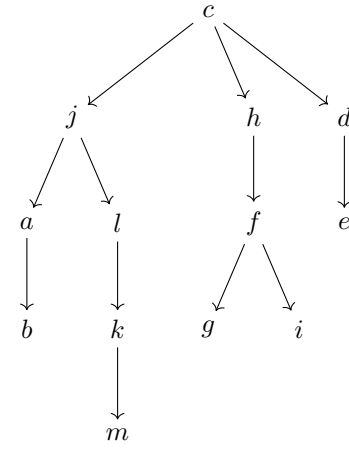


Figure 1.3

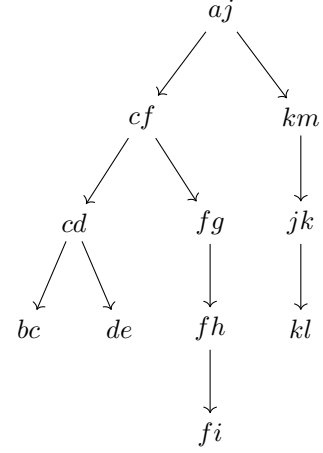


Figure 1.4

Figure 1.5: Sample input tree T (Figure 1.2) and two decision trees for T : one for the Vertex Tree Search Problem (Figure 1.3) and one for the Edge Tree Search Problem (Figure 1.4).

When the input tree is a path both problems become the classical binary search in the linearly ordered set.

Decision Trees and their cost A natural way to visualize encode a strategy is to represent it as a decision tree. A *decision tree* D is a rooted tree in which each vertex represents a query and each edge represents a possible response. The search is conducted by choosing as the next query the root of D . After receiving the response (If the search is not terminated) the searcher moves along the edge $e = (r, r_e)$ incident to r associated with the response. The process then recurses in D_{r_e} until the target is found. It should be noted, that this is far from the only viable way of encoding the search strategy. The choice of the data structure used is a matter of taste and often leads to simpler design and analysis of the algorithms.

In order to sensibly talk about the quality of such strategy we need to measure its cost. The cost of locating a vertex x using a strategy $\mathcal{A}$ is the amount of queries required to be performed to find x using $\mathcal{A}$. The two most intuitive ways to measure the cost of $\mathcal{A}$ are:

- The worst case search time which is maximum of costs of $\mathcal{A}$ over all vertices.
- The average case which is the sum of costs of $\mathcal{A}$ over all vertices³.

Even though similar, these two criterion often differ in their analysis and algorithms constructed for them usually exploit different properties of the input. It is often the case that greedy

²It should be pointed out that the target vertex might not always be the same across multiple searches.

³Note that the average search time and the sum of search times are equivalent up to a constant factor of n .

heuristics perform much better when we measure the average case cost of the decision trees created by them. In contrast, for the worst case, the best known solutions often require some intricate dynamic programming procedure as an essential subroutine. Interestingly enough, it is not hard to show that given two decision trees: one with good performance in the average case and one with good performance in the worst case, a simple algorithm can be used to create new decision tree with fairly good performance for both metrics [SLC14].

Costs So far, we assumed all queries to have uniform query costs. However it often might be a case that a particular query is more expensive than others. For example, determining the value of some complex comparison operation for two large objects may take a substantial amount of time. In such cases we associate with each query a cost function. To calculate the cost of finding a target x using a decision tree D , instead of measuring the amount of queries made by using D , we measure the sum of their costs. The worst case and the average case criteria are then defined according to these new values.

1.3 Motivations and applications

Fast retrieval of information in graph structures is a fundamental problem in computer science and related fields. Graphs are widely used to model complex relationships and structures in various domains, including social networks, transportation systems, biological networks, and communication networks. Efficiently searching for specific information within these graphs is crucial for applications such as data mining, network analysis, and decision-making processes. When the underlying search space provides non-local information about the target, the search process can be accelerated, since each query rules out a large set of possible target locations. The goal of designing search strategies is to find ways of exploiting this property as effectively as possible.

Searching arises in various practical problems, but it can also be reformulated, to fit a wide range of real-life applications. Searching in graphs can be used to model a variety of problems, that may initially appear unrelated. These include:

1. Automated bug detection in computer code [Riv+75; DW22; DW24; SB25]. This use case involves locating the source of an error in a program’s semantic structure by performing unit tests, each verifying whether the tested code segment contains the error. The resulting graph is a tree in which each child node represents a subsection of a given code segment. The objective is to minimize the total time required to identify the erroneous code segment by strategically selecting which segments to test.
2. Version control systems and regression bug identification [CDL22]. In software development projects using Version Control Systems (VCS) like git or mercurial, the project’s evolution forms a Directed Acyclic Graph (DAG) where vertices represent commits (project versions) and edges model changes between versions. When a regression bug is discovered, developers must identify the specific commit that introduced the error by performing queries on historical commits. Each query involves testing a commit to determine whether it contains the bug or is clean, which can be extremely time-consuming (potentially requiring compilation, extensive testing, or manual verification). When the input DAG is a tree, this problem reduces to the tree search problem.
3. Hierarchical clustering of data [Das16; DL05; CC17]. In this application, the objective is to construct a hierarchical clustering of data points by querying pairwise similarities. The data points are represented as nodes in a graph, and edges represent similarity measures. By querying specific pairs of nodes, one can partition the data points into separate clusters, thereby building a hierarchical clustering of data.
4. Bayesian active learning problem [BH08; CLS16; LLM; SLC14; BBS12; Car+04; KPB99; CLS14; KZ13]. In this scenario, the goal is to determine a correct hypothesis among the possibly very large set of possibilities. To do so the learner can execute tests which sample the features of the hypothesis revealing a partial information about the target. Each test is a partition of the input

space. By strategically selecting features to test, one can build a decision tree that minimizes the classification time. If the partitions incurred by the tests form a tree like hierarchy this problem reduces to the graph search problem. This problem has applications in various domains, including medical diagnosis, fault detection in systems, and criminal investigations.

5. VLSI routing and layer assignment [SDG92]. In VLSI circuit design, one is required to find a partition of *nets* connecting *terminals* into separate conductor layers in order to ensure that no such nets cross each other. In the *generalized river routing* variant of this problem, placing a net on each layer may incur different costs. Therefore, it is beneficial to assign as many nets to the cheapest layers as possible. Decision trees model the sequential layer assignment choices, where each level of the decision tree corresponds to the set of nets assigned to a layer. The objective is to minimize the total cost of layer assignments while ensuring that no nets cross each other.
6. Parallel assembly of multi-part products from their components [DK97; Der06]. The objective is to assemble complex products by performing assembly operations in parallel. The components of the product are represented as nodes in a graph, and hyperedges represent groups of such components which at some point need to be joined together. We assume that no component can undergo two join operations simultaneously. Therefore, the set of join operations performed at any given time is a matching. It can be showed that by reversing the ordering of queries given by the shallowest decision tree of the hyperedge search problem on the graph representing the query plan, one can obtain an optimal schedule for the join operations. A special case of this problem includes scheduling of parallel database join operations [DK06; Der01; Der00]. Here, the objective is to optimize the sequence of join operations required to process a database query. Each table is represented as a node, and edges represent join operations.
7. Parallel Cholesky factorization of sparse matrices [Der03]. In sparse linear algebra, the Cholesky factorization of a symmetric positive-definite matrix can be parallelized by exploiting its sparsity structure. The matrix is represented as a graph, where nodes correspond to matrix rows/columns and edges represent non-zero entries. The *elimination tree* captures column dependencies: if column i is the parent of column j in this tree, then i must be computed before j . Vertices belonging to the same layer of the decision tree can be processed in parallel. The objective is to construct a decision tree with minimum height, thereby minimizing the number of parallel steps required for factorization.
8. Investigative genetic genealogy and privacy-preserving search [EX21]. In forensic investigations and genealogical research, decision trees can model the sequential process of identifying unknown individuals through genetic database queries. The genealogical network represents family relationships, where each node corresponds to an individual and edges represent genetic relatedness. Each query involves collecting a genetic sample and comparing it to the target genome to determine genetic distance. The decision tree guides the strategic selection of individuals for genetic testing, balancing two competing objectives: minimizing the number of expensive genetic samples collected (search cost) while limiting privacy exposure of individuals whose genetic information becomes compromised.
9. Fast implementation of distance and shortest path queries [Ang18]. In this application, the goal is to efficiently answer distance and shortest path queries in large graphs. By constructing the so called *hub labeling* of a graph, one can preprocess the graph to enable fast query responses. The hub labeling is an assignment of labels to each vertex, where each label contains a subset of vertices (hubs) such that for any pair of vertices, their labels contain a common hub on the shortest path between them. When the input graph is a tree, the problem of constructing an optimal hub labeling is equivalent to the tree search problem, since it can be proven that any optimal hub labeling can be transformed to a hierarchical form, in which the labeling of each vertex v corresponds to the sequence of queries made when searching for v using the corresponding decision tree.
10. Locating sources of river pollution. In this application we are required to locate a hidden source of pollution in a river. To do so, we can probe the river in certain points. Each test reveals whether

the sampled location is polluted: since the pollution is carried downstream with the current its detection certifies that the source of the pollution must be located upstream. Otherwise, if the water is clean, the source must be located in a different creek. The river’s hierarchical structure naturally forms a tree where each confluence represents a branching point. This application extends to other hierarchical distribution systems such as water supply networks, natural gas pipelines, or electrical grids where fault localization follows similar principles.

1.4 The three field notation for the search problem

The multiplicity of variants for the problem motivates the following notation. Throughout this work we will use the following three field notation resembling the notation commonly used in task scheduling problems. Our notation will consist of the three fields: α , β and γ . The α field is the search space environment field resembling the machine environment. The β field is the query characteristics which resembles the job characteristics. The γ field is the objective function which we are trying to optimize. In order not to confuse the two notations we will separate the fields for the graph searching notation with double lines $||$ instead of $|$. The following table showcases example variants which may be considered:

α - search space	β - query characteristics	γ - objective value
P - paths	E - edge queries	C_{max} - maximum search time
T - trees	V - vertex queries	$\sum C_i$ - average search time
$POSET$ - POSETs	Q - any queries	$\sum U_i$ - throughput
G - graphs	c - cost function on queries	F_{max} - maximum flow time
HT - hypertrees	w - weight function on vertices	$\sum F_i$ - average flow time
HG - hypergraphs	d - due dates/deadlines	L_{max} - maximum lateness
	r - release times	$\sum L_i$ - average lateness
	$prec$ - precedences	T_{max} - maximum tardiness
		$\sum T_i$ - average tardiness

Table 1.1: Sample values for the three field notation for the search problem.

Note that due to the limited scope of this thesis most of the above variants will not be considered. However, we believe that the notation may prove useful in future works regarding the search problem. The striking resemblance between these two notations suggests that we can view the search problem as a specific variant of task scheduling, in which the search strategy is the schedule and the queries are the jobs. At the start of the schedule there is only one machine processing the queries. After completing each such job, the machine that was processing it is replaced by a new set of machines, one for each possible response to the query. Additionally, from this moment on each such machine only processes the queries to vertices belonging to response associated with it.

In contrast to the classical scheduling the search problem is not nearly as explored and most of the variants which can be constructed using the table above are not even mentioned in the literature. It also seems, that the search problem is usually harder than the corresponding scheduling variants. For example, the best algorithm known for the NP-hard variant $T||V, c||C_{max}$ achieves an $O(\sqrt{\log n})$ -approximation [Der+17]. A somewhat similar scheduling problem $P||C_{max}$ has a simple $\frac{4}{3}$ -approximation algorithm based on sorting the jobs according to their costs [Gra69], admits a PTAS for an unbounded number of machines [Leu89] and if the number of machines is bounded an FPTAS can be obtained [Sah76].

1.5 Many names, one problem

Throughout recent years the search problem has been continuously rediscovered under various names and definitions. The following list consists of different formulations under which the problem have been studied in the context of graphs:

- Binary Search [Knu73; OP06; Der+17; DMS19; EKS16; DW22; DW24; DLU25; DGW24; DLU21; DGP23],

- Tree Search Problem [Jac+10; Cic+14; Cic+16],
- Binary Identification Problem [Cic+12; KZ13],
- Ranking Colorings [Der06; Der08; DK06; DN06; LY98],
- Ordered Colorings [KMS95],
- Elimination Trees [Pot88],
- Hub Labeling [Ang18],
- Tree-Depth [NO06; BDO23],
- Partition Trees [Høg+21; Høg24],
- Hierarchical Clustering [CC17],
- Search Trees on Trees [BK22; Ber+22],
- LIFO-Search [GHT12].

Various different problem definitions stem from the learning theory including:

- Decision Tree [LN04; LLM; GNR10; SLC14],
- Bayesian Active Learning [GKR10; Das04; BH08],
- Discrete Function evaluation [CLS14],
- Tree Split [KPB99],
- Query Selection [BBS12].

1.6 The aim of the thesis

Hereby, we will be mostly concerned with the situation in which the input graph is tree. A motivation for this is twofold. Firstly, trees come up most often in the practical scenarios regarding the problem. Secondly, from the algorithmic perspective, the most interesting and structural results are obtained for trees. Beyond that, most of the algorithms with provable guarantees follow some simple greedy rule and the achieved approximations are far from the objective value. For example, the problem $T||V||C_{max}$ is solvable in linear time (the algorithm is non-trivial)[Sch89]. If we however allow arbitrary graphs $(G||V||C_{max})$ then the problem becomes NP-hard even in chordal graphs [DN06] and the best known approximation in general case is $O\left(\log^{\frac{3}{2}} n\right)$ which is trivially obtained via an almost blackbox use of the tree decomposition of the graph [Bod+98]. We will also be mostly concerned with the vertex query variant of the problem, since it is usually, the more general variant and we will only regard the worst case variant to not inflate the already wide scope of this thesis.

We conclude a series of experiments aimed at verifying whether the theoretical claims regarding the discussed algorithms are reflected in an experimental setup. In particular, we employ randomized techniques to generate various classes of inputs and test the performance of the implemented algorithms both in terms of running time and the quality of the solutions obtained.

1.7 Organization of the work

The second chapter serves as a more formal and detailed introduction necessary for further considerations. We formally restate all of the search models we are interested in and we recall the basic notions of graph theory required for the analysis.

The main part of the thesis is partitioned into two main chapters:

In the third chapter we focus ourselves on the formal analysis of the worst-case variants of the search problem on trees, including the presentation of the most interesting algorithmic results. We showcase exact and approximation algorithms and few hardness results for both uniform and non-uniform cost scenarios.

The fourth chapter is a description of the computer experiments conducted on four tree types (random, bounded degree, bounded diameter, and balanced) with three cost distributions (uniform, binomial, and exponential) in order to verify the theoretical claims regarding the performance of the previously presented algorithms.

The fifth chapter serves as a summary of our considerations and points the further research directions regarding this field.

Chapter 2

Notions and Definitions

2.1 Graph theory

A *graph* is a pair $G = (V(G), E(G))$ where $V(G)$ is the set of *vertices* and $E(G)$ is the set of *edges* which are unordered pairs of vertices. We denote $n(G) = |V(G)|$ and $m(G) = |E(G)|$, which are often abbreviated as n and m when the graph is clear from context. For $u, v \in V(G)$ by uv we denote the edge which connects them. A *subgraph* of a graph G is another graph G' formed from a subset of the vertices and edges of G . For any $V' \subseteq V(G)$ by $G[V']$ we denote the *subgraph induced* by V' in G (i. e. for every $u, v \in V'$ if $uv \in E(G)$, then also $uv \in E(G')$). Additionally, by $G - V'$ we denote the set of connected components occurring after deleting all vertices in V' from G . The set of *neighbors* of $v \in V(G)$ will be denoted as $N_G(v) = \{u \in V(G) \mid uv \in E(G)\}$ and the set of neighbors of subgraph G' of G as $N_G(G') = \bigcup_{v \in V(G')} N_G(v) - V(G')$. By $\deg_G(v) = |N_G(v)|$ we will denote the *degree* of v in G . By $\Delta(G) = \max_{v \in V(G)} \{\deg(v)\}$ we denote the degree of G .

A *cycle* is a non-empty sequence of vertices in which for every two consecutive vertices u, v : $uv \in E(G)$ and only the first and last vertices are equal. A *tree* T is a connected graph that contains no cycle. A *forest* is a (not necessarily connected) graph that contains no cycle. A *path* P is a tree such that $\Delta(P) = 2$. Let $v \in V(T)$. The set of *children* of v in T will be denoted as $\mathcal{C}_T(v)$. The *outdegree* of v in T will be denoted as $\deg_T^+(v) = |\mathcal{C}_T(v)|$. By $P_T(u, v) = T \setminus \{u, v\}$ we denote a path of vertices between u and v in T (excluding u and v). Analogously, for $V_1, V_2 \subseteq V(T)$ we define $P_T(V_1, V_2) = T \setminus (V_1 \cup V_2)$. For any V' we denote the minimal connected subtree of T containing all vertices from V' by $T[V']$.

2.2 Optimization problems

A *minimization problem* is one in which given an input I , the set of valid solutions S and a cost function $c : S \rightarrow \mathbb{R}^+$ we are required to find a solution $s^* \in S$ such that $c(s^*) = \min_{s \in S} \{c(s)\}$. Analogously, a *maximization problem* is one in which we are required to find a solution $s^* \in S$ such that $c(s^*) = \max_{s \in S} \{c(s)\}$. For both types of problems we define $\text{OPT}(I) = c(s^*)$. Given an instance (I, S, c) of a minimization problem such that $|I| = n$, an $\alpha(n)$ -approximation algorithm is an algorithm which always outputs a solution s such that:

$$\frac{c(s)}{\text{OPT}(I)} \leq \alpha(n)$$

Analogously, for a maximization problem such an $\alpha(n)$ -approximation algorithm is an algorithm which always outputs a solution s such that:

$$\frac{c(s)}{\text{OPT}(I)} \geq \alpha(n)$$

If $\alpha(n) = O(1)$ we say that the algorithm is a constant factor approximation algorithm for I . If $\alpha(n) = 1$ we say that the algorithm is an exact algorithm for I . For a minimization problem if for every $0 < \epsilon \leq 1$ the algorithm provides a $(1 + \epsilon)$ -approximation (or a $(1 - \epsilon)$ -approximation in case of a maximization problem) and:

- Runs in time $\text{poly}(n/\epsilon)$, then it is called a Fully-Polynomial Time Approximation Scheme (FPTAS).
- Runs in time $f(\epsilon) \cdot \text{poly}(n)$ for some computable function f , then it is called a Efficient-Polynomial

- Time Approximation Scheme (EPTAS).
- Runs in time $n^{O(1/\epsilon)}$, then it is called a Polynomial Time Approximation Scheme (PTAS).
- Runs in time $n^{\text{poly}(\log n/\epsilon)}$, then it is called a Quasi-Polynomial Time Approximation Scheme (QPTAS).

2.3 The graph search problem

Below we list the definitions regarding the search problem. Since the problem has a modular form and one can almost freely swap criteria and constraints, the number of separate variants is very large. Due to this we will present a general Graph Search Problem.

The *Graph Search Instance* consists of a pair $G = (V(G), E(G))$. Among $V(G)$ there is a unique hidden target element x which is required to be located. During the *Search Process* the searcher is allowed to iteratively perform a *query* which asks about chosen vertex v (or alternatively an edge e). By $G - v$ we denote the set of connected components of G after removing vertex v and all its incident edges. Similarly, $G - e$ denotes the set of connected components of G after removing edge e . If the answer to a query is affirmative, then v is the target, otherwise a connected component $H \in G - v$ is returned such that $x \in V(H)$ (for the edge version always $H \in G - e$ is returned). Based on this information the searcher narrows the subgraph of G which might contain x until there is only one possible option left.

Remark 2.3.0.1. *In the vertex query model we require that every vertex must be queried even when such vertex is the last among the candidate set. Note that it is sometimes assumed that in such case, this vertex does not need to be queried which may reduce the cost of the solution. Note that all of the algorithms showed in this work can be altered to take this assumption into account. For the sake of the brevity we do not include them but we encourage the reader to obtain them as an exercise.*

2.3.1 Additional input parameters

As a part of the input we will also allow the cost function. Let $\mathcal{Q}$ be the space of possible queries (either vertex or edge queries). The *cost* of query $q \in \mathcal{Q}$ is then denoted as $c : \mathcal{Q} \rightarrow \mathbb{R}^+$.

2.3.2 Decision trees, optimization criteria and the Graph Search Problem

Let G be a graph. A decision tree is a rooted tree $D = (V(D), E(D))$, where $V(D) = V(G)$ are the vertices of D and $E(D)$ are the edges of D . For a rooted tree T , by $r(T)$ we denote the root of T . It is required that each child of $q \in V(D)$ corresponds to a distinct response to the query at q , with respect to the subtree of candidate solutions that remain after performing all previous queries.

Let $Q_D(G, x)$ denote the sequence of queries made to locate a target $x \in V(G)$ using D , i. e., the sequence of vertices belonging to the unique path in D starting at $r(D)$ and ending at x . We define the worst case cost of a decision tree D in (G, c)

$$C_G(D, c) = \max_{x \in V(G)} \left\{ \sum_{q \in Q_D(G, x)} c(q) \right\}$$

By a slight abuse of notation we will also sometimes use $Q_D(G, x)$ as the set consisting of queries in sequence $Q_D(G, x)$. This is done in order to not inflate the amount of symbols and will not become problematic during the analysis of the solutions. Whenever clear from the context, for the clarity of the analysis, we will occasionally drop any of the subscripts or arguments of the C function. We are now ready to define the *Graph Search Problem*:

Generalized Search Problem

Input: Graph G , the query model and the optimization criterion

Output: A valid decision tree D for G with respect to the query model, which optimizes the criterion.

Chapter 3

Theoretical Analysis

The following chapter is concerned with the presentation and theoretical analysis of the algorithms for the Tree Search Problem in its worst-case variants. We partition the analysis into 2 main sections.

Firstly, we present a classic linear time algorithm for the uniform cost variant of the problem based on the equivalence of the ranking coloring and the Tree Search Problem [Sch89; OP06; MOW08].

Secondly, for the non-uniform cost model, we present previously known results including an $O(\log n / \log \log n)$ -approximation algorithm inspired by the algorithm for the edge query variant [Cic+16]. We also rewrite the QPTAS from [Der+17] using a somewhat different dynamic program inspired by the work on the average-case variant of the problem [Jac+10; Ang18]. As a result we also obtain an incremental improvement on the running of the QPTAS since the original construction yields an algorithm with running time roughly $n^{112898 \cdot \log n / \epsilon^2}$ while we managed to reduce this to approximately $n^{6965 \cdot \log n / \epsilon^2}$.

We then use this QPTAS to derive a polynomial time $O(\sqrt{\log n})$ -approximation algorithm and a $O(\log \log n)$ approximation algorithm running in $O(k^{O(\log n)} \cdot \text{poly}(n))$ time, where k is the k -up modularity of the cost function.

3.1 Trees, Worst Case, Uniform Costs

The *vertex ranking* of T is a labeling of vertices $l : V \rightarrow \{1, 2, \dots, \lceil \log n \rceil + 1\}$, which satisfies the following condition: for each pair of vertices $u, v \in V(T)$, whenever $l(u) = l(v)$, there exists $z \in \mathcal{P}_T(u, v)$ for which $l(z) > l(v)$. Such a labeling always exists and can be computed in linear time by means of dynamic programming [Sch89; OP06; MOW08]. For a visual example, see Figure 3.4

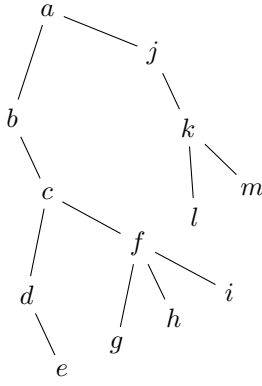


Figure 3.1

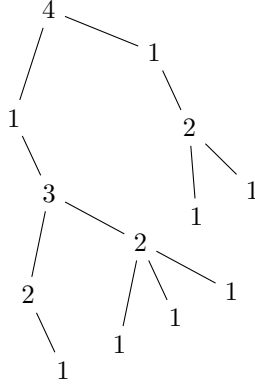


Figure 3.2

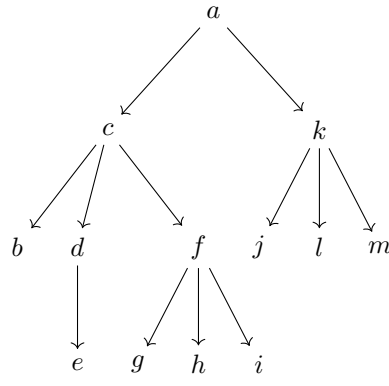


Figure 3.3

Figure 3.4: Sample input tree T (Figure 3.1), vertex ranking labeling l of T (Figure 3.2) and a decision tree D for T built using l (Figure 3.3).

Having a vertex ranking of T , one can easily obtain a decision tree for T using the following procedure:

1. Let $z \in V(T)$ be the unique vertex, such that for every $v \in V(T)$, $l(z) \geq l(v)$.
2. Schedule a query to z as the root of the decision tree D for T .
3. For each $T' \in T - z$, build a decision tree $D_{T'}$ recursively and hang it below the query to z in D .

When the input tree has uniform costs and the ranking uses the minimal number of labels, the decision tree built in this way is optimal and never uses more than $\lceil \log n \rceil + 1$ queries [OP06]. Let **RankingBasedDT** be the name of the latter procedure. We have the following corollary:

Corollary 3.1.0.1. *There exists an $O(n)$ time procedure **RankingBasedDT** that finds the optimal decision tree for the Tree Search Problem when all costs are uniform. Moreover, the depth of such a decision tree, i.e., the worst-case number of queries, is at most $\lceil \log n \rceil + 1$.*

We assume that the input tree is rooted at an arbitrary vertex. Below we show how to calculate the optimal vertex ranking of a given tree T . For any vertex $v \in V(T)$, and any coloring l we define the *visibility sequence* $S(v)$ as following: If there exists a vertex $u \in V(T)$, such that for every $z \in \mathcal{P}_T(u, v)$, $l(u) > l(z)$, then $l(u) \in S(v)$. We also demand that $S(v)$ is sorted decreasingly. For a given label k , $k \in S(v)$, we say that k is *visible* from v . In order to find a coloring using the minimal amount of colors, we will make use of the standard lexicographic order on visibility sequences. Notice, that for any vertex $v \in V(T)$, we have that $\max_{u \in V(T)} \{l(u)\} \in S(v)$. Therefore, a labeling with the lexicographically lowest $S(r(T))$ is also an optimal. We focus ourselves on finding such labeling. We devise the following dynamic programming procedure which calculates a labeling of T with a minimal visibility sequence of $r(T)$.

The following theorem is by [OP06]:

Theorem 3.1.0.2. *Let T be a tree. Algorithm 1 can be implemented in $O(n)$ time and returns a correct ranking labeling such that $S(r(T))$ is lexicographically optimal.*

Algorithm 1: The CalculateRanking procedure.

```

Procedure CalculateRanking( $T$ ):
  for  $1 \leq j \leq \deg_r^+(T)$  do
     $S(c_j) \leftarrow \text{CalculateRanking}(T_{c_j})$ .
   $m \leftarrow$  maximal value belonging to at least two distinct  $S(c_j)$ .
   $S(v) \leftarrow \bigcup_{j=1}^{\deg_r^+(T)} S(c_j)$ .
   $l(r(T)) \leftarrow \arg \min_{m < k \leq \log n + 1, k \notin S(v)} \{k\}$ .
  Append  $l(v)$  to  $S(v)$ .
  Remove any  $l < l(v)$  from  $S(v)$ .
  return  $S(v)$ .

```

3.2 Trees, worst case, non-uniform costs

The problem for non-uniform costs is NP-hard even when restricted to spiders of diameter 6 and binary trees [Cic+12; Cic+16]. A simple greedy heuristic which always queries the middle vertex of the graph achieves a $O(\log n)$ -approximation [Der06]. However one can obtain better results. We begin with the following simple lemma, which will become useful in few arguments:

Lemma 3.2.0.1. *Let T' be a connected subtree of T . Then, $\text{OPT}(T') \leq \text{OPT}(T)$.*

3.2.1 A warm up: $O(\log n / \log \log n)$ -approximation algorithm for $T||V, c||C$

This first algorithm is an adapted and simplified version of the algorithm due to [Cic+16] for the edge query model.

Theorem 3.2.1.1. *There exists a polynomial time, $O(\log n / \log \log n)$ -approximation algorithm for the $T||V, c||C_{\max}$ problem.*

Proof. To construct a decision tree we will use the following exact procedure:

Lemma 3.2.1.2. *There exists a $O(2^n n)$ algorithm for $T||V, c||C_{\max}$*

Proof. The algorithm is a general version of the dynamic programming procedure for paths. We have that:

$$\text{OPT}_{\max}(T) = \min_{v \in V(T)} \left\{ c(v) + \max_{H \in T-v} \{ \text{OPT}_{\max}(H) \} \right\}$$

There are at most $O(2^n)$ different subtrees of T to be checked. Additionally, for each $v \in V(T)$, there are at most $\deg_T(v)$ possible responses to check in the inner max function. Therefore, for each subproblem, there are at most $\sum_{v \in V(T)} \deg_T(v) = 2m = 2n - 2$

comparison operations to be performed. As at each level of the recursion the algorithm considers all possible choices of the next queried vertex v , it necessarily returns the optimal decision tree for T and the claim follows. $\square$

We will denote the above procedure. Let $k = 2^{\lceil \log \log n \rceil + 2}$. The basic idea of the Algorithm 2 is as follows: The algorithm is recursive. Let $\mathcal{T}$ be the tree currently processed by the algorithm. If $n(\mathcal{T}) \leq k$ then we call $\text{Exact}(\mathcal{T}, c)$ to find the optimal solution in time $2^k k = \text{poly}(n)$.

If otherwise, to build a solution we will firstly define a set $\mathcal{X} \subseteq V(\mathcal{T})$ which will be of size at most k . We build $\mathcal{X}$ iteratively. Starting with an empty set we pick the centroid x_1 of $\mathcal{T}$ which we add to $\mathcal{X}$. Then we take the forest $F = \mathcal{T} - x_1$, find the largest $H \in F$, pick its centroid x_2 and append it to $\mathcal{X}$. We continue this in $F - H + (H - x_2)$ until $|\mathcal{X}| = k$.

Lemma 3.2.1.3. *For every $H \in \mathcal{T} - \mathcal{X}$ we have that $n(H) \leq n(\mathcal{T}) / \log(n)$.*

Proof. We prove by induction on t that deleting first 2^t centroids from T each connected components H_t has size at most $n(H_t) \leq n(\mathcal{T})/2^{t-1}$. For the case when $t = 0$ we have that after 1 iteration every H_1 has size at most $n(\mathcal{T})/2 \leq 2(n)$ so the base of induction is complete.

Fix $t > 0$ and by assume by the induction hypothesis that after 2^{t-1} iterations all □

We also define set $\mathcal{Y} \subseteq V(\mathcal{T})$ which consists of vertices in $\mathcal{X}$ and all vertices in $v \in \mathcal{T}(X)$ such that $\deg_{\mathcal{T}(X)}(v) \geq 3$. Furthermore, we define set $\mathcal{Z} \subseteq V(\mathcal{T})$ as a set consisting of vertices in $\mathcal{Y}$ and for every $u, v \in \mathcal{Y}$ such that $\mathcal{P}_{\mathcal{T}}(u, v) \neq \emptyset$ and $\mathcal{P}_{\mathcal{T}}(u, v) \cap \mathcal{Y} = \emptyset$ we add to $\mathcal{Z}$ the vertex $\arg \min_{z \in \mathcal{P}_{\mathcal{T}}(u, v)} \{c(z)\}$ (for example see Figure 3.21). We then create an auxiliary tree $\mathcal{T}_{\mathcal{Z}} = (\mathcal{Z}, \{uv | \mathcal{P}_{\mathcal{T}}(u, v) \cap \mathcal{Z} = \emptyset\})$ (for example see Figure 3.22). The algorithm builds an optimal decision tree $D_{\mathcal{Z}}$ for $\mathcal{T}_{\mathcal{Z}}$ by applying the **Exact** procedure for $(\mathcal{T}_{\mathcal{Z}}, c)$. Observe, that $D_{\mathcal{Z}}$ is a partial decision tree for $\mathcal{T}$, so we get that:

Observation 3.2.1.4. $C_{D_{\mathcal{Z}}}(\mathcal{T}_{\mathcal{Z}}) = C_{D_{\mathcal{Z}}}(\mathcal{T})$.

Then for each $H \in \mathcal{T} - \mathcal{Z}$ we recursively apply the same algorithm to obtain the decision tree D_H and we hang it in $D_{\mathcal{Z}}$ below the unique last query to vertex in $N_{\mathcal{T}}(H)$ (By Observation 3.2.4.5).

Algorithm 2: Main recursive procedure (k is a global parameter)

```

Procedure DecisionTree( $\mathcal{T}, c$ ):
  if  $n(\mathcal{T}) \leq k$  then
     $D \leftarrow \text{Exact}(\mathcal{T}, c)$ .
    return  $D$ 
   $\mathcal{X} \leftarrow \emptyset$ .
   $\mathcal{F} \leftarrow \{\mathcal{T}\}$ .
  for  $1 \leq i \leq k$  do
    if  $\mathcal{F} = \emptyset$  then
      break
     $H \leftarrow \arg \max_{H \in \mathcal{F}} \{n(H)\}$ .
     $x \leftarrow$  the centroid of  $H$ .
     $\mathcal{X} \leftarrow \mathcal{X} \cup \{x\}$ .
     $\mathcal{F} \leftarrow \mathcal{F} \cup H - x$ .
   $\mathcal{Z} \leftarrow \mathcal{Y} \leftarrow \mathcal{X} \cup \{v \in \mathcal{T}(X) | \deg_{\mathcal{T}(X)}(v) \geq 3\}$ . // Branching vertices in  $\mathcal{T}(X)$ .
  foreach  $u, v \in \mathcal{Y}, \mathcal{P}_{\mathcal{T}}(u, v) \neq \emptyset, \mathcal{P}_{\mathcal{T}}(u, v) \cap \mathcal{Y} = \emptyset$  do
     $\mathcal{Z} \leftarrow \mathcal{Z} \cup \{\arg \min_{z \in \mathcal{P}_{\mathcal{T}}(u, v)} \{c(z)\}\}$ . // Lightest vertex on path  $\mathcal{P}_{\mathcal{T}}(u, v)$ .
   $\mathcal{T}_{\mathcal{Z}} = (\mathcal{Z}, \{uv | \mathcal{P}_{\mathcal{T}}(u, v) \cap \mathcal{Z} = \emptyset\})$ .
   $D \leftarrow D_{\mathcal{Z}} \leftarrow \text{Exact}(\mathcal{T}_{\mathcal{Z}}, c)$ .
  foreach  $H \in \mathcal{T} - \mathcal{Z}$  do
     $D_H \leftarrow \text{DecisionTree}(H, c)$ .
    Hang  $D_H$  in  $D$  below the last query to a vertex  $v \in N_{\mathcal{T}}(H)$ .
  return  $D$ 

```

Lemma 3.2.1.5. Let $\mathcal{T}_{\mathcal{Z}}$ be the auxiliary tree. Then, $|V(\mathcal{T}_{\mathcal{Z}})| \leq 4k - 3$.

Proof. We firstly show that $|\mathcal{Y}| \leq 2k - 1$. We use induction on the elements of set $\mathcal{X}$. For $1 \leq i \leq k$, let x_i denote the i -th centroid added to $\mathcal{X}$. We will construct a family of sets $\mathcal{X}_1, \mathcal{X}_2, \dots, \mathcal{X}_{|\mathcal{H}|}$, such that for every integer $1 \leq t \leq |\mathcal{X}|$: $|\mathcal{X}_t| = t$ and $\mathcal{X}_{|\mathcal{X}|} = \mathcal{X}$. For each $\mathcal{X}_t$, we will also construct a corresponding set $\mathcal{Y}_t$, eventually ensuring that $\mathcal{Y}_{|\mathcal{X}|} = \mathcal{Y}$. We will build the sets $\mathcal{Y}_t$ to ensure that $|\mathcal{Y}_t| \leq 2t - 1$.

Let $\mathcal{X}_1 = \{x_1\}$, $\mathcal{Y}_1 = \{x_1\}$. This establishes the base case. Assume by induction on $t \geq 1$ that $|\mathcal{Y}_t| \leq 2t - 1$ for some $t > 1$. Let $\mathcal{X}_{t+1} = \mathcal{X}_t \cup \{x_{t+1}\}$ and let $\mathcal{T}_t = \mathcal{T}(\mathcal{X}_t)$. If $x_t \in V(\mathcal{T}_t)$, then $\mathcal{Y}_{t+1} = \mathcal{Y}_t \cup \{x_t\}$. If otherwise, let $y_t \in V(\mathcal{T}_t)$ be the unique vertex, such that $P(x_t, y_t) \cap V(\mathcal{T}_t) = \emptyset$.

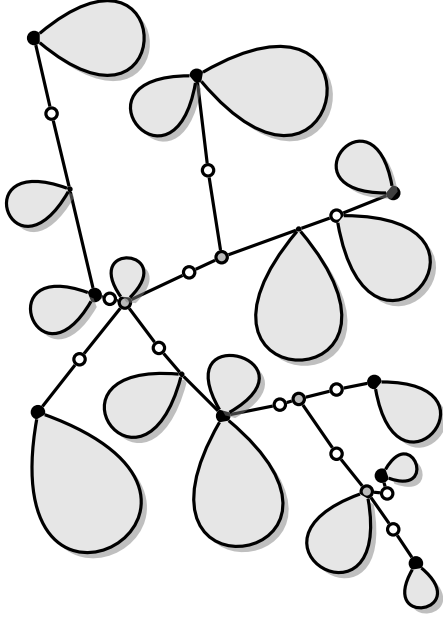


Figure 3.5: Example tree $\mathcal{T}$. Light grey regions represent light subtrees. Black vertices represent $\mathcal{X}$. Gray and black vertices represent $\mathcal{Y}$. White, gray and black vertices represent $\mathcal{Z}$. Lines represent paths of vertices between vertices of $\mathcal{Z}$.

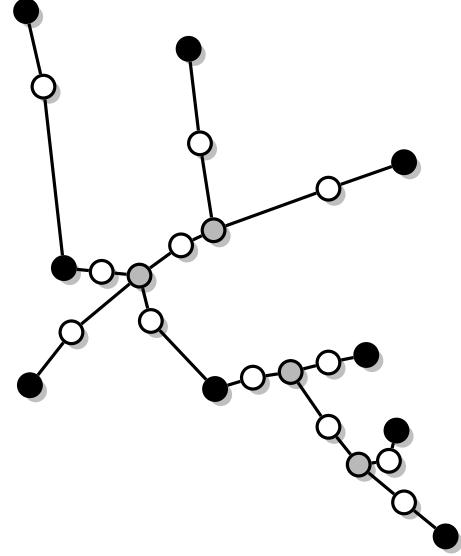


Figure 3.6: Auxiliary tree $\mathcal{T}_Z$ built from vertices of set $\mathcal{Z}$.

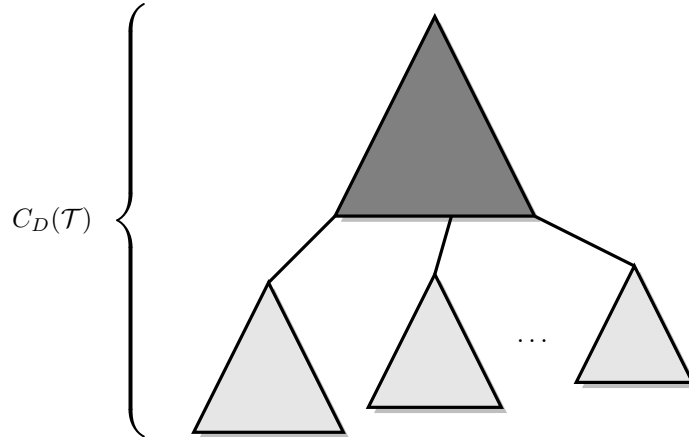


Figure 3.7: The structure of the decision tree D , built by the Algorithm 2. The dark gray subtree represents the decision tree D_Z , obtained by calling the **Exact** procedure for $\mathcal{T}_Z$ and c . Light gray subtrees represent decision trees D_L , built for each $H \in \mathcal{T} - \mathcal{Z}$, by recursively calling **DECISIONTREE** with H and c .

Then, $\mathcal{Y}_{t+1} = \mathcal{Y}_t \cup \{x_t, y_t\}$. As by induction, $|\mathcal{Y}_t| \leq 2t - 1$ and we add at most two vertices to it to obtain $\mathcal{Y}_{t+1}$, the induction step is complete.

By construction, $\mathcal{X}_{|\mathcal{X}|} = \mathcal{X}$ and $\mathcal{Y}_{|\mathcal{H}|} = \mathcal{Y}$, so $|\mathcal{Y}| \leq 2 \cdot |\mathcal{H}| - 1 \leq 2k - 1$. As paths between vertices in $\mathcal{Y}$ form a tree when contracted, at most $2k - 2$ additional vertices are added while constructing $\mathcal{Z}$ (at most one per path). The lemma follows. $\square$

Lemma 3.2.1.6. *Let $\mathcal{T}_{\mathcal{Z}}$ be the auxiliary tree. Then, $\text{OPT}(\mathcal{T}_{\mathcal{Z}}) \leq \text{OPT}(\mathcal{T})$.*

Proof. Let D^* be the optimal strategy for $\mathcal{T}(\mathcal{Z})$. We build a new decision tree $D'_{\mathcal{Z}}$ for $\mathcal{T}_{\mathcal{Z}}$ by transforming D^* : Let $u, v \in \mathcal{Y}$ such that $\mathcal{P}_{\mathcal{T}}(u, v) \neq \emptyset$ and $\mathcal{P}_{\mathcal{T}}(u, v) \cap \mathcal{Y} = \emptyset$. Let $q \in V(D^*)$ such that $q \in \mathcal{P}_{\mathcal{T}}(u, v)$ is the first query among vertices of $\mathcal{P}_{\mathcal{T}}(u, v)$. We replace q in D^* by the query to the distinct vertex $v_{u,v} \in \mathcal{P}_{\mathcal{T}}(u, v) \cap \mathcal{Z}$ and delete all queries to vertices $\mathcal{P}_{\mathcal{T}}(u, v) - v_{u,v}$ from D^* . By construction, $D'_{\mathcal{Z}}$ is a valid decision tree for $\mathcal{T}_{\mathcal{Z}}$ and as for every $z \in \mathcal{P}_{\mathcal{T}}(u, v)$: $c(v_{u,v}) \leq c(z)$ such strategy has cost at most $C_{D'_{\mathcal{Z}}}(\mathcal{T}_{\mathcal{Z}}) \leq \text{OPT}(\mathcal{T}(\mathcal{Z}))$. We get:

$$\text{OPT}(\mathcal{T}_{\mathcal{Z}}) \leq C_{D'_{\mathcal{Z}}}(\mathcal{T}_{\mathcal{Z}}) \leq \text{OPT}(\mathcal{T}(\mathcal{Z})) \leq \text{OPT}(\mathcal{T})$$

where the first inequality is due to the optimality and the last inequality is due to the fact that $\mathcal{T}(\mathcal{Z})$ is a subtree of $\mathcal{T}$ (by Lemma 3.2.0.1). The lemma follows. $\square$

Lemma 3.2.1.7. *Let $D_{\mathcal{T}}$ be the solution returned by the algorithm. Then the approximation factor of such solution is bounded by $\text{APP}_{D_{\mathcal{T}}}(\mathcal{T}) \leq \log n / \log \log n$.*

Proof. Let $\mathcal{T}$ be the tree processed at some level of the recursion and let $D_{\mathcal{T}}$ be the decision tree returned by the algorithm. The proof is by induction on the size of $\mathcal{T}$. We claim that $\text{APP}_{D_{\mathcal{T}}}(\mathcal{T}) \leq \max\{1, \log n(\mathcal{T}) / \log \log n\}$. If $n(\mathcal{T}) \leq k$ then $D_{\mathcal{T}}$ is the optimal decision tree for $\mathcal{T}$ which establishes the base case. Let $n(\mathcal{T}) > k$ and assume that claim holds for every $t < n(\mathcal{T})$. By construction, we have that:

$$\begin{aligned} \text{APP}_{D_{\mathcal{T}}}(\mathcal{T}) &= \frac{C_{D_{\mathcal{T}}}(\mathcal{T})}{\text{OPT}(\mathcal{T})} \\ &\leq \frac{C_{D_{\mathcal{Z}}}(\mathcal{T}) + \max_{H \in \mathcal{T} - \mathcal{Z}} \{C_{D_H}(H)\}}{\text{OPT}(\mathcal{T})} \\ &\leq \frac{C_{D_{\mathcal{Z}}}(\mathcal{T}_{\mathcal{Z}})}{\text{OPT}(\mathcal{T}_{\mathcal{Z}})} + \max_{H \in \mathcal{T} - \mathcal{Z}} \left\{ \frac{C_{D_H}(H)}{\text{OPT}(H)} \right\} \\ &\leq 1 + \frac{\log \left(\frac{n(\mathcal{T})}{\log n(\mathcal{T})} \right)}{\log \log n} = \frac{\log n(\mathcal{T})}{\log \log n} \end{aligned}$$

where the first inequality is by construction, the second is by usage of Observation 3.2.1.4, Lemma 3.2.1.6 and Lemma 3.2.0.1 and the last inequality is due to the Lemma 3.2.1.3 and the induction hypothesis. $\square$

Using the fact that the call to the exponential time procedure requires $O(2^{4k-3}(4k-3)) = \text{poly}(n)$ time (Due to Lemma 3.2.1.5), all other computations require polynomial time, and each $v \in V(T)$ belongs to $\mathcal{Z}$ at most once during the execution we get that the overall running time is polynomial in n . $\square$

In the above analysis we lose one factor of OPT per each level of recursion of which there are at most $O(\log n / \log \log n)$. Notice however, that we can allow some more loss (i. e. $c \cdot \text{OPT}$) without affecting the asymptotical approximation factor. As it turns out it is possible to obtain a constant factor approximation for this problem in quasipolynomial time. This is the main idea behind the improvement of the approximation factor for this problem as in such case the size of the set $\mathcal{Z}$ may be greater and less recursion levels are needed which directly improves the approximation.

3.2.2 An $O(\sqrt{\log n})$ -approximation algorithm for $T||V, c||C$

We begin with the following proposition [Der+17] about the existence of QPTAS for $T||V, c||C$:

Proposition 3.2.2.1. *For any $0 < \epsilon \leq 1$ there exists a $(1 + \epsilon)$ -approximation algorithm for the Tree Search Problem running in $2^{O(\frac{\log^2 n}{\epsilon^2})}$ time.*

In order to obtain the above QPTAS, we need to carefully analyze the structure of the decision trees used in the algorithm. The algorithm is very intricate and requires usage of an alternative notion of strategy, which is deferred to Section 3.2.3.

Theorem 3.2.2.2. *There exists a polynomial time, $O(\sqrt{\log n})$ -approximation algorithm for the $T||V, c||C_{max}$ problem .*

Proof. We use the same procedure as in the $O(\log n / \log \log n)$ -approximation algorithm, however we set $k = 2^{\lfloor \sqrt{\log n} \rfloor + 2}$ and we swap the exact procedure to the QPTAS with $\epsilon = 1$. The analysis of the algorithm is largely the same, except while evaluating the cost of the resulting decision tree.

Lemma 3.2.2.3. *Let D_T be the solution returned by the algorithm. Then the approximation factor of such solution is bounded by $APP_T(D_T) \leq 2\sqrt{\log n}$.*

Proof. Let $\mathcal{T}$ be the tree processed at some level of the recursion and let $D_{\mathcal{T}}$ be the decision tree returned by the algorithm. The proof is by induction on the size of $\mathcal{T}$. We claim that $APP_{\mathcal{T}}(D_{\mathcal{T}}) \leq \max\{1, 2\log n(\mathcal{T}) / \sqrt{\log n}\}$. If $n(\mathcal{T}) \leq k$ then $D_{\mathcal{T}}$ is the optimal decision tree for $\mathcal{T}$ which establishes the base case. Let $n(\mathcal{T}) > k$ and assume that claim holds for every $t < n(\mathcal{T})$. By construction, we have that:

$$\begin{aligned} APP_{D_{\mathcal{T}}}(\mathcal{T}) &= \frac{C_{D_{\mathcal{T}}}(\mathcal{T})}{OPT(\mathcal{T})} \\ &\leq \frac{C_{D_{\mathcal{Z}}}(\mathcal{T}) + \max_{H \in \mathcal{T}-\mathcal{Z}} \{C_{D_H}(H)\}}{OPT(\mathcal{T})} \\ &\leq \frac{C_{D_{\mathcal{Z}}}(\mathcal{T})}{OPT(\mathcal{T}_{\mathcal{Z}})} + \max_{H \in \mathcal{T}-\mathcal{Z}} \left\{ \frac{C_{D_H}(H)}{OPT(H)} \right\} \\ &\leq 2 + \frac{2\log\left(\frac{n(\mathcal{T})}{\sqrt{\log n}}\right)}{\sqrt{\log n}} = \frac{2\log n(\mathcal{T})}{\sqrt{\log n}} \end{aligned}$$

where the first inequality is by construction, the second is by usage of Observation 3.2.1.4, Lemma 3.2.1.6 and Lemma 3.2.0.1 and the last inequality is due to the Lemma 3.2.1.3 and the induction hypothesis. $\square$

$\square$

3.2.3 QPTAS for the $T||V, c||C$ problem

The following algorithm is a simplified version of the QPTAS provided in [Der+17]. The core idea of the algorithm is the same, however, our solution uses the language of decision trees instead of the language of sequence assignments, which makes the algorithm more intuitive.

For the rest of the analysis, without loss of the generality, we will assume that T is rooted in a vertex v minimizing $c(v)$. We will also assume that all costs are normalized, so that $\max_{v \in V(T)} \{c(v)\} = 1$. If not, the costs are scaled by dividing them by $\max_{v \in V(T)} \{c(v)\}$. Note that this operation does not affect the optimality of a strategy or the quality of an approximation.

Observation 3.2.3.1. *Let T be a tree such that $|V(T)| > 1$ and $c : V \rightarrow \mathbb{R}^+$ be a normalized weight function. Then, $1 \leq OPT(T) \leq \lfloor \log n \rfloor + 1$.*

Proof. The first inequality is due to the fact that there exists $v \in V(T)$, such that $c(v) = 1$ and for any decision tree D we have $v \in Q_D(T, v)$. The second inequality is due to the fact that we can always locate the target using $\lfloor \log n \rfloor + 1$ queries [OP06]. $\square$

Rounding

We will use the following rounding scheme which will allow us to discretize the space of possible solutions in order to process it efficiently. Let $p \in \mathbb{N}$, and $k = a/pn$ for some $a \in \mathbb{N}$. Define:

$$c'(v) = \begin{cases} \lceil c(v) \rceil_k, & \text{if } c(v) > pk, \text{ in which case the vertex will be called } \textit{heavy}, \\ \lceil c(v) \rceil_{\frac{1}{pn}}, & \text{otherwise, in which case the vertex will be called } \textit{light}. \end{cases}$$

Lemma 3.2.3.2.

$$\text{OPT}(T, c') \leq \left(1 + \frac{2}{p}\right) \cdot \text{OPT}(T, c)$$

Proof. Let D^* be an optimal strategy for (T, c) . By definition, we have that for every vertex $v \in V(T)$, $c'(v) \leq \left(1 + \frac{1}{p}\right) \cdot c(v) + \frac{1}{pn}$ and therefore:

$$\begin{aligned} \text{OPT}(T, c') &\leq C_{D^*}(T, c') = \max_{v \in V(T)} \left\{ \sum_{q \in Q_{D^*}(T, v)} c'(q) \right\} \\ &\leq \max_{v \in V(T)} \left\{ \sum_{q \in Q_{D^*}(T, v)} \left(\left(1 + \frac{1}{p}\right) \cdot c(v) + \frac{1}{pn} \right) \right\} \\ &\leq \frac{1}{p} + \left(1 + \frac{1}{p}\right) \cdot \max_{v \in V(T)} \left\{ \sum_{q \in Q_{D^*}(T, v)} c(v) \right\} \leq \left(1 + \frac{2}{p}\right) \text{OPT}(T, c) \end{aligned}$$

where in the third inequality we used the fact that for every $v \in V(T)$, $|Q_{D^*}(T, v)| \leq n$ and in the last inequality we used Observation 3.2.3.1. $\square$

While calculating the decision tree, we will divide the time into boxes of duration k , which will be further subdivided into a identical slots of length $\frac{1}{pn}$. Let t_q denote the start of some query in a decision tree D . Note that the numbers t_v provide a complete information about any decision tree, and are an equivalent representation of any strategy. We will assume that for any heavy vertex $v \in V(T)$, t_v is an integer multiple of c and for any light vertex $v \in V(T)$, t_v is an integer multiple of $\frac{1}{pn}$.¹ We have the following lemma:

Lemma 3.2.3.3. *There exists a decision tree D for (T, c') , such that $C_D(T, c') \leq \left(1 + \frac{3}{p}\right) \cdot \text{OPT}(T, c')$ and for every vertex $v \in V(T)$ we have:*

1. *if $c(v) > pk$, then $t_v/k \in \mathbb{N}$ (every heavy query is aligned to a multiple of c),*
2. *if $c(v) \leq pk$, then $t_v/pn \in \mathbb{N}$ (every light query is aligned to a multiple of $\frac{1}{pn}$).*

Proof. Let D^* be any optimal decision tree for (T, c') . For any $v \in V(T)$, let t_v^* be the start of query to v in D^* and let $t'_v = \left(1 + \frac{2}{p}\right) t_v^*$, thus construction a new decision tree D' . Since in this new decision tree D' , the ordering of vertices is exactly the same as in D^* , for any two consecutive queries v, u in D' we have:

$$t'_u - t'_v = \left(1 + \frac{2}{p}\right) \cdot (t_u - t_v) \geq \left(1 + \frac{2}{p}\right) \cdot c(v)$$

We now construct D as follows: If $v \in V(T)$ is heavy, we assign $t_v = \lceil t'_v \rceil_k$ and $t_v = t'_v$ otherwise. For any two consecutive queries v, u in D , such that v is heavy we have:

$$t_u - \lceil t'_v \rceil_k > t_u - t_v - k \geq \left(1 + \frac{2}{p}\right) \cdot c(v) - k > w(v) + k > c'(v)$$

¹Note that by doing so, we allow decision trees to contain idle time intervals, in which no queries are scheduled. However, if this occurs, after obtaining such decision tree, we simply delete the idle times, which results in a valid decision tree

So we conclude that no two queries overlap. To obtain the second part of the claim, we round up the starting time of each query to a light vertex in D to an integer multiple of $\frac{1}{pn}$. We have:

$$\begin{aligned} C_D(T, c') &\leq \max_{v \in V(T)} \left\{ \sum_{q \in Q_{D^*}(T, v)} \left(\left(1 + \frac{2}{p}\right) \cdot c'(v) + \frac{1}{pn} \right) \right\} \\ &\leq \frac{1}{p} + \left(1 + \frac{2}{p}\right) \cdot \max_{v \in V(T)} \left\{ \sum_{q \in Q_{D^*}(T, v)} c'(v) \right\} \leq \left(1 + \frac{3}{p}\right) \text{OPT}(T, c') \end{aligned}$$

where in the third inequality we used the fact that for every $v \in V(T)$, $|Q_D(T, v)| \leq n$ and in the last inequality we used Observation 3.2.3.1. $\square$

We will call a decision tree fulfilling above conditions *aligned*. In subsequent considerations, we will focus ourselves of finding such decision trees, whose properties will allow us to devise an efficient dynamic programming procedure finding an optimal, aligned decision tree.

Up and down responses, heavy module contraction, u

Recall the following definitions. Since our decision tree is rooted, we can reasonably talk about up and down responses to a query. An *up* response to a query to v in T occurs when the connected component $\in T - v$, which is the reply, happens to contain $r(T)$. If this is not the case, then such response is called a *down* response. As it will turn out, a repeating occurrence of light queries with down responses will become problematic for our algorithm. To account for this issue we will use the following notions:

We will define a new measure of cost for aligned decision trees called the *aligned cost*. Let D be any aligned strategy for (t, c') . For any vertex $v \in V(T)$ and query $q \in Q_D(T, v)$ the contribution $\kappa_{T, c, k}(q, v)$ of u is defined as:

$$\kappa_{T, c}(q, v) = \begin{cases} \lfloor t_v + c(q) \rfloor_k - t_v, & \text{if } c(v) \leq pk \text{ and the response to query } q \text{ in } T, \text{ towards } v \text{ is down,} \\ c(q), & \text{otherwise.} \end{cases}$$

Then, the *aligned cost* of D is defined as:

$$C'_D(T, c', k) = \max_{v \in V(T)} \left\{ \sum_{q \in Q_D(T, v)} \kappa_{T, c', k}(q, v) \right\}.$$

Let $\text{OPT}'(T, c', k)$ denote the optimal aligned cost among all aligned decision trees for (T, c', k) . Notice that in the above cost, we lose k query time per light query with a down response. Since of course, the amount of such queries may be of order $O(n)$, the difference between $C'_D(T, c', k)$ and $C_D(T, c', k)$ may grow almost arbitrarily large. However, we will make sure that this does not happen too often, which will give us the desired bound on the cost of the solution. We have the following simple observation:

Observation 3.2.3.4. *Let T' be a subtree of T . Then, $\text{OPT}'(T') \leq \text{OPT}'(T)$.*

We define a *heavy module* as $H \subseteq V(T)$ such that: $T[H]$ is connected, every $v \in H$ is heavy, i. e., $c(v) \geq pk$ and H is maximal - no vertex can be added to it without violating one of its properties. A *contraction* of a heavy module H is an operation which consists of deleting all of the vertices in H from T and connecting every vertex u which was a child of some vertex in H to the parent of $r(T \setminus H)$ if it exists. For example see Figure 3.8.

The main procedure

We will use the following propositions:

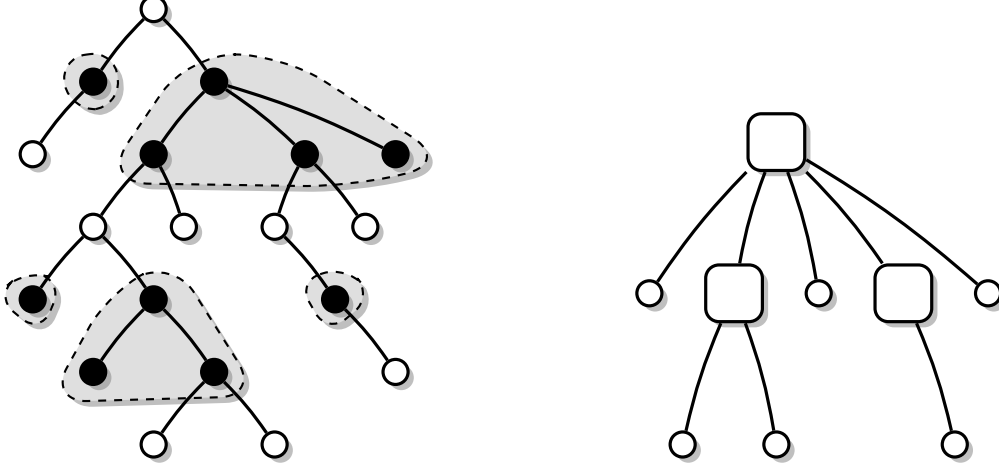


Figure 3.8: Example of contracting 5 heavy modules. Black vertices represent heavy vertices, white vertices represent light vertices and square vertices represent vertices which were a parent of at least one heavy module before contraction.

Proposition 3.2.3.5. *Let T be a tree, c' an aligned cost function, $p \in \mathbb{N}$, k the box size and $d \in \mathbb{N}$ be the depth. There exists a **DPTimelinesCosts** procedure, which (if it exists) calculates an aligned decision tree D for (T, c') of cost at most $C'_D(T, c', k) \leq kd$, running in $(pn)^{O(d)}$ time.*

Proposition 3.2.3.6. *Let T be a tree, c' be an aligned cost function, $p \in \mathbb{N}$, $k \in \mathbb{R}_{>0}$ be the box size, D_A be a decision tree for T and F_C be forest of decision trees for T with all heavy modules contracted. There exists a polynomial time **MergeDTs** procedure which returns a decision tree of cost at most:*

$$C_D(T, c', k) \leq C'_{D_A}(T, c', k) + 2pk \cdot C_{F_C}(T, 1).$$

The proofs will be provided in the further sections. We will now prove the Proposition 3.2.2.1.

Algorithm 3: The QPTAS for $T||V, c, w||C$.

Procedure QPTAS(T, c, ϵ):

- $p \leftarrow \lceil 59/\epsilon \rceil$.
- $d \leftarrow p^2 \cdot (\lfloor \log(n) \rfloor + 1)$.
- $k \leftarrow 0$.
- while** true **do**
 - $k \leftarrow k + \frac{1}{pn}$.
 - foreach** $v \in V(T)$ **do**
 - if** $c(v) > pk$ **then**
 - $c'(v) \leftarrow \lceil c(v) \rceil_k$.
 - else**
 - $c'(v) \leftarrow \lceil c(v) \rceil_{\frac{1}{pn}}$.
 - $D_A \leftarrow \text{DPTimelinesCosts}(T, c', p, k, d)$
 - if** $D_A \neq \emptyset$ **then**
 - $T_C \leftarrow T$ with all heavy modules contracted.
 - $D_C \leftarrow \text{RankingBasedDT}(T_C)$.
 - $D \leftarrow \text{MergeDTs}(T, D_A, D_C)$.
 - return** D .

Algorithm 3 starts by picking $p = \lceil \frac{59}{\epsilon} \rceil$, $d = p^2 \cdot (\lfloor \log n \rfloor + 1)$ and $k = 0$ and. At each iteration of the repeat loop, the algorithm picks k to be the next integer multiple of $\frac{1}{pn}$ and performs the

rounding operation. After that, the Proposition 3.2.3.5 is applied for (T, c', p, k, d) . If the returned decision tree $D_A \neq \emptyset$, then a new tree T_C is constructed by contracting all heavy modules of T . By calling the **RankingBasedDT** for T_C , the algorithm builds a second decision tree D_C , and then merges it with D_A by applying Proposition 3.2.3.6. Then, the algorithm returns the resulting decision tree D .

Let k' be the value of k for which D was built. Let $k'' = k - \frac{1}{pn}$ and c'' be the values of k and c' of the previous iteration of the while loop. Since we know that for k'' and c'' we had $D_A = \emptyset$, by Proposition 3.2.3.5 we have that $k''d \leq \text{OPT}'(T, c'')$. Hence, $k' \leq \frac{\text{OPT}'(T, c'', k'')}{d} + \frac{1}{pn} \leq \frac{2 \cdot \text{OPT}'(T, c'', k'')}{p^2 \cdot (\lfloor \log n \rfloor + 1)}$, so we have that:

$$\begin{aligned} C_D(T, c') &\leq \text{OPT}'(T, c', k') + 2pk' \cdot (\lfloor \log n \rfloor + 1) \\ &\leq \text{OPT}'(T, c'', k'') + 2p \cdot (\lfloor \log n \rfloor + 1) \cdot \frac{2 \cdot \text{OPT}'(T, c'', k'')}{p^2 \cdot (\lfloor \log n \rfloor + 1)} \\ &\leq \left(1 + \frac{4}{p}\right) \cdot \text{OPT}'(T, c'', k'') \leq \left(1 + \frac{2}{p}\right) \cdot \left(1 + \frac{3}{p}\right) \cdot \left(1 + \frac{4}{p}\right) \cdot \text{OPT}(T, c) \\ &\leq \left(1 + \frac{59}{p}\right) \cdot \text{OPT}(T, c) = \left(1 + \frac{59}{\lceil \frac{59}{\epsilon} \rceil}\right) \cdot \text{OPT}(T, c) \leq (1 + \epsilon) \cdot \text{OPT}(T, c) \end{aligned}$$

where the first inequality is by Proposition 3.2.3.6, the second inequality is by the fact that $\text{OPT}(T, c', k') \leq \text{OPT}(T, c', k'')$ and by applying Corollary 3.1.0.1 and the fourth inequality is by the fact that $\text{OPT}(T, c', k'') \leq \text{OPT}(T, c)$, Lemma 3.2.3.2 and Lemma 3.2.3.3.

We can assume that $c = \text{poly}(n)$, since beyond that the problem can be solved to optimality in $O(2^n n)$ time. Therefore the running time of the procedure is bounded by:

$$n^{O(d)} = n^{O(p^2 \log n)} = n^{O(\log n / \epsilon^2)}.$$

Proof of Proposition 3.2.3.6

Algorithm 4: The MergeDTs procedure.

```

Procedure MergeDTs( $T, D_A, F_C$ ):
  if  $F_C$  is connected then
     $r \leftarrow r(F_C)$ .
  else
     $r \leftarrow r(D_A)$ .
   $D \leftarrow (\{r\}, \emptyset)$ . foreach  $T' \in T - r$  do
     $D' \leftarrow \text{MergeDTs}(T, D_A|T', F_C|T')$ .
    Hang  $D'$  below  $r$  in  $D$ .
  return  $D$ .

```

Algorithm 4 takes as arguments a tree T , a decision tree D_A and a forest of decision trees F_C for T with heavy groups contracted. Since F_C may not be a valid partial decision tree for T , it may happen that it is disconnected. This because, after performing a light query to v according to F_C , if the response was down and v has heavy child modules, the children of v in T_C , may not be separated yet. Based on this, we derive two cases for our procedure:

1. F_C is connected. In such case we pick $r = r(F_C)$.
2. F_C is disconnected. In such case we pick $r = r(D_A)$.

We set $D = (\{r\}, \emptyset)$ and then, for each $T' \in T - r$, we build a decision tree D' for T' recursively by calling **MergeDTs** with arguments $(T, D_A|T', F_C|T')$ and hang D' below r in D .

Lemma 3.2.3.7. *Let D be the decision tree returned by the MergeDTs procedure. Then, $C_D(T, c', k) \leq C'_{D_A}(T, c', k) + 2pk \cdot C_{F_C}(T, 1)$.*

Proof. Let $x \in V(T)$. We have two cases:

1. r is heavy or r is light and the response is up. In this case the cost of query to r is incorporated into $C'_{DA}(T, c', k)$.
2. r is light and the response is down. Let r_1 and r_2 be two consecutive light queries in $Q_D(T, x)$ with a down response, and let $T'' \in T - \{r_1, r_2\}$, such that $x \in T''$. We will show that $C_{FC|T}(T, 1) \leq C_{FC}(T, 1) - 1$, so the additional cost of such queries is at most $2pk \cdot C_{FC}(T, 1)$.
 - (a) F_C is connected. In such case $r_1 = r(F_C)$ so after query to r_1 , by definition of the decision tree $C_{FC|T'}(T, 1) \leq C_{FC}(T, 1) - 1$.
 - (b) F_C is disconnected. Let T' be any down response to query to r . If $r_1 = r(F_C|T')$, then $C_{FC|T'}(T', 1) \leq C_{FC}(T, 1) - 1$. Otherwise, we argue that $F_C|T'$ is connected. Assume contrary. Firstly, we observe that $F_C|T_{r_1}$ is connected. This is because, r_1 is light, so the tree $T_{r_1, C}$ denoting T_{r_1} with all heavy modules contracted is connected. By assumption, there at least two connected components $D_1, D_2, \dots, D_j$ of $F_C|T'$, with roots $d_1, d_2, \dots, d_j$ respectively. Since every other query $q \in V(F_C|T')$ is a descendant of some $d_1, \dots, d_j$, the only query which could be the parent of $d_1, \dots, d_j$ in $F_C|T'$ is r_1 , which is a contradiction. Therefore, the next query r_2 of D in T' is $r(F_C|T')$. We get that by definition of the decision tree $C_{FC|T''}(T, 1) \leq C_{FC}(T, 1) - 1$.

□

Dynamic programming procedure for a fixed box size

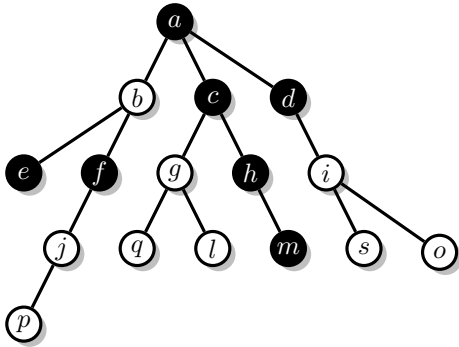


Figure 3.9

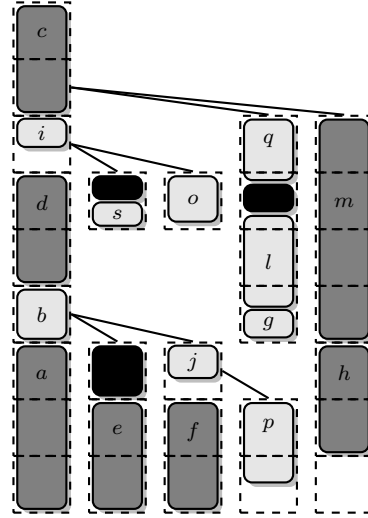


Figure 3.10

Figure 3.11: Structure of a boxed decision tree. Figure 3.9 shows example input tree T , black vertices are heavy and white vertices are white. Figure 3.10 shows example boxed decision tree D for T , dark gray queries are heavy, light gray queries are light, and black spaces represent additional load of boxes.

To devise our dynamic programming procedure, we will need the following generalization of hierarchical decision trees. A *boxed decision tree* $D = (V(D), E(D), u, l)$ for the tree T is a tuple, in which $V(D)$ are the nodes of the decision trees, which we will call *boxes*, $E(D)$ are edges of the decision tree, $u : V(T) \times V(D) \rightarrow \{0, 1/pn, 2/pn, \dots, k\}$ is the *usage* function and $l : V(D) \rightarrow \{0, 1/pn, 2/pn, \dots, k\}$ is the *load* function. Based on u , for every $b \in V(D)$ we will also define the *query assignment* as following: $Q(b) = \{v \in V(T) \mid u(v, b) > 0\}$, these are the vertices of T , such that queries to them overlap with the box b .

The boxed decision tree is defined analogously as ordinary decision tree, however, to each query in $b \in V(D)$ instead of one vertex of $V(T)$, we assign an arbitrary subset of $V(T)$, via the query

assignment function Q . Let $b \in V(D)$, and let T' be a candidate subtree of T right before any vertex in $Q(b)$ was queried. A box p is a left child of b , if either b corresponds to a subtree $T'' \in T - Q(b)$ such that $r(T') \in V(T'')$ or $Q(b) \cup Q(p) \neq \emptyset$. If otherwise, then p is a right child of b .

We demand that for every $v \in V(T)$, all boxes $b \in V(D)$, such that $v \in Q(b)$ form a connected path in D , in which every child is a left child, for any interior box b of this path, $l(b) = 0$ and $u(v, b) = k$. We will also require that: for every $q \in V(D)$, $l(b) + \sum_{v \in V(T)} u(v, b) \leq k$ and for every $v \in V(T)$ either $\sum_{b \in V(D)} (v, b) = 0$, in which case such query is called *unassigned* or $\sum_{b \in V(D)} (v, b) = c(v)$ and the query is called *assigned*. Since we want the boxed decision tree to also be aligned, we will demand that if for any vertex $v \in V(T)$, $c(v) > pk$, then for every $b \in V(D)$, either $u(v, b) = 0$ or $u(v, b) = k$.

We now define how to search in T using a boxed decision tree D . Firstly, the query process stops for time $l(u)$. If $Q(r(D)) \cup V(T) = \emptyset$, then we recurse on the appropriate child of $r(T)$ (which corresponds to our current candidate subtree). Otherwise, we pick the least costly vertex v in $Q(r(D))$, we remove v from $Q(q)$ for any $q \in V(D)$ and we recurse on the last box which contained a query to v (note that this can also be $r(v)$). This is done in order to ensure that the start of the query t_v of each vertex v happens in the box containing it. The aligned cost of searching using a boxed decision tree is defined analogously as the aligned cost of searching using ordinary decision tree. Note that any boxed decision tree, can also be transformed to an equivalent aligned decision tree with the same aligned cost of searching. Conversely, any aligned decision tree can be transformed into a boxed decision tree, by subdividing the queries into boxes according to their starting points. Note that since we do not count the cost of light down responses, by sorting all queries starting in a given box according to their cost, we cannot increase the aligned cost.

We define a *boxline* $B\langle(b_1, \tau_1), (b_2, \tau_2), \dots, (b_d, \tau_d)\rangle$ to be a sequence of pairs, each consisting of box and additional boolean flag, such that for every box b in B , $Q(b) = \emptyset$. We will build our decision trees around boxlines. Define the *left box-path* $B_D = \langle q_1, f_1, (q_2, f_2), \dots, (q_h, f_h) \rangle$ of D as a sequence of pairs. Each such pair consists of the overall loads of consecutive boxes obtained by traversing D starting from root $r(D)$, and stepping to the left child until there is none (for each such box b with index j , $q_j = l(b) + \sum_{v \in Q(b)} q(v, b)$), and boolean values denoting whether there exists a query transcending the current box unto the next one (f_h is always false, however we include it for convenience). We will say that a decision tree D with a left box-path B_D is *box-compatible* with a boxline B , such that $h \leq d$, if for every integer $1 \leq j \leq h$, $l(q_j) \geq l(b_j)$ and if τ_j implies f_j . To build a decision D tree using B , we simply create a boxed decision tree D consisting of path of vertices $\langle q_1, \dots, q_h \rangle$, such that $l(q_j) = l(b_j)$ and $Q(q_j) = \emptyset$.

We will also use the following operations:

- Putting a query to vertex v at s -th slot of a box b :

1. $\sigma(v) \leftarrow c(v)$;
2. **while** $\sigma(v) > 0$:
 - (a) $u(v, b) \leftarrow \min\{k - s/pn, \sigma(v)\}$.
 - (b) $\sigma(v) \leftarrow \sigma(v) - u(v, b)$.
 - (c) $b \leftarrow$ left child of b .

If such operation violates the definition of D or query to v transcends any box b_j , such that τ_j we mark D as *conflicted*.

- Building a decision tree D based on boxline B :

1. $D \leftarrow \emptyset$.
2. **for** $1 \leq j \leq |B|$:
 - (a) Create box q_j in D .
 - (b) $l(j) \leftarrow b_j$.
 - (c) $Q(q_j) \leftarrow \emptyset$.
 - (d) **if** $j > 1$ **then**: hang q_j as the left child of q_{j-1} .
3. **return** D .

Algorithm 5: The dynamic programming procedure finding $\text{OPT}'(T_{v,i}, B)$ (k, p, c, n and d are global parameters).

```

Procedure DPTimelinesCosts( $T_{v,i}, B$ ):
  if  $i = 0$  then
    for  $1 \leq b \leq d$  and  $0 \leq s \leq (k/pn \text{ if } c(v) > pk \text{ else } 0)$  do
       $D \leftarrow$  a decision tree based on  $B$ .
      Put query to  $v$  at the  $s$ -th slot of  $q_b$ .
      if  $D$  is not conflicted. then
        if  $C'_D(T_{v,i}, c', k) \leq dk$  then: return  $D$ , else: return  $\emptyset$ .
    return  $\emptyset$ .
   $\mathcal{D} \leftarrow \emptyset$ .
  if  $i = 1$  then
    for  $1 \leq b \leq d$  and  $0 \leq s \leq (k/pn \text{ if } c(v) > pk \text{ else } 0)$  do
       $D \leftarrow$  a decision tree based on  $B$ .
      Put query to  $v$  at the  $s$ -th slot of  $q_b$ .
      if  $D$  is not conflicted and  $C'_D(T_{v,i}, c', k) \leq dk$  then
        if  $C'_D(T_{v,i}, c, k) \leq dk$  then
           $B' \leftarrow$  left box-path of  $D$ .
           $h \leftarrow$  index of the last box  $q_h$  occupied by the query to  $v$ .
          for  $h < j \leq d$  do
             $b'_j \leftarrow 0$ .
             $t'_j \leftarrow \text{false}$ .
           $D' \leftarrow \text{DPTimelinesCosts}(T_{c_1}, c', B')$ .
          Put query to  $v$  at the  $s$ -th slot of  $q'_b$ .
          Rotate  $D'$  around  $v$ .
           $\mathcal{D} \leftarrow \mathcal{D} \cup \{D \text{ and } D' \text{ with their left paths aligned}\}$ .
  else
    foreach bipartition  $(B_1, B_2)$  of  $B$  do
       $D_1 \leftarrow \text{DPTimelinesCosts}(T_{v,i-1}, c', B_1)$ .
       $h \leftarrow$  index the last box  $q_{1,h}$  occupied by the query to  $v$ .
      for  $h \leq j \leq d$  do
         $b_{2,j} \leftarrow 0$ .
         $t_{2,j} \leftarrow \text{false}$ 
       $D_2 \leftarrow \text{DPTimelinesCosts}(T_{c_i}, c', B_2)$ .
      Put query to  $v$  at the  $s$ -th slot of  $q_{2,b}$ .
      Rotate  $D_2$  around  $v$ .
       $\mathcal{D} \leftarrow \mathcal{D} \cup \{D_1 \text{ and } D_2 \text{ with their left paths aligned}\}$ .
  return  $\arg \min_{D \in \mathcal{D}} \{C'_D(T_{v,i}, c, k)\}$ .

```

- Rotating a decision tree D around vertex $v \in V(D)$:
 1. $q_h \leftarrow$ the box containing the end of query to v .
 2. Sort queries starting in $Q(q_h)$ according to c' .
 3. Create box q .
 4. Move queries from $Q(q_h)$ to $Q(q)$, so that all queries after v are in $Q(q)$.
 5. Hang q'_h as a right child of q_h .
 6. Rehang left child of q_h as the left child of q .
- Bipartitioning of B . A bipartition of a boxline B consists of a pair of boxlines (B_1, B_2) such that:
 - $|B| = |B_1| = |B_2|$.
 - $l(b_{1,j}) + l(b_{2,j}) - k = l(b_j)$.
 - $(\tau_{1,j} \wedge \tau_{2,j} \iff \tau_j)$.
 - $(\tau_{1,j} \vee \tau_{2,j})$.
- Aligning D_1 and D_2 by their left paths to create a new decision tree D :
 1. $D \leftarrow \emptyset$.
 2. **for** $1 \leq j \leq |B|$:
 - (a) Create box q_j in D .
 - (b) $l(j) \leftarrow l(q_{1,j}) + l(q_{2,j}) - k$.
 - (c) **for** $v \in V(T)$: $u(v, q_j) \leftarrow \max\{u(v, q_{1,j}), u(v, q_{2,j})\}$.
 - (d) Hang all right children of $q_{1,j}$ and $q_{2,j}$ below q_j .
 - (e) **if** $j > 1$ **then**: hang q_j as the left child of q_{j-1} .
 3. **return** D .

We now introduce the subproblems which our dynamic programming solves. A problem $\text{OPT}'(T_{v,i}, B)$ consists of finding an optimal boxed decision tree for the tree $T_{v,i}$, which is box-compatible with B . If no B is given, we assume that $B = \langle (b_1, \tau_1), \dots, (b_d, \tau_d) \rangle$, where for every $1 \leq j \leq d$, $b_j = 0$ and τ_j is false. The algorithm computes the solutions in a bottom-up, left-to-right manner. If at any point there is no way to create an extended decision tree with given parameters we simply declare such instance *unfeasible*. We will now show how to compute $\text{OPT}(T_{v,i}, B)$ efficiently. The Algorithm 5 consists of 3 cases:

1. $T_{v,0}$. We start by building a decision tree D based on B . Then, we greedily pick the box with smallest index $1 \leq b \leq d$, such that there exists slot s for which it is possible to place query to v at the s -th slot of q_b , without introducing conflicts. If there is no such index, we declare the subproblem unfeasible. In other case, the solution obtained by taking timeline P and setting $p_k = v$.
2. $T_{v,1}$. Assume that we have already solved all the subproblems of T_u . We again start by building a decision tree D based on B . Then, for any $1 \leq b \leq d$, such that there exists slot s for which it is possible to place query to v at the s -th slot of q_b , without introducing conflicts we put such query. Let h be the index of last box q_h occupied by v . We set the load of each next consecutive box in B' to be 0 and we set all further boolean flags to false. We then retrieve the optimal decision tree D' for (T_{c_1}, B') , put query to v in D' at the s -th slot of box q'_b , rotate D' around v and align D' with D .

Then, among all of such obtained decision trees we pick the one which minimizes the aligned cost.

3. $T_{v,i}$ for $i > 1$, we assume that we have already solved all the subproblems of $T_{v,i-1}$ and T_{c_i} . We consider all bipartitions (B_1, B_2) of B . We retrieve the optimal decision tree D_1 for $(T_{v,i-1}, B_1)$. Let h be the index of last box q_h occupied by v in D_1 . We set the load of each next consecutive

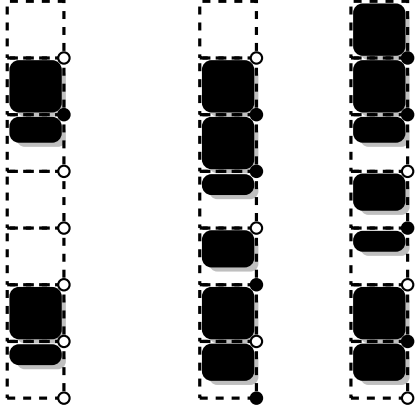


Figure 3.12



Figure 3.13

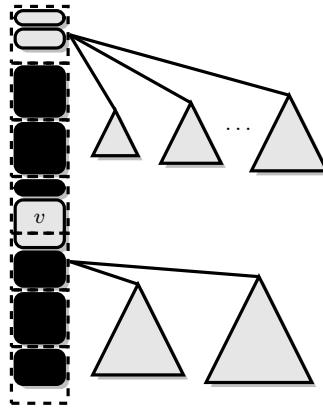


Figure 3.14



Figure 3.15

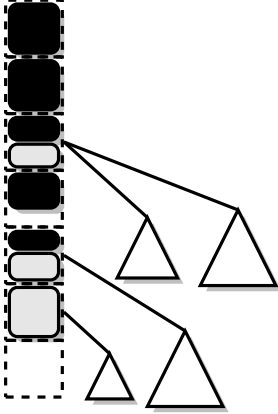


Figure 3.16

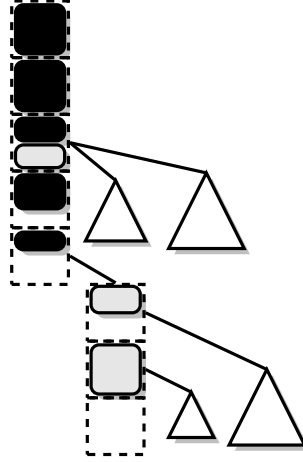


Figure 3.17

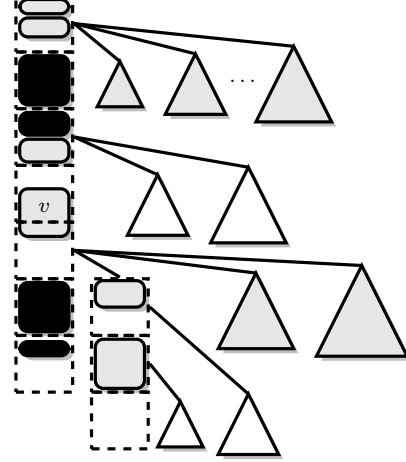


Figure 3.18

Figure 3.19: Basic steps of the case when $i > 1$. The black regions represent the load of a box. Fig. 3.12: An example of a boxline B . Fig. 3.13: boxlines B_1 and B_2 induced by bipartitioning B . Fig 3.14: decision tree D_1 box-compatible with timeline B_1 . Fig ???: timeline B_2 after deleting loads of nodes with indices below k . Fig 3.16: a decision tree D_2 compatible with B_2 . Fig 3.17: D_2 with left child of q_k rehanged to right. Fig 3.18: D_1 and D_2 aligned by their left box-paths.

box in B_2 to be 0 and we set all further boolean flags to false. Then, we retrieve the optimal decision tree D_2 for (T_{c_i}, B_2) , put query to v in D_2 at the s -th slot of box q'_b , rotate D_2 around v and align D_1 with D_2 .

Then, among all of such obtained decision trees we pick the one which minimizes the aligned cost.

3.2.4 An $O(\log \log n)$ -approximation algorithm parametrized by the k -up-modularity of the cost function

k -up-modularity

The main algorithmic difficulty in dealing with the problem arises when the values of the cost function vary drastically. We would like to measure this "irregularity" in a quantifiable way. To do so, we introduce the notion of k -up-modularity.

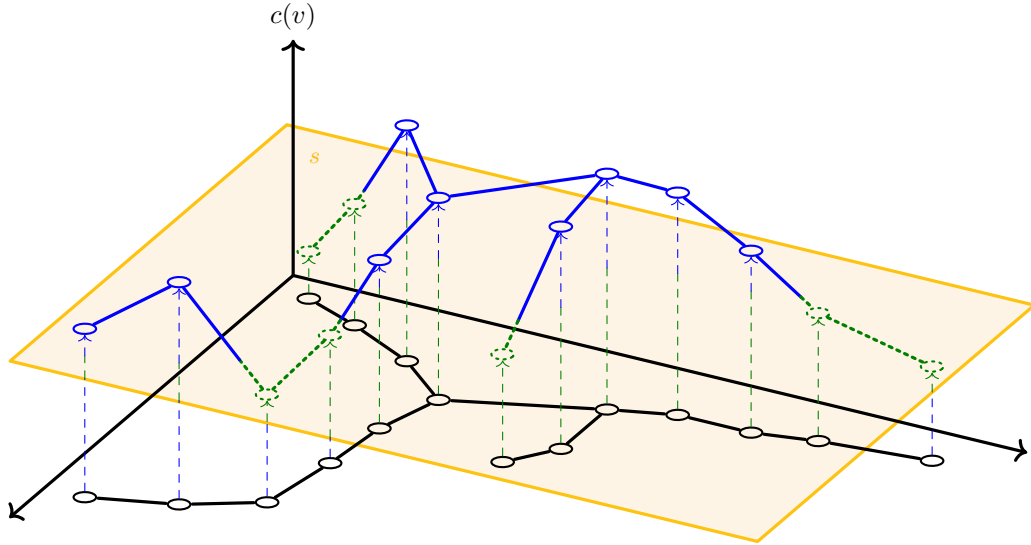


Figure 3.20: A visual depiction of a tree T with a 3-up-modular cost function c . Each vertex of a tree is mapped onto some value of c . The yellow plane represents some threshold value $t \in \mathbb{R}_{\geq 0}$ (in this particular example $k(T, c, t) = 2$). The (two) blue subtrees represent members of $\mathcal{H}_{T,c}(t)$.

Let $t \in \mathbb{R}_{\geq 0}$. We define a *heavy module* with respect to t as $H \subseteq V(T)$ such that, $T[H]$ is connected, for every $v \in H$, $c(v) > t$, and H is maximal - no vertex can be added to it without violating one of its properties. We then define the *heavy module set* with respect to t in (T, c) as:

$$\mathcal{H}_{T,c}(t) = \{H \subseteq V(T) \mid H \text{ is a heavy module w.r.t. } t\},$$

Let $k(T, c, t) = |\mathcal{H}_{T,c}(t)|$ be the size of the heavy module set, and finally let:

$$k(T, c) = \max_{t \in \mathbb{R}_{\geq 0}} \{k(T, c, t)\}$$

We say that a function c is k -up-modular in T when $k \geq k(T, c)$. Whenever clear from the context, we will use $k(T, c)$, $k(T)$, or k to denote the lowest value such that c is k -up-modular in T . To illustrate the notion of k -up-modularity, see Figure 3.20.

The concept of k -up-modularity is a direct generalization of the notion of up-monotonicity of the cost function introduced in [DW22] (as monotonicity) and in [DW24] (as up-monotonicity). Let $z = \arg \max_{v \in V(T)} \{c(v)\}$. A function c is *up-monotonic* in T if for every $v, u \in V(T)$, whenever v lies on the path between z and u , we have $c(v) \geq c(u)$.

It is easy to see that 1-up-modularity is equivalent to up-monotonicity. Observe that if c is

up-monotonic in T , then for every $t \in \mathbb{R}_{\geq 0}$, $T[V(T) - \{v \in V(T) \mid c(v) \leq t\}]$ is connected and forms a single heavy module. Conversely, let $r = \arg \max_{v \in V(T)} \{c(v)\}$ and u be any other vertex. If c is 1-up-modular in T , then there is no vertex v on the path between r and u such that $c(v) < c(u)$. Otherwise, for any $t \in (c(v), c(u))$, v does not belong to any heavy module, but u and r do. Since v lies between them, $|\mathcal{H}_{T,c}(t)| > 1$, a contradiction.

The parametrized $O(\log \log n)$ -approx. solution

Cost levels

The main idea of the algorithm is to partition vertices into intervals called *cost levels* and process them in a top-down manner. At each level of the recursion, the algorithm schedules all necessary queries to vertices belonging to the given cost level. The rest of the decision tree is then built recursively. We consider the following intervals²:

1. Firstly, an interval $(0, 1/\log n]$.
2. Then, each subsequent interval $\mathcal{I}' = (a', b']$ starts at the left endpoint of the previous interval $\mathcal{I} = (a, b]$, that is, $a' = b$, and ends with $b' = \min\{2b, 1\}$.

This results in the following sequence of intervals, which partitions the interval $(0, 1]$:

$$(0, 1/\log n], (1/\log n, 2/\log n], (2/\log n, 4/\log n], \dots, (2^{\lceil \log \log n \rceil - 1}/\log n, 1].$$

We will ensure that when we call our procedure with parameters $(T, c, (2^{\lceil \log \log n \rceil - 1}/\log n, 1])$, the returned decision tree will be a valid decision tree for T .

We are now ready to introduce the notions of heavy and light vertices (and queries to them). We say that a vertex v (or a query to it) is *heavy* with respect to the interval $\mathcal{I} = (a, b]$ when $c(v) > a$. Otherwise, i.e., if $c(v) \leq a$, the vertex (and the query to it) is *light* with respect to $\mathcal{I}$. Note that each heavy vertex belongs to some heavy module. Whenever clear from the context, we will omit the phrase "with respect to" and simply call the vertices and queries heavy and light.

The main recursive procedure

We are ready to present the main recursive procedure. To avoid ambiguity, let $\mathcal{T}$ be the subtree of T processed at some level of the recursion. Alongside $\mathcal{T}$ and a cost function c , the algorithm takes as input an interval $(a, b]$, such that for every $v \in V(\mathcal{T})$, $c(v) \leq b$ and $2a \geq b$. The basic steps of Algorithm 6 are as follows:

1. If every vertex is heavy, return a decision tree built by calling the **RankingBasedDT** procedure for $\mathcal{T}$.
2. Otherwise, find a set $\mathcal{Z}$, such that each connected component of $\mathcal{T}' \in \mathcal{T} - \mathcal{Z}$ contains at most one heavy module.
3. Create an auxiliary tree $T_{\mathcal{Z}}$ using the vertices of $\mathcal{Z}$ and create a new decision tree $D_{\mathcal{Z}}$ for $T_{\mathcal{Z}}$, using the QPTAS from [Der+17].
4. For each $\mathcal{T}' \in \mathcal{T} - \mathcal{Z}$, build a decision tree D_H , by calling the **RankingBasedDT** procedure for $\mathcal{T}'(H)$. Then, hang D_H below the last query to $v \in N_{\mathcal{T}}(\mathcal{T}')$ in $D_{\mathcal{Z}}$.
5. For each $L \in \mathcal{T}' - H$, build a decision tree recursively. Then, hang D_L below the last query to a vertex $v \in N_{\mathcal{T}'}(L)$ in $D_{\mathcal{Z}}$.
6. Return the resulting decision tree D .

²We present the intervals in the ascending order in which a complete solution for each of them is obtained. However, since the procedure is recursive, the order in which the recursive calls are made is reverse.

Before providing a detailed description and analysis of the above procedure, we first present some basic properties necessary for the subsequent considerations. In particular, we will make use of the following well-known lemma [CLS16]:

Lemma 3.2.4.1. *Let T' be a subtree of T . Then, $OPT(T') \leq OPT(T)$.*

Algorithm 6: The main recursive procedure

```

Procedure CreateDecisionTree( $\mathcal{T}, c, (a, b]$ ):
  if  $b \leq 1/\log n$  or for every  $v \in V(\mathcal{T}), c(v) > a$  ;           // Every  $v \in \mathcal{T}$  is heavy
  then
    return RankingBasedDT( $\mathcal{T}$ ) ;                                     // Apply Corollary 3.1.0.1
  else
     $\mathcal{X} \leftarrow \emptyset$ .
    foreach  $H \in \mathcal{H}_{\mathcal{T},c}(a)$  do
      Pick arbitrary  $v \in H$ .
       $\mathcal{X} \leftarrow \mathcal{X} \cup \{v\}$ .
     $\mathcal{Z} \leftarrow \mathcal{Y} \leftarrow \mathcal{X} \cup \{v \in V(\mathcal{T}(\mathcal{X})) \mid \deg_{\mathcal{T}(\mathcal{X})}(v) \geq 3\}$ .
    foreach  $u, v \in \mathcal{Y}$  with  $\mathcal{P}_{\mathcal{T}}(u, v) \neq \emptyset$  and  $\mathcal{P}_{\mathcal{T}}(u, v) \cap \mathcal{Y} = \emptyset$  do
       $\mathcal{Z} \leftarrow \mathcal{Z} \cup \{\arg \min_{z \in \mathcal{P}_{\mathcal{T}}(u, v)} \{c(z)\}\}$  ;           // Lightest vertex on path
     $\mathcal{T}_{\mathcal{Z}} \leftarrow (\mathcal{Z}, \{uv \mid \mathcal{P}_{\mathcal{T}}(u, v) \cap \mathcal{Z} = \emptyset\})$  ;           // Build auxiliary tree
     $D \leftarrow D_{\mathcal{Z}} \leftarrow \text{QPTAS}(\mathcal{T}_{\mathcal{Z}}, c, \epsilon = 1)$  ;           // Apply Theorem 3.2.2.1
    foreach  $\mathcal{T}' \in \mathcal{T} - \mathcal{Z}$  do
       $H \leftarrow$  the unique heavy module in  $\mathcal{T}'$ .
       $D_H \leftarrow \text{RankingBasedDT}(\mathcal{T}' \setminus H)$  ;           // Apply Corollary 3.1.0.1
      Hang  $D_H$  in  $D$  below the last query to  $v \in N_{\mathcal{T}}(\mathcal{T}')$  ;           // By Obs. 3.2.4.5
      foreach  $L \in \mathcal{T}' - H$  do
         $D_L \leftarrow \text{CreateDecisionTree}(L, c, (a/2, a])$ .
        Hang  $D_L$  in  $D$  below the last query to  $v \in N_{\mathcal{T}'}(L)$  ;           // By Obs. 3.2.4.5
    return  $D$ .

```

For the rest of the analysis, fix $\mathcal{H} = \mathcal{H}_{\mathcal{T},c}(a)$ to be the set of heavy modules in $\mathcal{T}$. We have the following observations, which will be useful in the description and analysis of the algorithm:

Observation 3.2.4.2. *Let $\mathcal{H}$ be the set of heavy modules in T . Then, $|\mathcal{H}| \leq k(T)$.*

Proof. Since $\mathcal{H} = \mathcal{H}_{\mathcal{T},c}(a)$, we have $|\mathcal{H}| = k(\mathcal{T}, c, a) \leq \max_{t \in \mathbb{R}_{\geq 0}} k(T, c, t) = k(T, c)$. $\square$

Observation 3.2.4.3. *Let T' be a subtree of T . Then, $k(T') \leq k(T)$.*

Proof. Fix any $t \in \mathbb{R}_{\geq 0}$ and let $H \in \mathcal{H}_{T,c}(t)$. We show that each such H contributes at most 1 to $k(T', c, t)$. If $H \cap V(T') = \emptyset$, then H contributes 0. Otherwise, $H \cap V(T')$ forms a connected subtree of T' , and thus contributes at most 1. The lemma follows. $\square$

Observation 3.2.4.4. *Let T' be a subtree of a tree T and let D' be a decision tree for T' . Then, D' is a partial decision tree for T .*

Observation 3.2.4.5. *Let T' be a subtree of a tree T and let D be a partial decision tree for T having no queries to vertices of T' , but containing at least one query to the vertices of $N_T(V(T'))$. Let Q denote the set of all such queries to vertices of $N_T(V(T'))$ in D . Then, $D \setminus Q$ forms a path in D .*

Proof. Let q be any query in D . There are two cases:

1. $q \in V(T - V(T') - N_T(V(T')))$. Then, for every $x \in N_T(V(T'))$ being the target, x belongs to the same connected component of $T - q$. Thus, no matter which vertex is the target, the answer is always the same. Therefore, q has at most one child u in D , such that $V(D_u) \cap Q \neq \emptyset$.
2. $q \in N_T(V(T'))$. After a query to q , the situation is as in the first case, except when $x = q$. Then, the response is x itself, so no further queries are needed, and again q has at most one child u in D , such that $V(D_u) \cap Q \neq \emptyset$.

Since each $q \in Q$ has at most one child u in D , with $D_u \cup Q \neq \emptyset$, $D \langle Q \rangle$ forms a path and the claim follows. $\square$

Base of the recursion

We begin the description of our algorithm with the recursion base, which occurs whenever $b \leq 1/\log n$ or for every $v \in V(\mathcal{T})$, $c(v) > a$, i.e., every vertex is heavy. In such a situation, a solution is built by disregarding the costs of vertices and constructing a decision tree using the vertex ranking of $\mathcal{T}$.

Lemma 3.2.4.6. *Let D be a decision tree built, by calling `RankingBasedDT` ($\mathcal{T}$) in line 6 of the `CreateDecisionTree` procedure. Then,*

$$C_D(\mathcal{T}) \leq 2 \cdot \text{OPT}(\mathcal{T}).$$

Proof. There are two cases:

1. If $b \leq \frac{1}{\log n}$, then:

$$C_D(\mathcal{T}) \leq \frac{\lfloor \log n \rfloor + 1}{\log n} \leq \frac{\log n + 1}{\log n} \leq 2 \leq 2 \cdot \text{OPT}(\mathcal{T}) \leq 2 \cdot \text{OPT}(\mathcal{T}),$$

where the first inequality is due to Corollary 3.1.0.1, the fourth inequality follows from Observation 3.2.3.1, and the last inequality is due to Observation 3.2.4.1.

2. If for every $v \in V(\mathcal{T})$, we have $c(v) > a$, then, define $c'(v) = a$ for all $v \in V(\mathcal{T})$ (note that any value could be chosen here, since we treat each query as unitary). As $2c'(v) = 2a \geq b \geq c(v)$, we obtain $2 \cdot C_D(\mathcal{T}, c') \geq C_D(\mathcal{T}, c)$. Additionally, using the fact that $c'(v) \leq c(v)$, we have $\text{OPT}(\mathcal{T}, c') \leq \text{OPT}(\mathcal{T}, c)$. Therefore:

$$C_D(\mathcal{T}, c) \leq 2 \cdot C_D(\mathcal{T}, c') = 2 \cdot \text{OPT}(\mathcal{T}, c') \leq 2 \cdot \text{OPT}(\mathcal{T}, c) \leq 2 \cdot \text{OPT}(\mathcal{T}, c),$$

where the equality is due to Corollary 3.1.0.1 and the last inequality is due to Observation 3.2.4.1. The lemma follows. $\square$

Construction of the Auxiliary Tree

To obtain the solution for the non-base case of our algorithm, we first construct the so-called *auxiliary tree*. To do so, we begin by defining a set $\mathcal{X} \subseteq V(\mathcal{T})$. For every heavy module $H \in \mathcal{H}$, we pick an arbitrary $v \in H$ and add it to $\mathcal{X}$. We also define a set $\mathcal{Y} = \mathcal{X} \cup \left\{ v \in V(\mathcal{T} \langle \mathcal{X} \rangle) \mid \deg_{\mathcal{T} \langle \mathcal{X} \rangle}(v) \geq 3 \right\}$, by extending $\mathcal{X}$ to contain all vertices with degree at least 3 in $\mathcal{T} \langle \mathcal{X} \rangle$. Furthermore, we define a set $\mathcal{Z} \subseteq V(\mathcal{T})$ consisting of the vertices in $\mathcal{Y}$ and, for every $u, v \in \mathcal{Y}$, such that $\mathcal{P}_{\mathcal{T}}(u, v) \neq \emptyset$ and $\mathcal{P}_{\mathcal{T}}(u, v) \cap \mathcal{Y} = \emptyset$, we add to $\mathcal{Z}$ the lightest vertex between them, i. e., $v_{u,v} = \arg \min_{z \in \mathcal{P}_{\mathcal{T}}(u,v)} \{c(z)\}$. To see an example of construction of the sets $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$, see Figure 3.21.

We then create the auxiliary tree $\mathcal{T}_{\mathcal{Z}} = (\mathcal{Z}, \{uv \mid \mathcal{P}_{\mathcal{T}}(u, v) \cap \mathcal{Z} = \emptyset\})$ (for an example, see Figure 3.22). Our algorithm starts by building a decision tree $D_{\mathcal{Z}}$ for $\mathcal{T}_{\mathcal{Z}}$, by taking $\epsilon = 1$ and applying the QPTAS from Theorem 3.2.2.1. Observe that, since $D_{\mathcal{Z}}$ is a partial decision tree for $\mathcal{T}$ and corresponding vertices in $\mathcal{T}$ and $\mathcal{T}_{\mathcal{Z}}$ have the same costs, we have that:

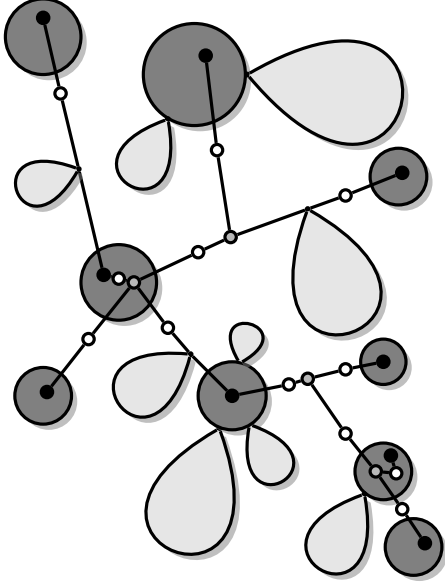


Figure 3.21: Example tree $\mathcal{T}$. Dark grey circles represent heavy modules. Light grey regions represent light subtrees. Black vertices represent $\mathcal{X}$. Gray and black vertices represent $\mathcal{Y}$. White, gray and black vertices represent $\mathcal{Z}$. Lines represent paths of vertices between vertices of $\mathcal{Z}$.

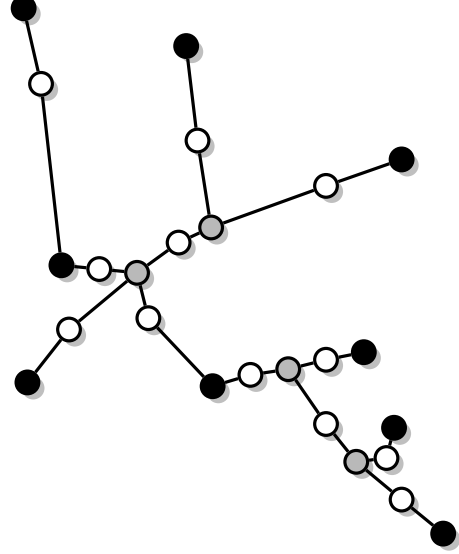


Figure 3.22: Auxiliary tree $\mathcal{T}_Z$ built from vertices of set $\mathcal{Z}$.

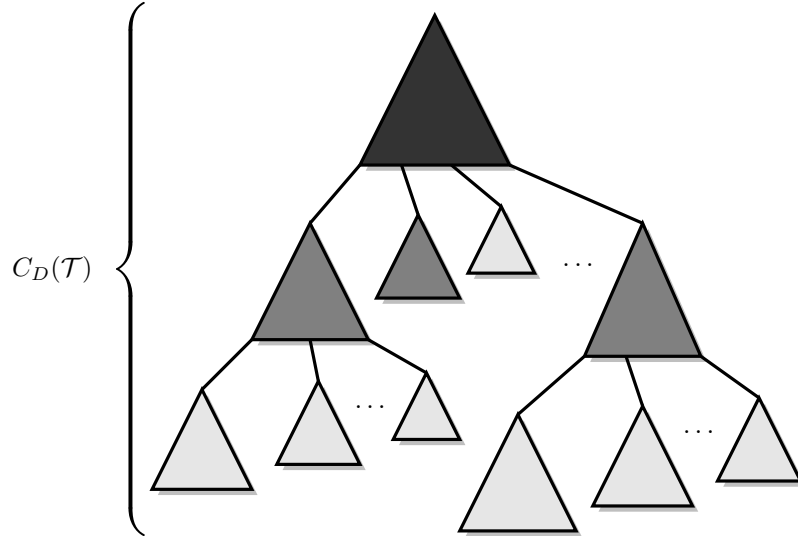


Figure 3.23: The structure of the decision tree D , built by the Algorithm 6. The dark gray subtree represents the decision tree D_Z , obtained by calling the QPTAS for $\mathcal{T}_Z$, c and $\epsilon = 1$. Gray subtrees represent decision trees D_H , each built for a unique heavy module $H \subseteq V(\mathcal{T}')$ of every $\mathcal{T}' \in \mathcal{T} - \mathcal{Z}$, by calling the RANKINGBASEDDT procedure for $\mathcal{T}' \langle H \rangle$. Light gray subtrees represent decision trees D_L , built for each $L \in \mathcal{T}' - H$, by recursively calling CREATEDECISIONTREE with L , c and $(a/2, a]$.

Observation 3.2.4.7. $C_{D_Z}(\mathcal{T}_Z) = C_{D_Z}(\mathcal{T})$.

Let $D = D_Z$. For each connected component $\mathcal{T}' \in \mathcal{T} - \mathcal{Z}$, we build a new decision tree as follows: By the construction of $\mathcal{Z}$, all heavy vertices in $V(\mathcal{T}')$ form a single heavy module $H \subseteq V(\mathcal{T}')$. We create a new decision tree D_H for $\mathcal{T}' \langle H \rangle$, by calling the **RankingBasedDT** procedure with argument $\mathcal{T}' \langle H \rangle$ and we hang D_H in D below the unique last query to a vertex in $N_{\mathcal{T}'}(\mathcal{T}')$ (which is possible due to Observation 3.2.4.5). As, by Observation 3.2.4.4, D_H is a partial decision tree for $\mathcal{T}'$, it follows that D is also a partial decision tree for $\mathcal{T}$.

Now notice that for each $L \in \mathcal{T}' - H$, there is no $v \in V(L)$, such that $c(v) > a$. This allows us to create a decision tree D_L recursively, by calling the **CreateDecisionTree** procedure with arguments L , c and $(a/2, a]$. Next, we hang D_L in D below the unique last query to a vertex in $N_{\mathcal{T}'}(L)$ (again, using Observation 3.2.4.5). Since after all such operations, every vertex $v \in V(\mathcal{T})$ also belongs to D , we obtain a valid decision tree D for $\mathcal{T}$. To see example structure of such solution, see Figure 3.23.

Analysis of the algorithm

Lemma 3.2.4.8. *Let $\mathcal{T}_Z$ be the auxiliary tree. Then, $|V(\mathcal{T}_Z)| \leq 4k - 3$.*

Proof. First, we show that $|\mathcal{Y}| \leq 2k - 1$. We use induction on the elements of $\mathcal{H}$. We construct a family of sets $\mathcal{H}_1, \mathcal{H}_2, \dots, \mathcal{H}_{|\mathcal{H}|}$, such that for every integer $1 \leq h \leq |\mathcal{H}|$, $|\mathcal{H}_h| = h$ and $\mathcal{H}_{|\mathcal{H}|} = \mathcal{H}$. For each $\mathcal{H}_h$, we also construct a corresponding set $\mathcal{Y}_h$, eventually ensuring that $\mathcal{Y}_{|\mathcal{H}|} = \mathcal{Y}$.

Let $\mathcal{H}_1 = \emptyset$, $\mathcal{Y}_1 = \emptyset$. Pick any heavy module $H \subseteq V(\mathcal{T})$ and add it to $\mathcal{H}_1$. Add the unique vertex v , such that $v \in H \cap \mathcal{X}$ to $\mathcal{Y}_1$, so that $|\mathcal{Y}_1| = 1$. Assume by induction that for some $h \geq 1$, $|\mathcal{Y}_h| \leq 2h - 1$. Two heavy modules $H_1, H_2 \subseteq V(\mathcal{T})$ will be called *neighbors* if for every $H_3 \subseteq V(\mathcal{T})$ with $H_3 \neq H_1, H_2$, we have $\mathcal{P}_{\mathcal{T}}(H_1, H_2) \cap H_3 = \emptyset$. Pick $H \in \mathcal{H}$, such that $H \notin \mathcal{H}_h$ to be a heavy module that is a neighbor of some member of $\mathcal{H}_h$. We define $\mathcal{H}_{h+1} = \mathcal{H}_h \cup \{H\}$. Let z be the unique vertex, such that $v \in H \cap \mathcal{X}$, and let $\mathcal{Y}_{h+1} = \mathcal{Y}_h \cup \{z\}$. Define $\mathcal{T}_{h+1} = \mathcal{T} \langle \{v \in \mathcal{Y}_{h+1} | \mathcal{P}_{\mathcal{T}}(v, z) \cap \mathcal{Y}_{h+1} = \emptyset\} \rangle$. Note that $\mathcal{T}_{h+1}$ is a spider (a tree with at most one vertex of degree above 2). Add to $\mathcal{Y}_{h+1}$ the unique vertex $v \in V(\mathcal{T}_{h+1})$, such that $\deg_{\mathcal{T}_{h+1}}(v) \geq 3$, if it exists. Clearly, $|\mathcal{Y}_{h+1}| \leq 2h + 1$, completing the induction.

By construction, $\mathcal{H}_{|\mathcal{H}|} = \mathcal{H}$ and $\mathcal{Y}_{|\mathcal{H}|} = \mathcal{Y}$, so $|\mathcal{Y}| \leq 2 \cdot |\mathcal{H}| - 1 \leq 2k - 1$ where the last inequality is by Observation 3.2.4.2. As paths between vertices in $\mathcal{Y}$ form a tree when contracted, at most $2k - 2$ additional vertices are added while constructing $\mathcal{Z}$ (at most one per path). The lemma follows. $\square$

Lemma 3.2.4.9. *Let $\mathcal{T}_Z$ be the auxiliary tree. Then, $\text{OPT}(\mathcal{T}_Z) \leq \text{OPT}(\mathcal{T})$.*

Proof. Let D^* be the decision tree for $\mathcal{T} \langle \mathcal{Z} \rangle$. We build a new decision tree D'_Z for $\mathcal{T}_Z$ by transforming D^* as follows:

Let $u, v \in \mathcal{Y}$, such that $\mathcal{P}_{\mathcal{T}}(u, v) \neq \emptyset$ and $\mathcal{P}_{\mathcal{T}}(u, v) \cap \mathcal{Y} = \emptyset$. Let $q \in V(D^*)$ be the first query to a vertex among $\mathcal{P}_{\mathcal{T}}(u, v)$. Recall that we picked $v_{u,v} = \arg \min_{z \in \mathcal{P}_{\mathcal{T}}(u,v)} \{c(z)\}$, so $c(v_{u,v}) \leq c(q)$. We replace q in D^* with the query to $v_{u,v}$ and delete all queries to vertices in $\mathcal{P}_{\mathcal{T}}(u, v) - v_{u,v}$. By construction, D'_Z is a valid decision tree for $\mathcal{T}_Z$, and by choosing $v_{u,v}$ to minimize c , we did not increase the cost, so we have that:

$$C_{D'_Z}(\mathcal{T}_Z) \leq \text{OPT}(\mathcal{T} \langle \mathcal{Z} \rangle).$$

Therefore, we have:

$$\text{OPT}(\mathcal{T}_Z) \leq C_{D'_Z}(\mathcal{T}_Z) \leq \text{OPT}(\mathcal{T} \langle \mathcal{Z} \rangle) \leq \text{OPT}(\mathcal{T}),$$

where the first inequality is due to the definition of optimality and the last inequality follows by Lemma 3.2.4.1. $\square$

Lemma 3.2.4.10. *Let H be the unique heavy module of $\mathcal{T}' \in \mathcal{T} - \mathcal{Z}$. Then, the decision tree D_H is of cost at most:*

$$C_{D_H}(\mathcal{T}' \langle H \rangle) \leq 2 \cdot \text{OPT}(\mathcal{T}).$$

Proof. For every $v \in H$ let $c'(v) = a$. We have $2c'(v) \geq bc'(v)/a = b \geq c(v)$ so we get that $2 \cdot C_{D_H}(\mathcal{T}'\langle H \rangle, c') \geq C_{D_H}(\mathcal{T}'\langle H \rangle, c)$. Additionally, using the fact that $c'(v) \leq c(v)$ we have that $\text{OPT}(\mathcal{T}'\langle H \rangle, c') \leq \text{OPT}(\mathcal{T}'\langle H \rangle, c)$. Hence:

$$\begin{aligned} C_{D_H}(\mathcal{T}'\langle H \rangle, c) &\leq 2 \cdot C_{D_H}(\mathcal{T}'\langle H \rangle, c') = 2 \cdot \text{OPT}(\mathcal{T}'\langle H \rangle, c') \\ &\leq 2 \cdot \text{OPT}(\mathcal{T}'\langle H \rangle, c) \leq 2 \cdot \text{OPT}(\mathcal{T}, c) \end{aligned}$$

where the equality is by the Corollary 3.1.0.1 and the last inequality is due to the fact that $\mathcal{T}'\langle H \rangle$ is a subtree of $\mathcal{T}'$, which is a subtree of $\mathcal{T}$ (Lemma 3.2.4.1). $\square$

The main result

Let d be the remaining depth of the recursion call performed in Line 6 of the algorithm, i.e., the number of recursive steps from the current call to the base case (for the base case, this value is equal to $d = 0$). We show that at each level of the recursion we pay $O(\text{OPT}(T))$, so the approximation ratio of the algorithm is bounded by $O(d)$:

Lemma 3.2.4.11. $C_D(\mathcal{T}) \leq (4d + 2) \cdot \text{OPT}(T)$.

Proof. Let $Q_D(\mathcal{T}, x)$ be the sequence of queries performed in order to find $x \in V(\mathcal{T})$. By construction of Algorithm 6, $Q_D(\mathcal{T}, x)$ consists of at most three distinct subsequences of queries (see Figure 3.23):

1. Firstly, there is a sequence of queries belonging to $Q_{D_Z}(\mathcal{T}_Z, x)$.
2. If $x \notin Z$, then, there is a sequence of queries belonging to $Q_{D_H}(\mathcal{T}'\langle H \rangle, x)$ for a unique heavy group $H \subseteq V(\mathcal{T}')$ of $\mathcal{T}' \in \mathcal{T} - Z$, such that $x \in \mathcal{T}'$.
3. At last, if $x \notin H$, there is a sequence of queries belonging to $Q_{D_L}(L, x)$ for $L \in \mathcal{T}' - H$, such that $x \in V(L)$.

Note that it sometimes may happen that some of the above sequences are empty.

We prove by induction that $C_D(\mathcal{T}) \leq (4d + 2) \cdot \text{OPT}(T)$. When $d = 0$ (the base case), the induction hypothesis is true, due to the Lemma 3.2.4.6. For $d > 0$, assume by induction that the cost of the decision tree built for each L , is at most $C_{D_L}(L) \leq (4(d - 1) + 2) \cdot \text{OPT}(T)$. We have:

$$\begin{aligned} C_D(\mathcal{T}) &\leq C_{D_Z}(\mathcal{T}) + \max_{\mathcal{T}' \in \mathcal{T} - Z} \left\{ C_{D_H}(\mathcal{T}'\langle H \rangle) + \max_{L \in \mathcal{T}' - H} \{C_{D_L}(L)\} \right\} \\ &\leq C_{D_Z}(\mathcal{T}_Z) + \max_{\mathcal{T}' \in \mathcal{T} - Z} \{2 \cdot \text{OPT}(\mathcal{T}) + (4(d - 1) + 2) \cdot \text{OPT}(T)\} \\ &\leq 2 \cdot \text{OPT}(\mathcal{T}_Z) + 2 \cdot \text{OPT}(T) + (4(d - 1) + 2) \cdot \text{OPT}(T) \\ &\leq 2 \cdot \text{OPT}(\mathcal{T}) + 4d \cdot \text{OPT}(T) = (4d + 2) \cdot \text{OPT}(T) \end{aligned}$$

where the first inequality is due to the construction of the decision tree returned by the Algorithm 6, the second inequality is by Observation 3.2.4.7, Observation 3.2.4.10 and by the induction hypothesis, the third inequality is due to Theorem 3.2.2.1 and using the fact that $\mathcal{T}$ is a subtree of T (Lemma 3.2.4.1) and the last inequality is due to Lemma 3.2.4.9. $\square$

We are now ready to prove our main theorem:

Theorem 3.2.4.12. *There exists an $O(\log \log n)$ -approximation algorithm for the Tree Search Problem running in $k^{O(\log k)} \cdot \text{poly}(n)$ time.*

Proof. Let $D = \text{CreateDecisionTree}(T, (2^{\lceil \log \log n \rceil - 1} / \log n, 1])$. Since there are at most $\lceil \log \log n \rceil + 1$ intervals processed, the depth of the recursion is bounded by $d \leq \lceil \log \log n \rceil \leq \log \log n + 1$. Hence, using Lemma 3.2.4.11 we get that:

$$C_D(T) \leq (4 \cdot \log \log n + 6) \cdot \text{OPT}(T) = O(\log \log n \cdot \text{OPT}(T)).$$

By Observation 3.2.4.3, for every subtree $\mathcal{T}$ of T , processed at some level of the recursion, we have $k(\mathcal{T}) \leq k(T)$. Using Lemma 3.2.4.8, at each such level the call to the QPTAS from Theorem 3.2.2.1 (line 6 of the CREATEDECISIONTREE) runs in time bounded by:

$$k(\mathcal{T})^{O(\log(4 \cdot k(\mathcal{T})))} = k(T)^{O(\log k(T))}.$$

Since $d = O(\text{poly}(n))$ and all other computation can be performed in polynomial time, the overall running time is bounded by $k^{O(\log k)} \cdot \text{poly}(n)$, as required. $\square$

Chapter 4

Experimental Results

The following chapter is concerned with a series of experiments conducted in order to evaluate the performance of the algorithms for the Tree Search Problem presented in the previous chapter.

Our main goal was to compare the quality of the solutions produced by the different algorithms on various classes of trees and cost functions and the computation time required to obtain such results. We measured the performance of our algorithms on randomly generated trees of three main classes: random trees, trees with bounded degree and trees with a bounded diameter. For the two latter, we have chosen trees of degree at most 3 and diameter at most 6, since these are minimal values of these parameters for which the Tree Search Problem with non uniform costs is known to be NP-hard [Cic+12]. We also used three different ways of generating the cost functions: random uniform distribution, binomial distribution and exponential distribution.

We implemented the previously described algorithms using python programming language. The experiments were conducted on a machine with 32 x AMD Ryzen 9 7950X 16-Core Processor and 128 GB of RAM.

4.1 Overview of the experiments

The experimental analysis is divided into two sections concerning uniform and non-uniform query costs. In the first section we measured the running time of the **RankingBasedDT** algorithm from Section 3.1 and the depth of the decision trees produced by it. The second section was concerned with the non-uniform cost model and included the evaluation of the $O(\log n / \log \log n)$ -approximation algorithm from Section 3.2.1, the $O(\sqrt{\log n})$ -approximation algorithm from Section 3.2.2 based on the QPTAS and the $O(\log \log n)$ -approximation algorithm from Section 3.2.4. We compared the quality of the solutions produced by these algorithms on various classes of trees and cost functions and the computation time required to obtain these results. Additionally, for small input sizes we managed to calculate the optimal solutions using the exact exponential time algorithm. This allowed us to compare the approximation ratios of the algorithms with respect to the optimal solutions.

We have also employed a few heuristic speed-ups, particularly for the QPTAS. Firstly, instead of using $\epsilon = 1$ we set it to $\epsilon = 59$, which reduces most of the computational complexity of each QPTAS call. For both the $O(\sqrt{\log n})$ -approximation algorithm and the $O(\log \log n)$ -approximation algorithm this does not impact the asymptotic guarantee of the algorithms. Secondly, a careful reading of the proof of the QPTAS reveals that the depth of the considered boxed decision trees can be limited to $\text{rank}(T)$, where $\text{rank}(T)$ is the number of labels used by the optimal ranking of T . This reduces the running time of the QPTAS to $n^{O(\log \text{rank}(T)/\epsilon^2)}$, which in practice is usually significantly better than the upper bound $\text{rank}(T) = O(\log n)$.

Lastly we decided to simplify the implementation of the linear time algorithm for $T||V||C_{max}$ based on calculating the ranking coloring. We implemented this algorithm naively with $O(n \log n)$ time complexity. This is done in order to simplify the already complex implementation. Additionally, this running time is negligible compared to the other algorithms and we are mainly focused on the quality of the solutions produced by it. Also note that although we performed experiments for the uniform costs variant of the problem using this algorithm, the main focus of this thesis is the non-uniform costs model and this algorithm serves as an essential subroutine for both QPTAS and the $O(\log \log n)$ -approximation algorithm, rather than a standalone procedure.

Tree Generation Methods

We used three different methods of generating random trees for our experiments: random trees, trees with bounded degree and trees with bounded diameter. Table 4.1 summarizes all tree generation methods used in our experiments. We decided to measure the performance of our algorithms on relatively small instances, 10 to 100 vertices per instance, since the complex dynamic programming subroutines of our algorithm require significant computation time even for such sizes.

Table 4.1: Tree generation methods and their properties

Method	Parameters	Description
Random	n	Uniformly random labeled tree generated using Prüfer sequence
Bounded Degree	n, Δ	Tree with maximum degree constraint Δ (each vertex has degree $\leq \Delta$)
Bounded Diameter	n, D	Tree with maximum diameter constraint D (maximum distance between any two vertices $\leq D$)

Detailed Generation Algorithms

Random Trees. Generated using NetworkX's `random_labeled_tree` function, which employs Prüfer sequences to sample uniformly from all labeled trees on n vertices. A random vertex is chosen as root, and the tree is oriented via BFS traversal from this root.

Bounded Degree Trees. Constructed incrementally by maintaining a list of available parent nodes with fewer than $\Delta - 1$ children. Each new vertex is attached to a uniformly random available parent.

When a parent reaches its degree limit, it is removed from the available list. This ensures every vertex has total degree at most Δ .

Bounded Diameter Trees. For diameter bound D , vertices are distributed across $h = \lfloor D/2 \rfloor + 1$ levels. The root occupies level 0, and remaining vertices are randomly assigned to levels $1, \dots, h - 1$. Each non-root vertex connects to a random parent in the previous level, ensuring the diameter does not exceed D .

Cost Function Generation

Table 4.2 lists all probability distributions used for vertex weights (w) and edge costs (c).

Table 4.2: Probability distributions for edge costs

Distribution	Formula	Properties
Uniform	$\mathcal{U}(0, 1)$	All values equally likely in $[0, 1]$, mean $\mathbb{E}[X] = 0.5$
Binomial	$\text{Binomial}(n, p)$	Discrete distribution with parameters $n = 10$, $p = 0.5$, mean $\mathbb{E}[X] = np = 5$
Exponential	$\mathcal{E}(\lambda = 1)$	Memoryless, continuous distribution with rate $\lambda = 1$, mean $\mathbb{E}[X] = 1$

Extended Three-Field Notation for Experiments

To systematically describe our experimental configurations, we extend the theoretical three-field notation $\alpha||\beta||\gamma$ to include specific tree generation methods and distribution types:

Field α (Search Space): In experiments, we specify the specific tree generation method using subscripts. For this study, we focus on three representative tree classes:

- T_R - Random trees (uniformly random labeled trees via Prüfer sequence)
- T_Δ - Bounded degree trees (maximum degree constraint Δ)
- T_D - Bounded diameter trees (maximum diameter constraint D)

Field β (Query Characteristics): All experiments use vertex queries (V). For worst-case optimization, we consider four cost scenarios:

- V - vertex queries with uniform unit costs ($c(v) = 1$ for all v)
- $V, c = \mathcal{U}$ - vertex queries with costs from uniform distribution $\mathcal{U}(0, 1)$
- $V, c = \mathcal{B}$ - vertex queries with costs from binomial distribution $\text{Binomial}(n = 10, p = 0.5)$
- $V, c = \mathcal{E}$ - vertex queries with costs from exponential distribution $\mathcal{E}(\lambda = 1)$

Experiment Coverage Matrix

Table 4.3 shows which combinations of tree types and problem variants are tested. Each cell contains a reference to the corresponding subsection with detailed results.

Table 4.3: Experiment coverage: tree generation methods vs. problem variants

Tree Method	$T V C_{max}$	$T V, c C_{max}$
Random	4.2.1	4.3.1
Bounded Degree	4.2.2	—
Bounded Diameter	4.2.3	—

4.2 Trees with Uniform Costs (Worst Case)

This section presents experimental results for the basic tree search problem with uniform query costs. The objective is to construct a decision tree that minimizes the worst-case number of queries needed to identify any target vertex.

4.2.1 Random Trees ($T_R||V||C$)

Table 4.4: Performance on random trees with uniform query costs

n	Avg Depth	Max Depth	Avg Time (s)	Max Time (s)
10	3.80	4	0.001514	0.001936
20	4.60	5	0.003452	0.003635
30	5.00	5	0.006719	0.007023
40	5.10	6	0.009460	0.009888
50	5.70	6	0.012409	0.012886
60	6.00	6	0.015446	0.016239
100	6.50	7	0.021648	0.046397
150	6.90	7	0.034571	0.036309
200	7.10	8	0.051168	0.077717
250	7.90	8	0.088678	0.106789
300	8.00	8	0.110067	0.140505
350	8.00	8	0.122534	0.150200
400	8.10	9	0.139465	0.174724
450	8.20	9	0.110219	0.125244
500	8.80	9	0.129039	0.147685
550	8.40	9	0.212270	0.234442
600	8.90	9	0.168215	0.177733
650	8.70	9	0.182155	0.190271
700	9.00	9	0.270885	0.309166
750	9.00	9	0.299953	0.355814
800	9.00	9	0.275221	0.295997
850	9.10	10	0.241985	0.250178
900	9.10	10	0.359294	0.398942
950	9.20	10	0.395425	0.455887
1000	9.30	10	0.430645	0.459383

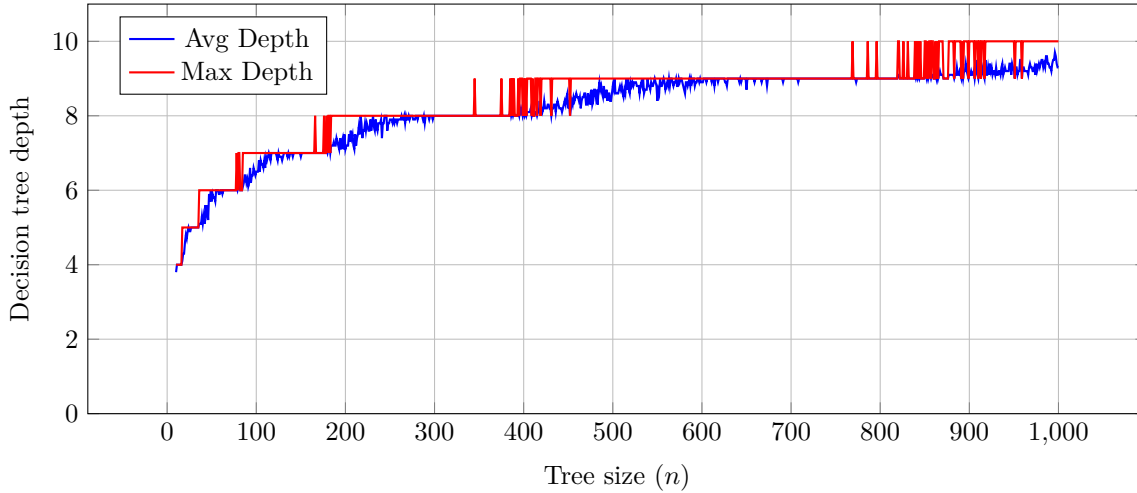


Figure 4.1: Decision tree depth for random trees (uniform query costs)

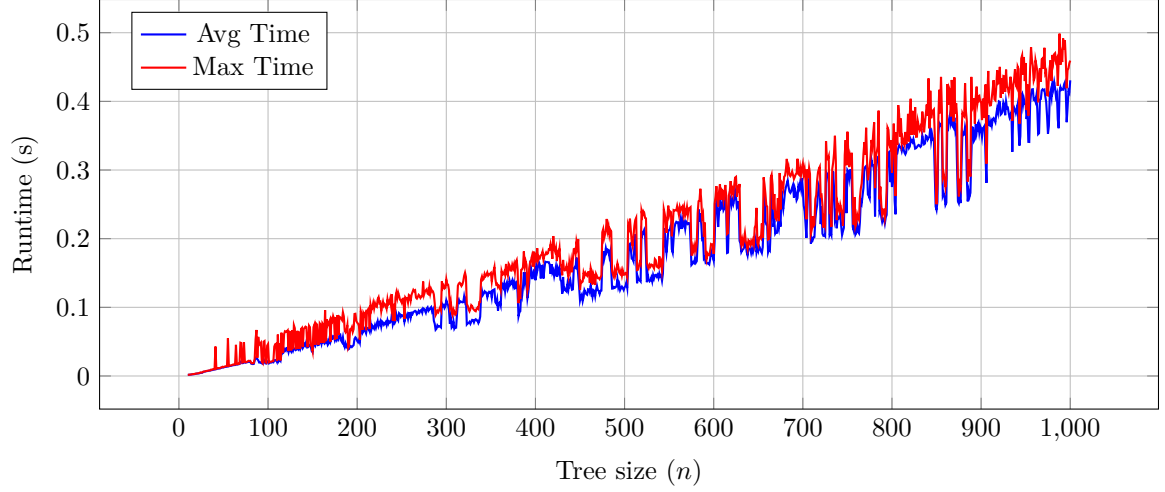


Figure 4.2: Runtime for random trees (uniform query costs)

4.2.2 Bounded Degree Trees ($T_{\Delta}||V||C$)

Table 4.5: Performance on bounded degree trees with uniform query costs

n	Avg Depth	Max Depth	Avg Time (s)	Max Time (s)
10	3.80	4	0.001883	0.002315
20	4.40	5	0.004077	0.004300
30	5.00	5	0.006379	0.006953
40	5.60	6	0.009117	0.009902
50	5.60	6	0.008258	0.008587
60	6.00	6	0.014716	0.015088
100	6.30	7	0.028718	0.054380
150	7.00	7	0.040778	0.041550
200	7.50	8	0.057506	0.058477
250	8.00	8	0.081548	0.106034
300	8.00	8	0.105412	0.130591
350	8.00	8	0.121891	0.140366
400	8.00	8	0.087627	0.090278
450	8.10	9	0.107712	0.123486
500	8.40	9	0.175684	0.199389
550	8.80	9	0.167661	0.215635
600	9.00	9	0.191693	0.206806
650	9.00	9	0.233231	0.263469
700	9.00	9	0.230719	0.241494
750	9.00	9	0.279499	0.292872
800	9.00	9	0.291323	0.305180
850	9.00	9	0.316835	0.327101
900	9.10	10	0.341218	0.350593
950	9.20	10	0.315355	0.353623
1000	9.00	9	0.262918	0.273126

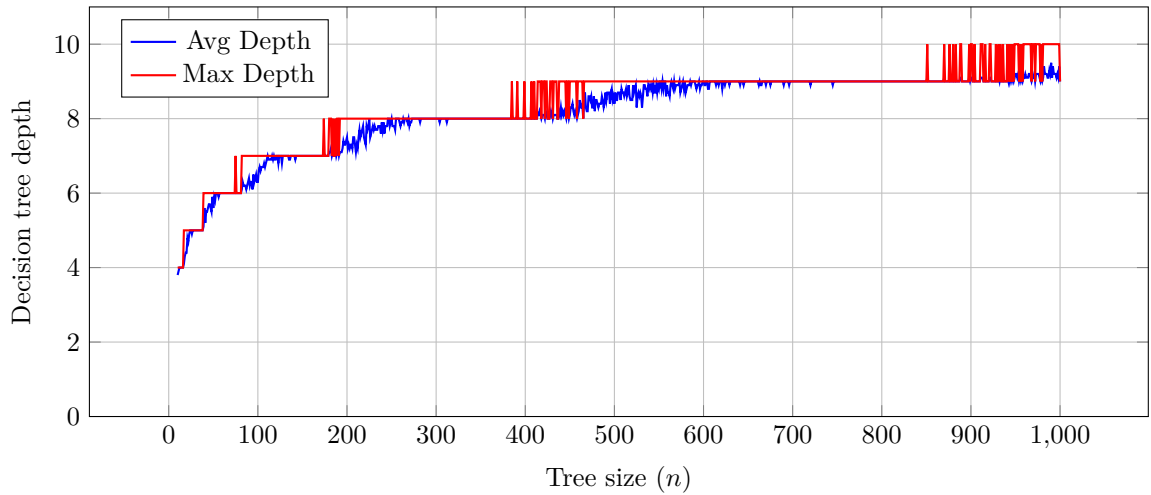


Figure 4.3: Decision tree depth for bounded degree trees (uniform query costs)

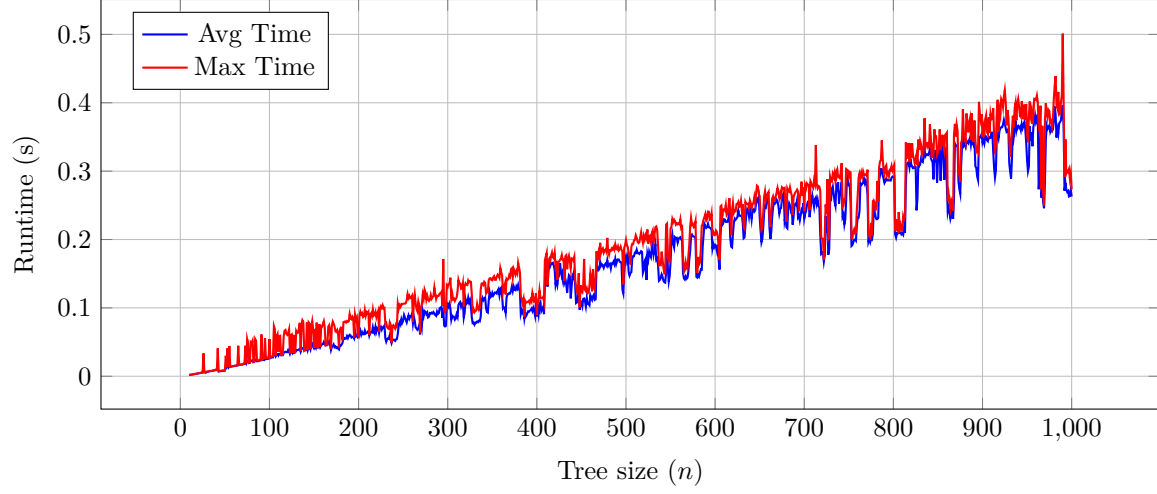


Figure 4.4: Runtime for bounded degree trees (uniform query costs)

4.2.3 Bounded Diameter Trees ($T_D || V || C$, $D = 6$)

Table 4.6: Performance on bounded diameter trees with uniform query costs

n	Avg Depth	Max Depth	Avg Time (s)	Max Time (s)
10	3.00	4	0.001891	0.002284
20	3.40	4	0.004504	0.004940
30	3.30	4	0.007768	0.009012
40	3.20	4	0.014947	0.045265
50	3.50	4	0.015760	0.020563
60	3.60	4	0.021442	0.027050
100	3.80	4	0.047224	0.085641
150	3.50	4	0.118843	0.148323
200	3.80	4	0.183898	0.304099
250	3.90	4	0.195326	0.339757
300	3.70	4	0.337928	0.602597
350	3.60	4	0.421854	0.613242
400	3.90	4	0.357429	0.727468
450	3.80	4	0.468534	0.910761
500	3.70	4	0.773654	1.458311
550	3.80	4	0.908509	1.504898
600	3.60	4	1.529075	2.472266
650	3.80	4	1.540004	2.624646
700	4.00	4	0.893710	2.124919
750	3.80	4	1.477481	2.154393
800	3.90	4	2.113809	4.169857
850	3.70	4	2.906338	4.655112
900	3.70	4	2.625933	4.967109
950	3.70	4	3.366969	6.116271
1000	3.60	4	2.875865	4.676517

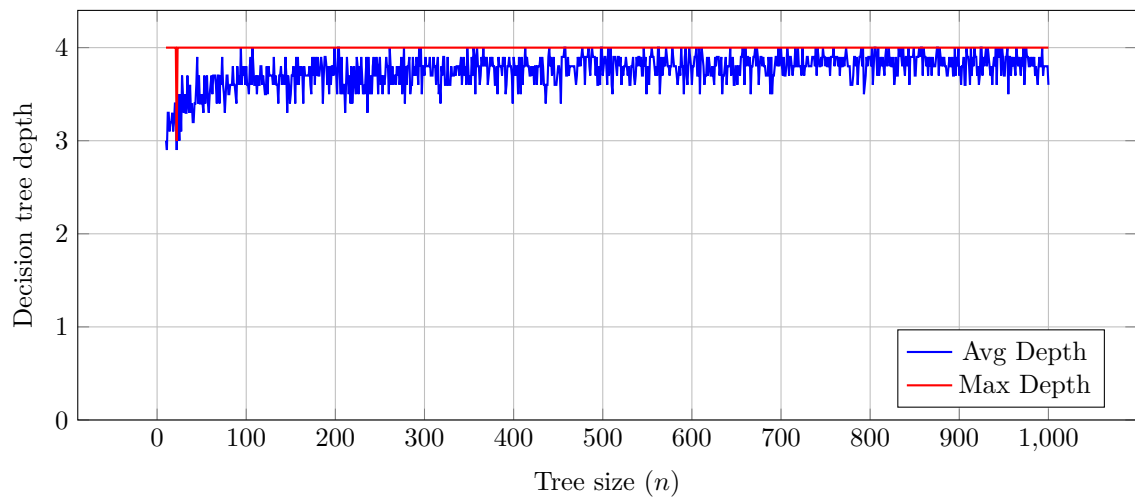


Figure 4.5: Decision tree depth for bounded diameter trees (uniform query costs)

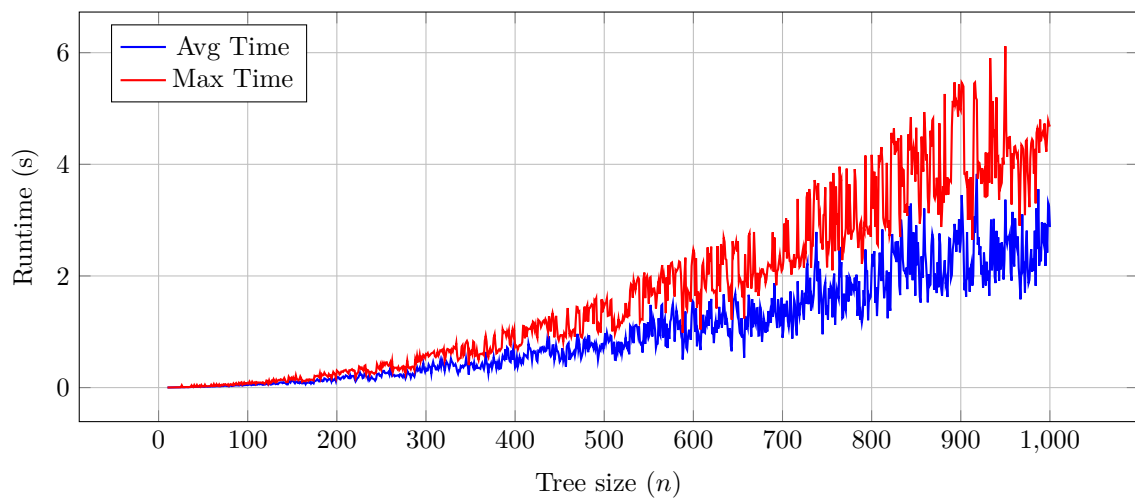


Figure 4.6: Runtime for bounded diameter trees (uniform query costs)

4.3 Trees with Non-Uniform Costs (Worst Case)

This section evaluates algorithms for the worst-case search problem with non-uniform query costs, where different queries have different costs.

4.3.1 Random Trees with Uniform Cost Distribution ($T_R || V, c \sim \mathcal{U} || C$)

DP Exact Algorithm

Table 4.7: Random trees with uniform cost distribution - DP exact algorithm

n	Avg Time (s)	Max Time (s)	Avg Cost	Max Cost
10	0.515344	0.996765	2.11	2.87
11	0.688948	1.871738	2.07	2.74
12	1.118181	1.820036	2.18	2.60
13	3.752255	5.853787	2.35	2.75
14	3.924208	5.459482	2.31	2.84
15	6.742663	12.371523	2.21	3.07
16	8.338989	17.136759	2.34	3.10
17	13.161961	19.730425	2.60	3.40
18	49.783415	217.751785	2.59	3.20
19	53.941062	108.616709	2.53	2.99
20	72.510728	279.192467	2.56	3.17
21	126.797035	323.067169	2.48	3.23
22	146.939623	328.078981	2.51	2.88
23	159.220462	248.346399	2.39	2.84
24	444.063066	1208.271180	2.62	3.22
25	948.392856	3652.773897	2.56	3.16
26	919.064031	2638.620480	2.54	3.05
27	2219.712304	4494.591545	2.69	3.57
28	4474.570226	15483.063044	2.66	3.21
29	6210.535691	22857.351522	2.73	3.39
30	11405.857347	51794.339840	2.64	3.15
31	8092.897929	24723.579952	2.91	3.48
32	19858.948001	50624.244158	2.80	3.51
33	17082.196660	22448.280074	2.64	2.85

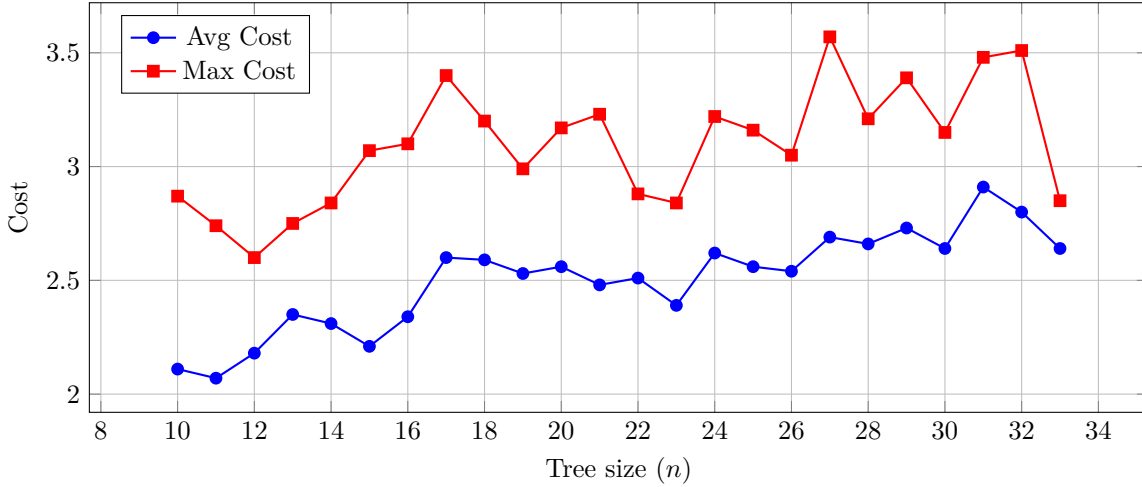


Figure 4.7: Random trees with uniform costs - DP exact algorithm: cost

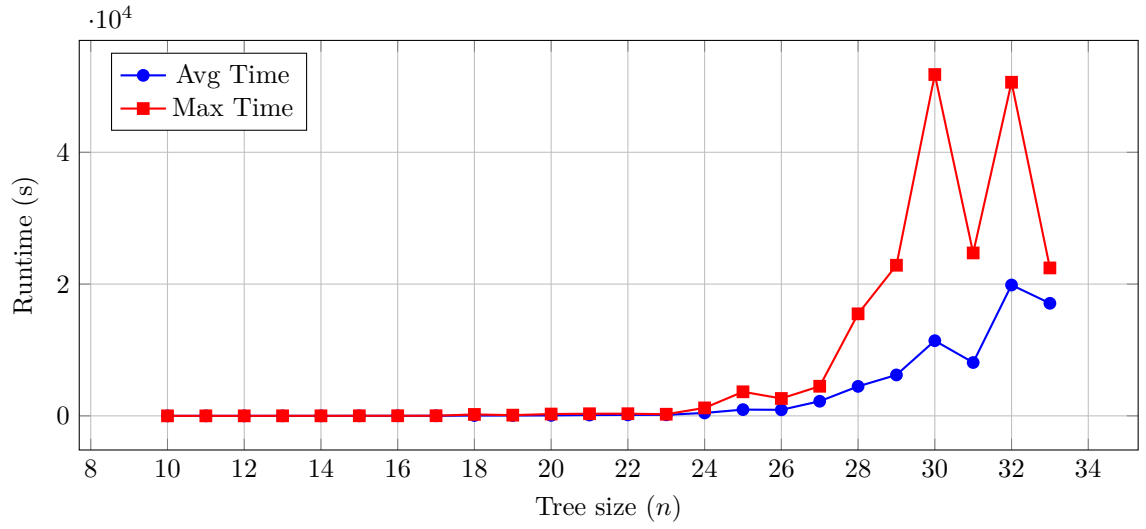


Figure 4.8: Random trees with uniform costs - DP exact algorithm: runtime

Cicalese et al. $O(\log n / \log \log n)$ -Approximation Algorithm

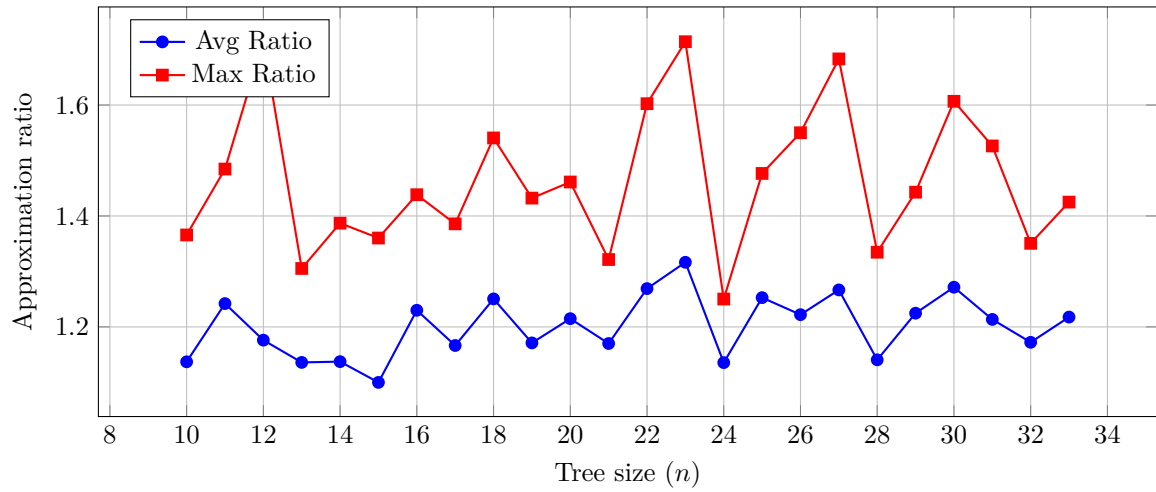


Figure 4.9: Random trees with uniform costs - Cicalese algorithm: approximation quality

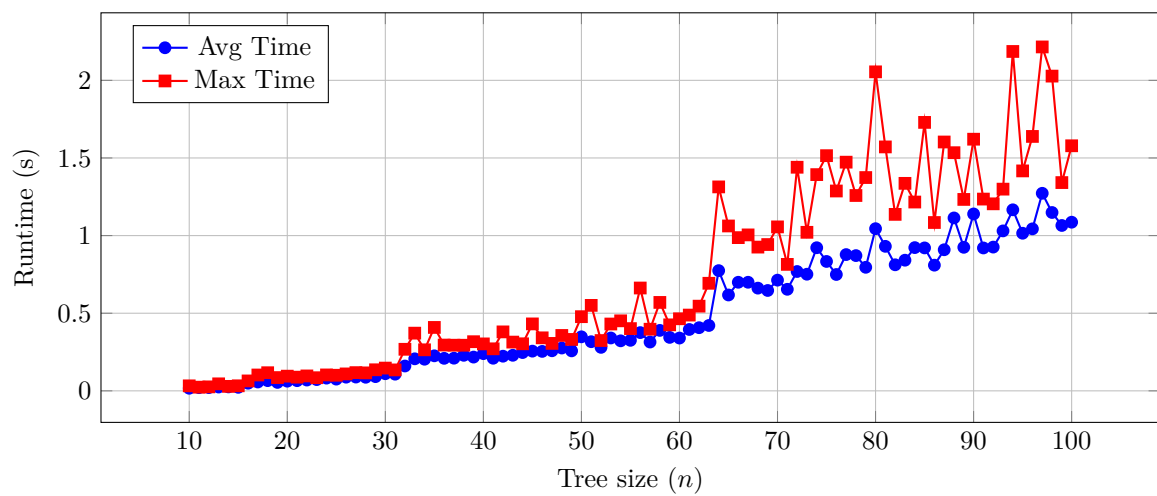


Figure 4.10: Random trees with uniform costs - Cicalese algorithm: runtime

Table 4.8: Random trees with uniform cost distribution - Cicalese algorithm (every 2nd instance)

n	Avg Time (s)	Max Time (s)	Avg Cost	Max Cost	Avg Ratio	Max Ratio
10	0.016762	0.032318	2.3955	3.1361	1.137075	1.365714
11	0.020501	0.023684	2.5411	3.3119	1.242042	1.484599
12	0.020812	0.025657	2.5090	2.9782	1.176145	1.715289
13	0.024581	0.045187	2.6730	3.5875	1.135985	1.305341
14	0.025502	0.028926	2.5913	2.8914	1.137403	1.387021
15	0.022680	0.032554	2.4654	3.6102	1.099938	1.360228
16	0.049179	0.063968	2.8590	3.9065	1.229984	1.438320
17	0.057024	0.102183	3.0322	4.4976	1.166470	1.385779
18	0.065770	0.116763	3.2151	3.6839	1.250493	1.540655
19	0.054388	0.085611	2.9521	3.5943	1.171139	1.432209
20	0.062404	0.094678	3.0932	3.8894	1.214856	1.461230
21	0.065561	0.088133	2.9010	3.8122	1.170052	1.321474
22	0.069631	0.096130	3.2054	4.5964	1.269055	1.602414
23	0.072472	0.084640	3.1571	4.8598	1.316487	1.714126
24	0.081970	0.102568	2.9581	3.5771	1.135594	1.250227
25	0.075609	0.100487	3.2128	4.0764	1.252776	1.476635
26	0.088151	0.109135	3.0840	3.8475	1.220257	1.550057
27	0.088868	0.117435	3.3649	3.9552	1.266685	1.683139
28	0.087563	0.115817	3.0404	3.6023	1.140615	1.334636
29	0.091547	0.135715	3.3287	3.9535	1.224709	1.442870
30	0.110397	0.146635	3.3719	4.2595	1.271571	1.606712
31	0.107470	0.134955	3.5149	4.7702	1.213483	1.526203
32	0.160086	0.267672	3.2690	4.0821	1.172232	1.350648
33	0.206370	0.371636	3.4796	4.4312	1.217703	1.424988
34	0.203727	0.264368	3.4051	4.3033	N/A	N/A
35	0.225650	0.408465	3.3490	4.6967	N/A	N/A
36	0.209532	0.294949	3.5425	4.4719	N/A	N/A
37	0.210439	0.294024	3.5497	4.6951	N/A	N/A
38	0.228890	0.293844	3.5003	4.4850	N/A	N/A
39	0.217514	0.316559	3.5849	4.5623	N/A	N/A
40	0.239622	0.302497	3.3435	4.1278	N/A	N/A
41	0.210024	0.270788	3.7290	4.9261	N/A	N/A
42	0.223183	0.379812	3.1765	4.2009	N/A	N/A
43	0.229074	0.314054	3.6954	4.2974	N/A	N/A
44	0.246577	0.303195	3.2966	3.7578	N/A	N/A
45	0.255775	0.431221	3.4435	5.5378	N/A	N/A
46	0.254112	0.341533	3.6033	4.2704	N/A	N/A
47	0.258181	0.305618	3.8509	5.0594	N/A	N/A
48	0.275816	0.356788	3.7386	4.8264	N/A	N/A
49	0.257603	0.331242	3.4366	4.4493	N/A	N/A
50	0.348037	0.477731	3.5857	4.3841	N/A	N/A
51	0.316554	0.550524	3.8813	5.3098	N/A	N/A
52	0.280130	0.324085	3.7065	4.5545	N/A	N/A
53	0.340562	0.430524	3.5367	4.4463	N/A	N/A
54	0.322247	0.451032	3.6614	4.0527	N/A	N/A
55	0.325494	0.400998	3.2778	4.1324	N/A	N/A
56	0.375180	0.662389	3.6973	4.3144	N/A	N/A
57	0.314435	0.396946	3.5694	4.3770	N/A	N/A
58	0.389405	0.569548	3.9511	4.7930	N/A	N/A
59	0.344359	0.425114	3.7978	4.5595	N/A	N/A
60	0.340081	0.464291	3.5671	4.3226	N/A	N/A
61	0.395388	0.488097	3.9104	4.5430	N/A	N/A
62	0.406840	0.545518	3.7279	5.2662	N/A	N/A
63	0.421006	0.692861	3.9630	4.9307	N/A	N/A
64	0.774373	1.313631	3.6539	4.5179	N/A	N/A
65	0.617531	1.061992	3.9063	4.6499	N/A	N/A
66	0.699391	0.986953	3.6481	4.2931	N/A	N/A
67	0.699400	1.004483	3.8057	4.5491	N/A	N/A
68	0.661586	0.925920	3.7493	5.4706	N/A	N/A
69	0.646546	0.942351	3.7461	4.3904	N/A	N/A
70	0.713039	1.056106	3.9947	4.8039	N/A	N/A
71	0.654112	0.814631	3.7397	5.1063	N/A	N/A
72	0.768729	1.440199	3.9641	5.2356	N/A	N/A
73	0.751237	1.021116	4.1910	5.3329	N/A	N/A
74	0.921879	1.392781	4.1473	6.5996	N/A	N/A
75	0.833485	1.514807	3.3756	4.1316	N/A	N/A
76	0.749446	1.287268	4.2359	5.0890	N/A	N/A
77	0.877686	1.472915	3.7374	4.5212	N/A	N/A
78	0.871158	1.258194	4.2788	5.6682	N/A	N/A
79	0.795813	1.373866	4.0119	5.1522	N/A	N/A
80	1.044580	2.054690	3.9946	4.6330	N/A	N/A
81	0.930796	1.571128	4.4896	5.6123	N/A	N/A
82	0.811977	1.136639	4.1623	5.4257	N/A	N/A
83	0.841508	1.336485	3.9659	4.6923	N/A	N/A
84	0.922589	1.216132	3.8927	4.5882	N/A	N/A
85	0.920700	1.729080	3.9099	4.7718	N/A	N/A
86	0.809507	1.084260	4.1819	5.4446	N/A	N/A
87	0.908831	1.602874	4.0625	5.0116	N/A	N/A
88	1.113665	1.533967	4.1221	5.7956	N/A	N/A
89	0.924536	1.232939	3.9472	4.5441	N/A	N/A
90	1.139403	1.620814	3.9961	4.6164	N/A	N/A
91	0.920372	1.235162	4.1071	5.3134	N/A	N/A
92	0.925961	1.204636	3.9849	5.2190	N/A	N/A
93	1.030268	1.299180	4.3475	4.9864	N/A	N/A
94	1.166125	2.185943	4.6044	5.3951	N/A	N/A
95	1.015151	1.416649	3.7895	4.1652	N/A	N/A
96	1.043433	1.638646	4.1189	5.3262	N/A	N/A
97	1.272435	2.215122	4.0738	4.9698	N/A	N/A
98	1.148774	2.027226	4.2355	5.0905	N/A	N/A
99	1.065212	1.341485	3.7770	4.9176	N/A	N/A
100	1.086270	1.578486	4.1955	4.9503	N/A	N/A

Dereniowski $O(\sqrt{\log n})$ -Approximation Algorithm

Table 4.9: Random trees with uniform cost distribution - Dereniowski algorithm (every 2nd instance)

n	Avg Time (s)	Max Time (s)	Avg Cost	Max Cost	Avg Ratio	Max Ratio
10	3.667458	12.756751	2.5010	3.2296	1.175826	1.386895
11	6.128544	29.399862	2.6884	3.9218	1.305199	1.484599
12	4.272992	12.674806	2.7705	3.7881	1.305226	1.715289
13	2.433804	17.536711	2.7154	3.3657	1.169089	1.552337
14	0.718830	5.269130	2.7575	3.5176	1.207746	1.476948
15	0.869210	6.602548	2.5724	3.6206	1.136520	1.334345
16	27.517307	156.347909	2.8909	3.9065	1.242290	1.527108
17	128.062403	792.909551	3.0567	4.3032	1.175246	1.410590
18	24.850276	106.738707	3.2770	3.6839	1.234498	1.395855
19	11.442593	36.674916	3.0969	3.7111	1.224265	1.494932
20	27.091810	142.684030	3.2803	4.1737	1.288483	1.470591
21	117.702028	889.725199	3.1512	4.0302	1.272082	1.782426
22	61.510348	238.084481	3.4905	4.5964	1.382578	1.602414
23	42.097233	106.463449	3.1512	3.8740	1.326100	1.569625
24	39.732011	149.680999	3.3651	4.5315	1.289401	1.560927
25	27.877790	88.513900	3.2326	3.9459	1.246003	1.426526
26	129.002198	827.922871	3.4034	4.2511	1.383815	1.731408
27	21.298450	45.592222	3.5469	4.1581	1.337913	1.612733
28	32.335598	89.950481	3.1076	4.0093	1.166253	1.414333
29	138.223743	1199.429222	3.6682	4.5527	1.345866	1.661554
30	38.511334	65.983931	3.3417	4.2340	1.270631	1.479690
31	45.612598	124.638447	3.6357	4.3272	1.262941	1.665712
32	41.170141	83.916606	3.4353	4.0821	1.213553	1.440375
33	59.477436	271.004911	3.6244	4.7029	1.132198	1.201444
34	10.454132	28.285260	3.5311	4.2430	N/A	N/A
35	56.178653	189.509749	3.4741	4.5948	N/A	N/A
36	23.174181	55.223033	3.6154	4.4719	N/A	N/A
37	420.367692	3115.145578	3.7468	4.2312	N/A	N/A
38	77.981897	523.390589	3.8508	4.9392	N/A	N/A
39	20.521917	82.864569	3.5730	4.5864	N/A	N/A
40	80.364828	258.686817	3.2656	4.0902	N/A	N/A
41	29.889535	66.581582	3.5451	4.2329	N/A	N/A
42	1408.819309	7446.854561	3.0848	4.2009	N/A	N/A
43	762.760202	2743.887134	3.6147	4.1267	N/A	N/A
44	1170.757902	2836.208023	3.4973	4.1143	N/A	N/A
45	3497.560223	14352.968259	3.4963	5.1732	N/A	N/A
46	395.566427	910.341421	3.6523	4.2395	N/A	N/A
47	3811.398716	16410.480209	3.9294	4.8548	N/A	N/A
48	3495.496766	27287.483563	3.7671	5.0592	N/A	N/A
49	2659.659927	15202.043831	3.6172	4.4903	N/A	N/A
50	8457.697716	42358.960755	3.6696	4.4136	N/A	N/A
51	3506.404116	22816.730345	4.2133	5.6921	N/A	N/A
52	561.841314	1359.119127	3.6460	3.9423	N/A	N/A
53	1328.843090	2958.886347	3.5827	4.4842	N/A	N/A
54	321.809560	1099.148587	3.7848	4.5348	N/A	N/A
55	16207.413238	70117.505497	3.4477	4.4387	N/A	N/A
56	1235.476850	4147.176084	3.6509	4.0335	N/A	N/A

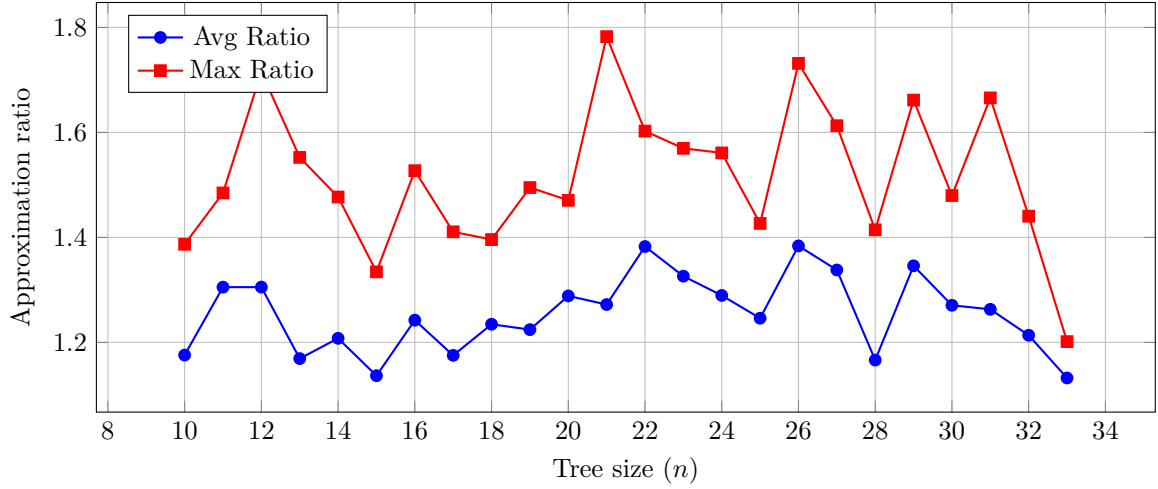


Figure 4.11: Random trees with uniform costs - Dereniowski algorithm: approximation quality

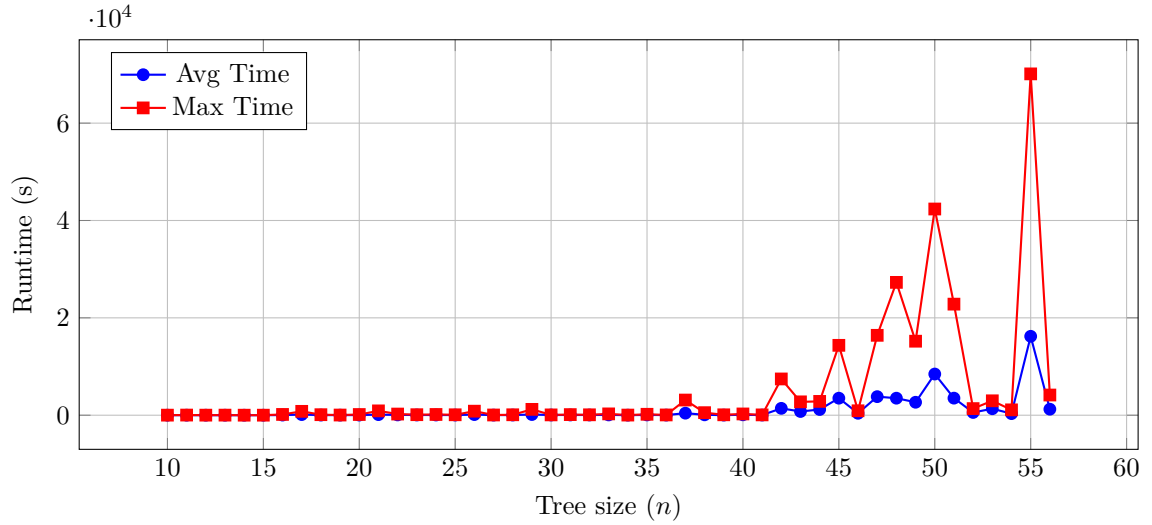


Figure 4.12: Random trees with uniform costs - Dereniowski algorithm: runtime

K-up $O(\log \log n)$ -Approximation Algorithm

Table 4.10: Random trees with uniform cost distribution - K-up algorithm (every 2nd instance)

n	Avg Time (s)	Max Time (s)	Avg Cost	Max Cost	Avg Ratio	Max Ratio
10	121.404046	756.719610	2.3900	3.3096	1.127822	1.319998
11	260.592758	2163.498792	2.5370	3.0668	1.228647	1.484599
12	5.358513	28.820250	2.7337	3.5471	1.243215	1.385444
13	11611.709343	116056.363491	2.8209	4.0259	1.195636	1.479760
14	3892.703673	34297.438517	2.8904	4.3107	1.256194	1.519283
15	4188.206773	19999.133534	2.9683	3.8046	1.365824	1.875606
16	98515.257964	98515.257964	2.3474	2.3474	1.000000	1.000000

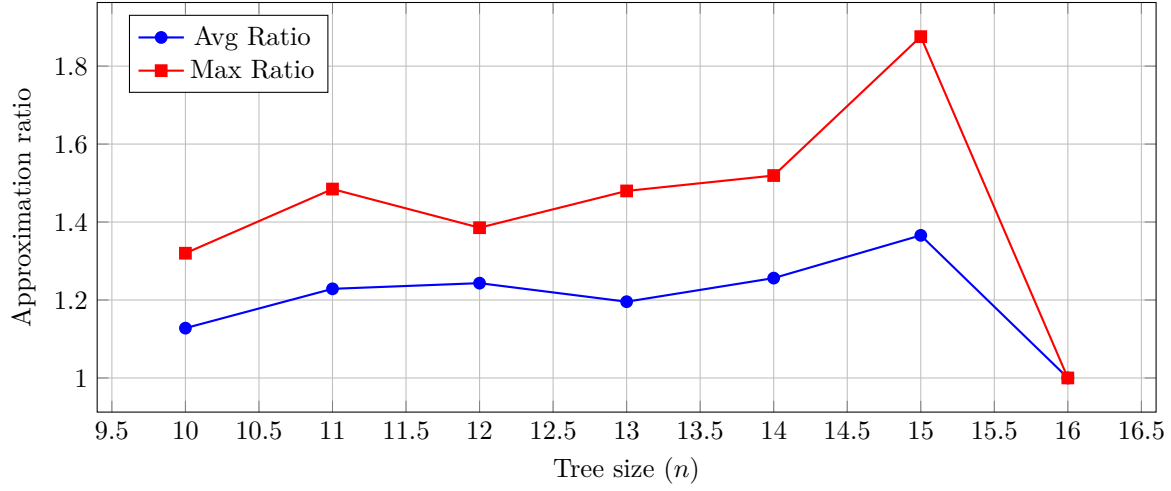


Figure 4.13: Random trees with uniform costs - K-up algorithm: approximation quality

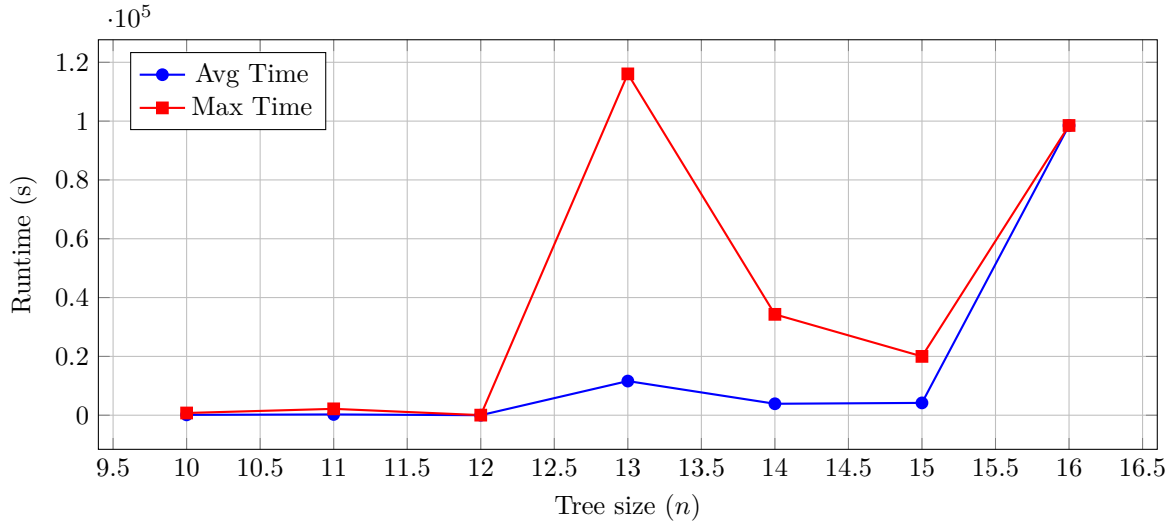


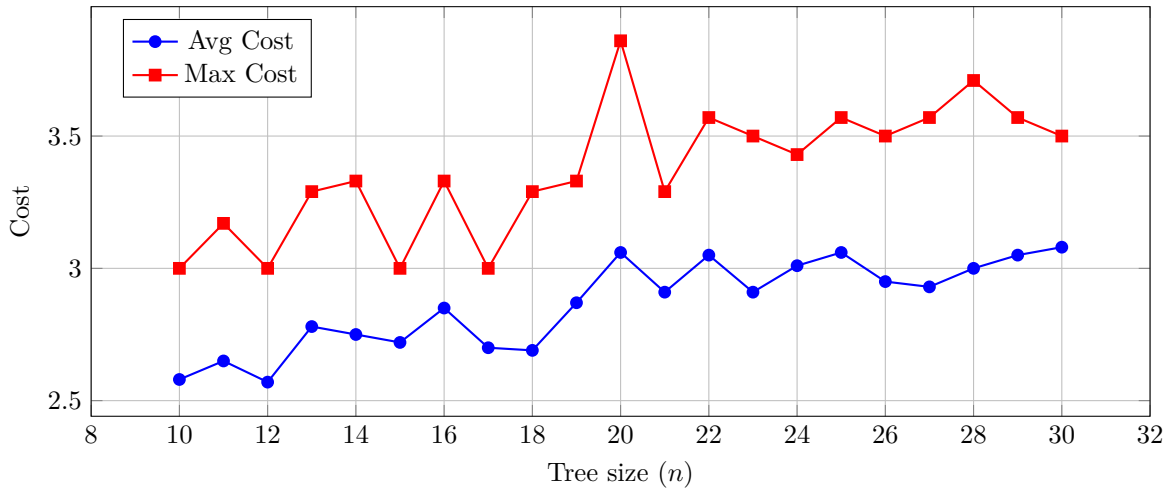
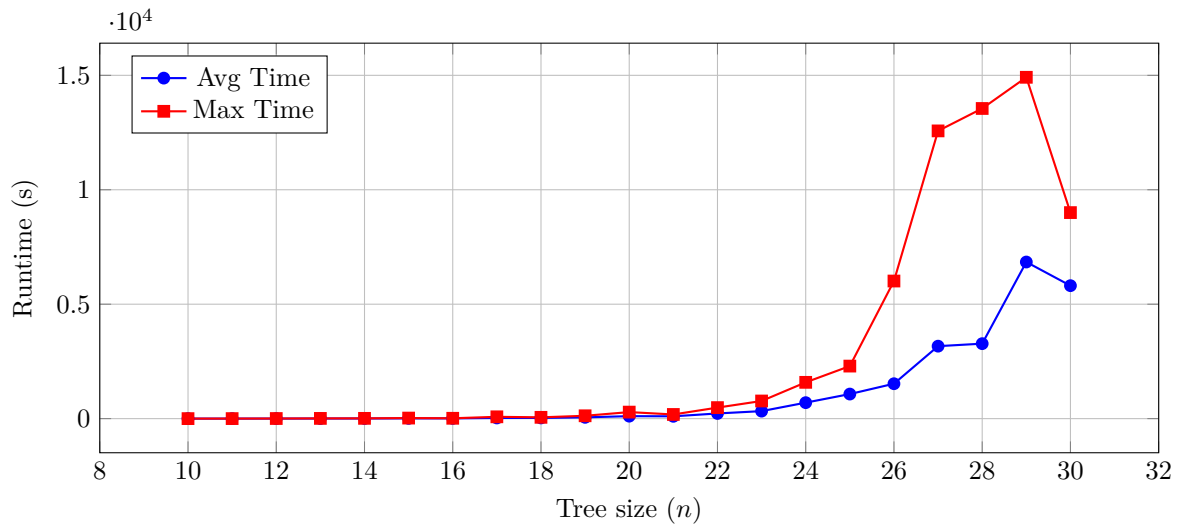
Figure 4.14: Random trees with uniform costs - K-up algorithm: runtime

4.3.2 Random Trees with Binomial Cost Distribution ($T_R || V, c \sim \mathcal{B} || C$)

DP Exact Algorithm

Table 4.11: Random trees with binomial cost distribution - DP exact algorithm

n	Avg Time (s)	Max Time (s)	Avg Cost	Max Cost
10	0.556375	0.927114	2.58	3.00
11	0.988541	1.709204	2.65	3.17
12	2.284303	4.845210	2.57	3.00
13	4.457306	9.871454	2.78	3.29
14	6.074104	11.887673	2.75	3.33
15	10.764245	24.311487	2.72	3.00
16	10.557692	15.240707	2.85	3.33
17	28.881488	79.292225	2.70	3.00
18	30.493508	56.675099	2.69	3.29
19	56.074162	124.334870	2.87	3.33
20	107.637201	284.935946	3.06	3.86
21	101.401123	182.020454	2.91	3.29
22	225.660437	481.720823	3.05	3.57
23	327.566712	770.705944	2.91	3.50
24	697.875948	1585.505406	3.01	3.43
25	1073.534832	2295.687021	3.06	3.57
26	1525.110602	6012.207639	2.95	3.50
27	3166.287653	12571.961075	2.93	3.57
28	3275.602415	13551.198494	3.00	3.71
29	6842.866619	14913.855759	3.05	3.57
30	5809.227155	9002.505390	3.08	3.50

**Figure 4.15:** Random trees with binomial costs - DP exact algorithm: cost comparison**Figure 4.16:** Random trees with binomial costs - DP exact algorithm: runtime comparison

Cicalese et al. $O(\log n / \log \log n)$ -Approximation Algorithm

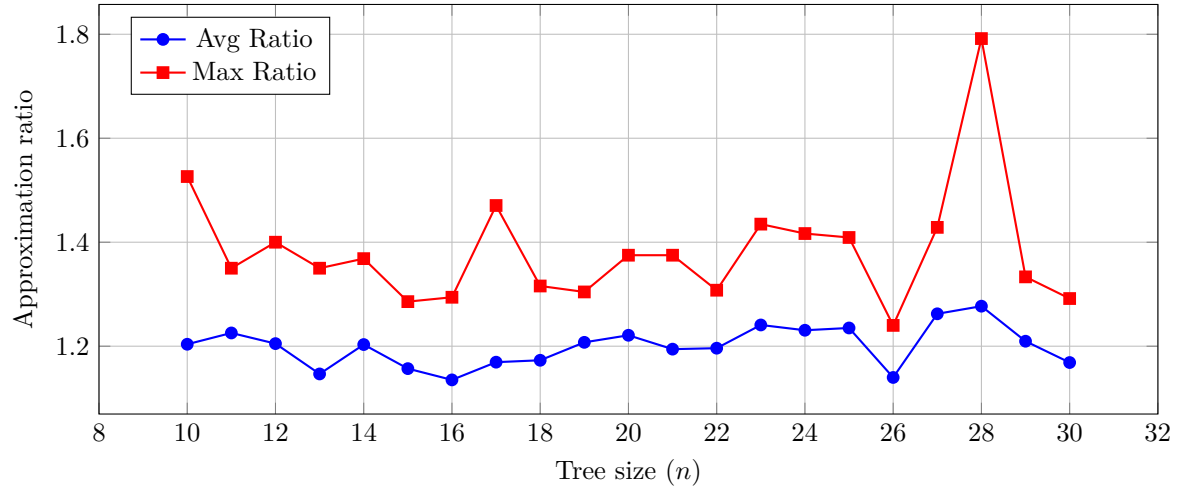


Figure 4.17: Random trees with binomial costs - Cicalese algorithm: approximation quality

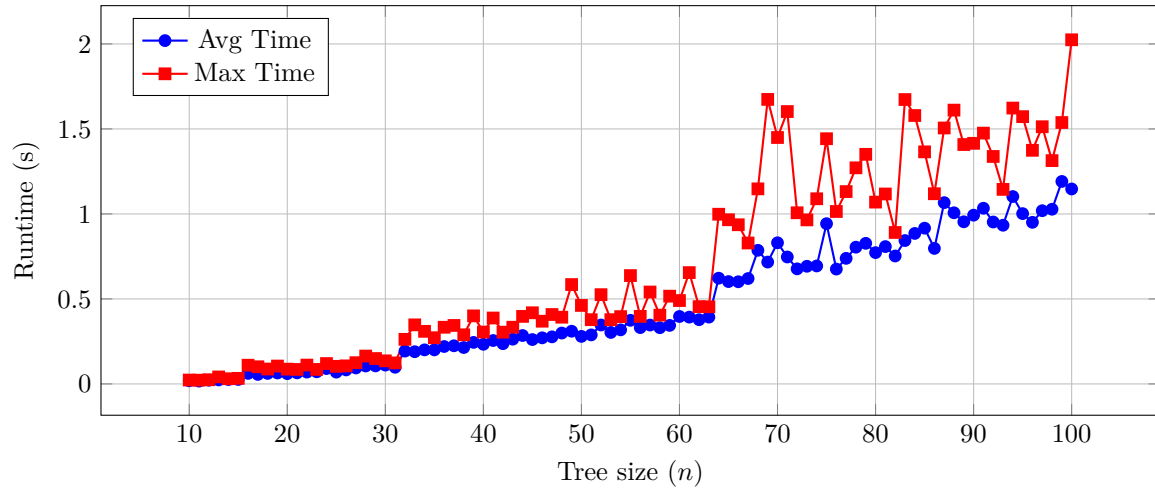


Figure 4.18: Random trees with binomial costs - Cicalese algorithm: runtime

Table 4.12: Random trees with binomial cost distribution - Cicalese algorithm (every 2nd instance)

n	Avg Time (s)	Max Time (s)	Avg Cost	Max Cost	Avg Ratio	Max Ratio
10	0.018569	0.022960	3.0950	4.1429	1.203734	1.526316
11	0.016663	0.022207	3.2399	4.1667	1.225465	1.350000
12	0.021850	0.024641	3.0845	3.7143	1.204960	1.400000
13	0.023680	0.041130	3.1821	3.5714	1.146632	1.350000
14	0.025879	0.030532	3.3019	4.1667	1.203148	1.368421
15	0.026162	0.032810	3.1452	3.8571	1.156865	1.285714
16	0.061919	0.110077	3.2125	4.0000	1.135182	1.294118
17	0.055949	0.101016	3.1308	3.5714	1.169257	1.470588
18	0.061837	0.087722	3.1532	4.0000	1.172924	1.315789
19	0.064491	0.105249	3.4496	4.0000	1.207410	1.304348
20	0.060418	0.087146	3.7143	4.2857	1.221015	1.375000
21	0.065168	0.085291	3.4685	4.1250	1.194317	1.375000
22	0.069730	0.110900	3.6357	4.1429	1.196165	1.307692
23	0.071404	0.085374	3.6167	4.3750	1.240822	1.434783
24	0.090381	0.119865	3.7004	4.2500	1.230664	1.416667
25	0.069459	0.103622	3.7879	4.7143	1.235042	1.409091
26	0.082531	0.106300	3.3647	4.0000	1.139867	1.240000
27	0.094121	0.125063	3.7028	4.8571	1.262302	1.428571
28	0.105554	0.164045	3.8099	4.7778	1.276963	1.791667
29	0.106147	0.149406	3.6984	4.7143	1.209574	1.333333
30	0.110902	0.135836	3.7730	4.2857	1.168583	1.291667
31	0.098121	0.123841	3.7861	4.6250	N/A	N/A
32	0.193028	0.262267	3.8365	4.1429	N/A	N/A
33	0.189587	0.347502	3.6373	4.6250	N/A	N/A
34	0.200383	0.309303	4.1887	5.1429	N/A	N/A
35	0.200387	0.271846	4.1000	4.7500	N/A	N/A
36	0.220161	0.333407	3.7250	4.7500	N/A	N/A
37	0.224506	0.344000	4.0871	4.8571	N/A	N/A
38	0.214219	0.289047	4.1190	4.8571	N/A	N/A
39	0.244387	0.400370	4.1522	5.0000	N/A	N/A
40	0.233333	0.305361	4.0108	5.2857	N/A	N/A
41	0.255417	0.387852	3.8694	4.5000	N/A	N/A
42	0.236747	0.304515	4.2767	5.2857	N/A	N/A
43	0.263362	0.333614	4.3161	5.2857	N/A	N/A
44	0.284715	0.397314	4.1347	5.3750	N/A	N/A
45	0.261028	0.418599	4.1025	5.1250	N/A	N/A
46	0.271188	0.369017	4.1264	4.6250	N/A	N/A
47	0.277001	0.408348	3.9500	4.5000	N/A	N/A
48	0.299343	0.392631	4.4736	5.6250	N/A	N/A
49	0.310086	0.584580	4.3097	4.8750	N/A	N/A
50	0.279730	0.461822	4.1040	5.4286	N/A	N/A
51	0.288490	0.377989	4.4371	5.2500	N/A	N/A
52	0.346704	0.525220	4.3319	5.6250	N/A	N/A
53	0.302892	0.378068	4.4000	5.0000	N/A	N/A
54	0.318150	0.395956	4.3444	5.1111	N/A	N/A
55	0.374280	0.637381	4.4926	6.0000	N/A	N/A
56	0.331411	0.397564	4.4208	5.0000	N/A	N/A
57	0.346383	0.540732	4.4865	5.1429	N/A	N/A
58	0.330891	0.404741	4.0968	4.8571	N/A	N/A
59	0.344452	0.516313	4.4069	5.0000	N/A	N/A
60	0.396750	0.490501	4.0633	4.6667	N/A	N/A
61	0.393049	0.654955	4.6899	6.3333	N/A	N/A
62	0.377914	0.454795	4.5514	6.0000	N/A	N/A
63	0.392999	0.453872	4.5014	5.5000	N/A	N/A
64	0.622009	0.998268	4.4056	5.2222	N/A	N/A
65	0.602131	0.965777	4.6806	5.3750	N/A	N/A
66	0.601055	0.936481	4.5473	5.4286	N/A	N/A
67	0.619979	0.829050	4.3717	5.8750	N/A	N/A
68	0.786046	1.147479	4.5740	5.3750	N/A	N/A
69	0.716865	1.673666	4.5047	5.3750	N/A	N/A
70	0.830285	1.449599	4.4514	5.3750	N/A	N/A
71	0.746811	1.602311	4.0550	4.7500	N/A	N/A
72	0.677003	1.006876	4.0522	5.0000	N/A	N/A
73	0.692125	0.965051	4.4597	5.2222	N/A	N/A
74	0.694072	1.089183	4.6856	5.7500	N/A	N/A
75	0.942688	1.442461	4.5486	5.2222	N/A	N/A
76	0.675616	1.014416	4.5903	5.5000	N/A	N/A
77	0.738686	1.131518	4.4708	5.6250	N/A	N/A
78	0.804489	1.271243	4.3222	6.0000	N/A	N/A
79	0.827115	1.350617	4.6319	6.1250	N/A	N/A
80	0.772503	1.069667	4.7014	5.1250	N/A	N/A
81	0.807888	1.117379	4.3583	5.6250	N/A	N/A
82	0.752628	0.891018	4.3946	5.5714	N/A	N/A
83	0.843558	1.672940	4.6194	5.4444	N/A	N/A
84	0.885294	1.578962	4.7444	5.3750	N/A	N/A
85	0.916309	1.365215	4.4406	6.0000	N/A	N/A
86	0.797230	1.118827	4.3847	6.0000	N/A	N/A
87	1.066447	1.505697	4.4931	5.0000	N/A	N/A
88	1.006799	1.610752	4.3719	5.0000	N/A	N/A
89	0.953852	1.408214	4.5742	5.4444	N/A	N/A
90	0.992794	1.415388	4.9639	5.7500	N/A	N/A
91	1.033525	1.475431	4.3475	4.8750	N/A	N/A
92	0.952505	1.338109	4.7014	6.0000	N/A	N/A
93	0.933517	1.144395	4.3536	4.8889	N/A	N/A
94	1.102392	1.622867	4.7156	6.2500	N/A	N/A
95	1.001937	1.572406	5.0700	6.2500	N/A	N/A
96	0.950718	1.374386	4.7661	5.2500	N/A	N/A
97	1.018995	1.512993	4.5375	5.3333	N/A	N/A
98	1.027694	1.313775	4.5117	5.2222	N/A	N/A
99	1.191126	1.537766	4.8014	6.3750	N/A	N/A
100	1.146860	2.024471	4.6328	5.5556	N/A	N/A

Dereniowski $O(\sqrt{\log n})$ -Approximation Algorithm

Table 4.13: Random trees with binomial cost distribution - Dereniowski algorithm (every 2nd instance)

n	Avg Time (s)	Max Time (s)	Avg Cost	Max Cost	Avg Ratio	Max Ratio
10	2.766419	12.594700	3.3401	4.1429	1.292396	1.571429
11	2.086002	14.903139	3.3429	4.1667	1.265143	1.444444
12	2.153725	10.921436	3.3010	3.8571	1.287693	1.400000
13	2.613087	7.186258	3.3679	4.0000	1.213860	1.454545
14	5.149202	28.974054	3.3388	4.5000	1.207946	1.368421
15	1.701935	7.544428	3.3972	3.8571	1.254578	1.450000
16	31.955564	215.138099	3.3500	4.0000	1.178475	1.476190
17	173.457108	1603.917324	3.4433	4.3750	1.282322	1.666667
18	36.867875	74.429148	3.1513	3.4286	1.190714	1.315789
19	159.390315	980.178797	3.3651	4.0000	1.210639	1.304348
20	21.763832	51.343457	3.6888	4.2857	1.212980	1.450000
21	20.482378	69.922542	3.4206	3.7500	1.176545	1.375000
22	29.476656	73.267557	3.8889	4.8571	1.282130	1.416667
23	25.331472	54.049392	3.8347	4.5000	1.322905	1.523810
24	45.758897	226.747744	3.8049	4.2857	1.257332	1.375000
25	27.503702	90.087763	3.8988	4.7143	1.273956	1.409091
26	24.759023	70.813797	3.5870	4.2222	1.202938	1.407407
27	19.324728	75.035363	3.9431	5.0000	1.328577	1.500000
28	344.503286	2423.304454	4.0135	5.2857	1.343051	1.791667
29	28.559347	60.115264	3.9452	5.5714	1.293771	1.571429
30	42.816815	189.415031	3.8849	4.7500	1.198378	1.461538
31	24.652245	91.068090	3.8580	4.6250	N/A	N/A
32	42.166849	131.750438	3.7951	4.1250	N/A	N/A
33	182.292682	1195.174706	3.7666	4.8750	N/A	N/A
34	27.921171	61.951006	4.4401	5.1429	N/A	N/A
35	29.155326	148.524253	4.3611	5.5000	N/A	N/A
36	472.628700	3956.030065	3.9568	4.7500	N/A	N/A
37	49.684235	247.921100	4.3062	5.0000	N/A	N/A
38	39.706982	130.295335	4.3476	5.1429	N/A	N/A
39	4106.072874	40689.805987	4.3978	5.5714	N/A	N/A
40	78.306776	521.714127	4.2023	5.7143	N/A	N/A
41	441.544937	2842.636941	4.2562	5.0000	N/A	N/A
42	1985.783616	10832.178223	4.0703	5.4286	N/A	N/A
43	2032.209634	15993.999531	4.4253	5.1429	N/A	N/A
44	49843.050655	392703.720958	4.3628	4.6667	N/A	N/A
45	1756.696046	4066.276252	4.3210	5.0000	N/A	N/A
46	1637.604034	8195.422064	4.2083	5.1250	N/A	N/A
47	4518.566281	24557.124074	4.2075	4.8889	N/A	N/A
48	730.966484	2768.719063	4.6134	5.6250	N/A	N/A
49	1398.716203	7138.296084	4.3847	5.0000	N/A	N/A
50	1469.494227	3253.316360	4.3385	4.6250	N/A	N/A
51	6672.336573	44175.934440	4.4511	5.5000	N/A	N/A
52	2005.430994	5185.360214	4.4213	5.4444	N/A	N/A

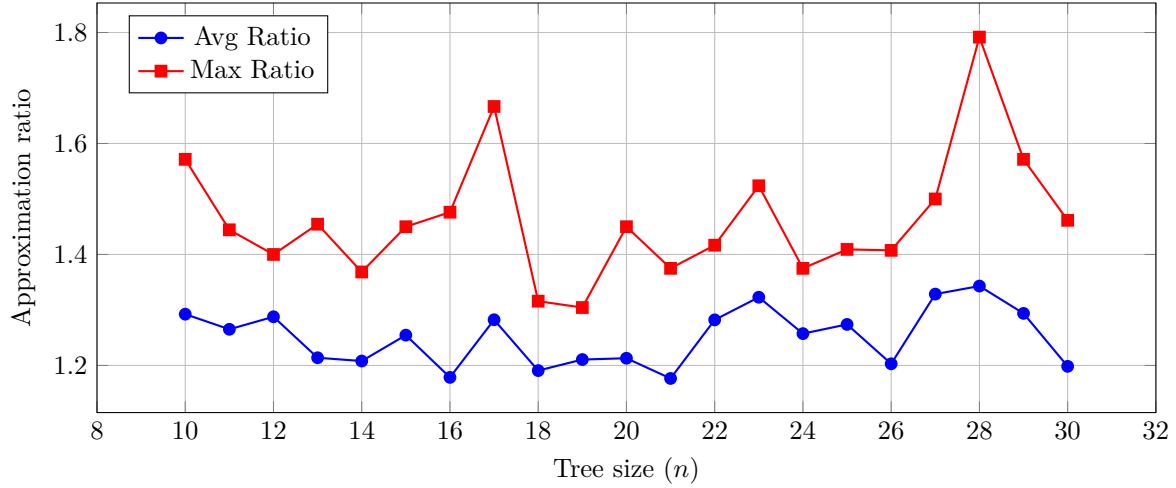


Figure 4.19: Random trees with binomial costs - Dereniowski algorithm: approximation quality

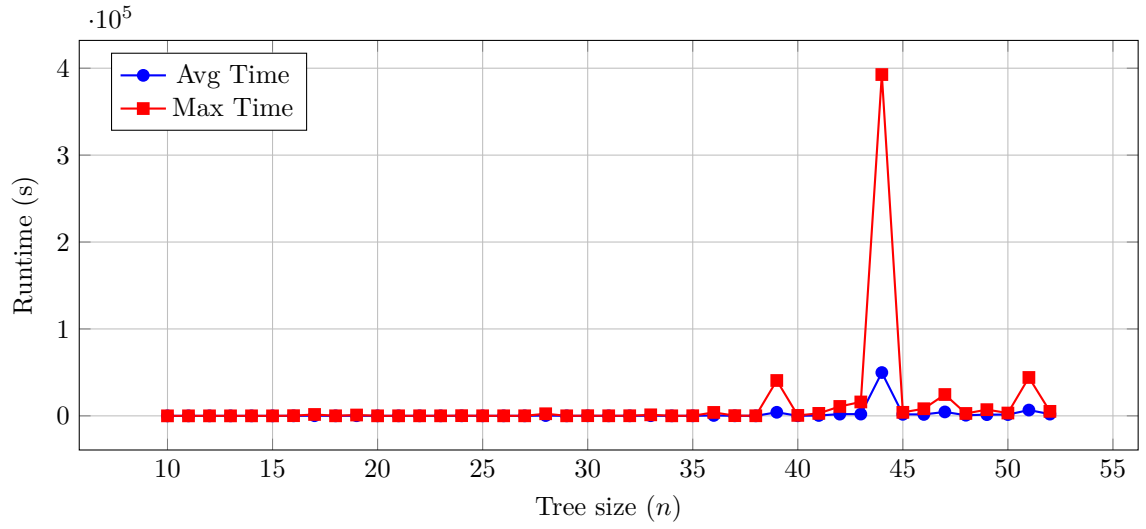


Figure 4.20: Random trees with binomial costs - Dereniowski algorithm: runtime

K-up $O(\log \log n)$ -Approximation Algorithm

Table 4.14: Random trees with binomial cost distribution - K-up algorithm (every 2nd instance)

n	Avg Time (s)	Max Time (s)	Avg Cost	Max Cost	Avg Ratio	Max Ratio
10	270.171032	1845.239705	3.1917	4.2857	1.242521	1.647059
11	273.610037	2390.518776	3.5163	4.8571	1.325272	1.888889
12	162.207568	958.610187	3.0595	3.5000	1.195159	1.400000
13	108.180016	731.949716	3.4089	3.8571	1.231742	1.380952
14	8520.517828	56561.592049	3.4552	4.3333	1.281385	1.736842
15	493.213468	1619.288856	3.6667	4.1429	1.297056	1.526316
16	369.386095	1452.884733	3.6000	3.8750	1.293589	1.428571

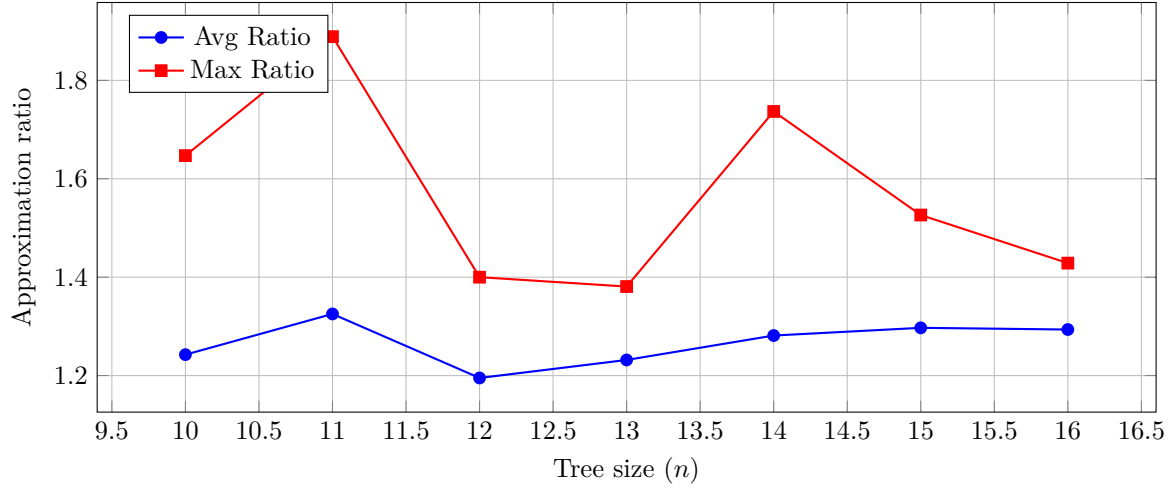


Figure 4.21: Random trees with binomial costs - K-up algorithm: approximation quality

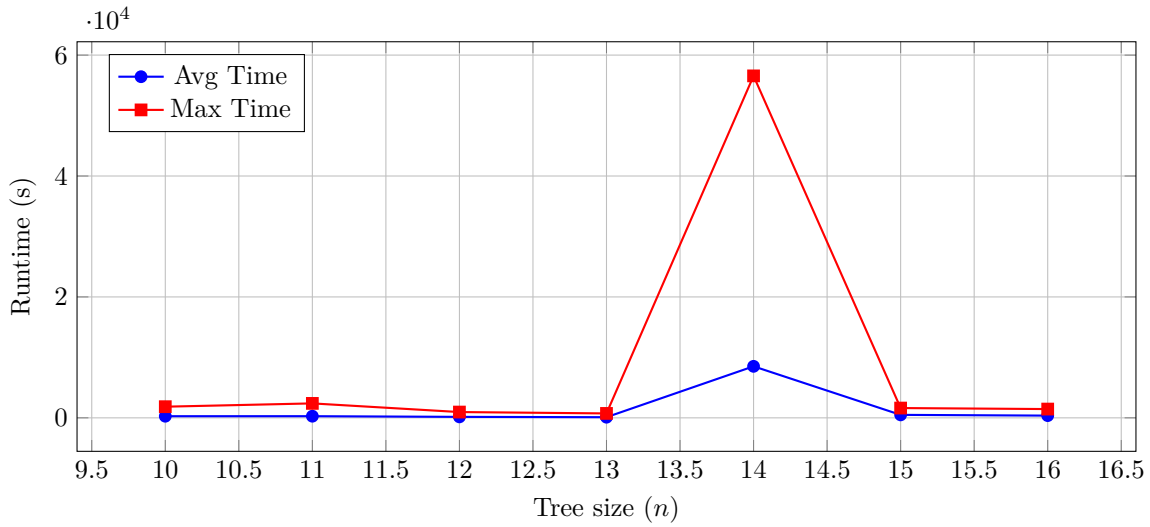


Figure 4.22: Random trees with binomial costs - K-up algorithm: runtime

4.3.3 Random Trees with Exponential Cost Distribution ($T_R || V, c \sim \mathcal{E} || C$)

DP Exact Algorithm

Table 4.15: Random trees with exponential cost distribution - DP exact algorithm

n	Avg Time (s)	Max Time (s)	Avg Cost	Max Cost
10	0.397258	0.549761	1.56	2.08
11	0.881495	1.185577	1.52	2.16
12	1.160748	1.885142	1.55	2.10
13	1.826028	3.421310	1.56	2.14
14	4.676321	11.680625	1.51	2.41
15	7.653875	15.009777	1.53	1.93
16	12.748250	44.517421	1.43	1.86
17	23.283714	40.561355	1.26	1.74
18	21.641541	51.801028	1.40	1.91
19	62.381975	140.589124	1.41	2.29
20	101.497002	175.680065	1.65	2.53
21	131.955652	233.756469	1.62	2.01
22	163.721968	432.129969	1.67	2.74
23	180.511257	460.316967	1.61	2.41
24	734.623643	2883.680292	1.86	2.55
25	948.515655	2682.398239	1.54	2.53
26	1253.883607	2169.719410	1.62	2.05
27	2585.928873	7403.012652	1.52	1.99
28	1985.006110	3254.015142	1.64	2.32
29	5615.846933	11391.047865	1.68	2.58
30	13563.202799	29000.578610	1.68	2.33
31	16919.006757	38392.842757	1.46	2.16
32	25802.145074	69527.490872	1.40	2.08
33	6209.514245	6209.514245	1.57	1.57

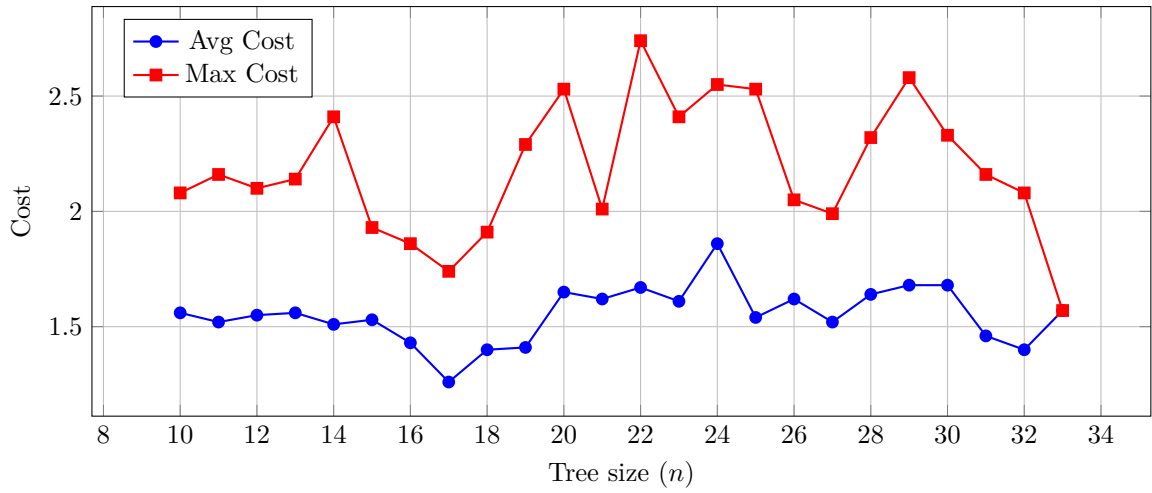


Figure 4.23: Random trees with exponential costs - DP exact algorithm: cost comparison

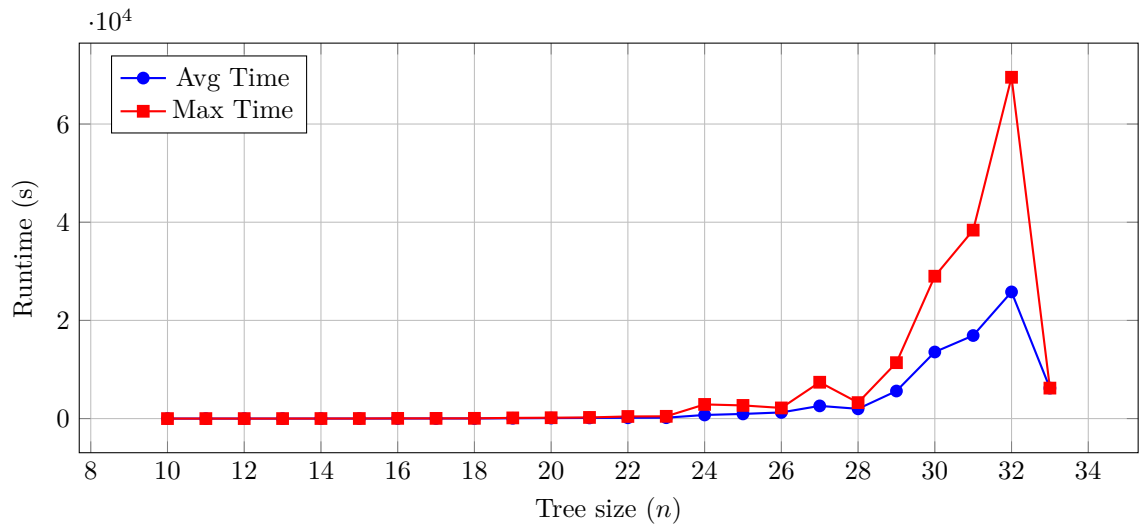


Figure 4.24: Random trees with exponential costs - DP exact algorithm: runtime comparison

Cicalese et al. $O(\log n / \log \log n)$ -Approximation Algorithm

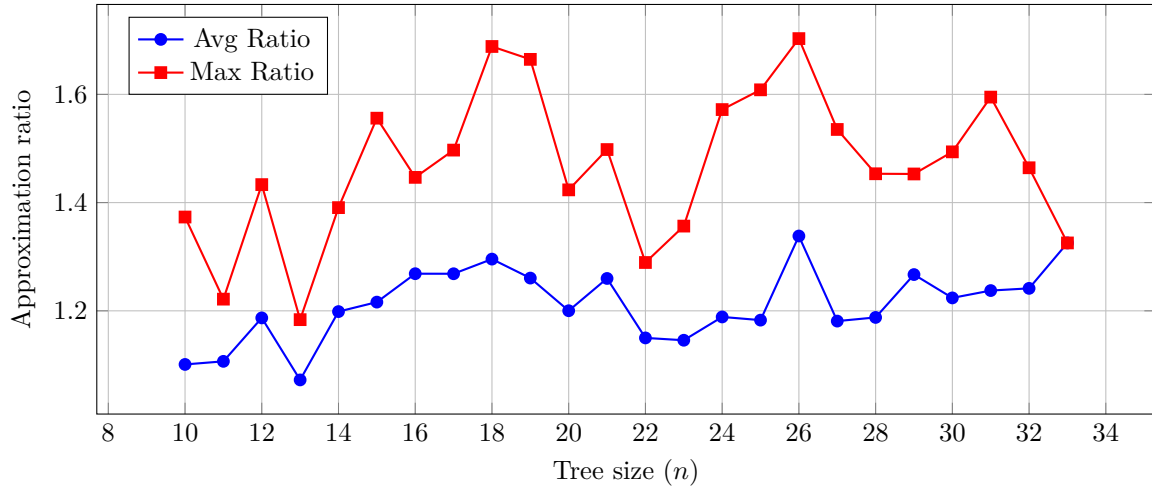


Figure 4.25: Random trees with exponential costs - Cicalese algorithm: approximation quality

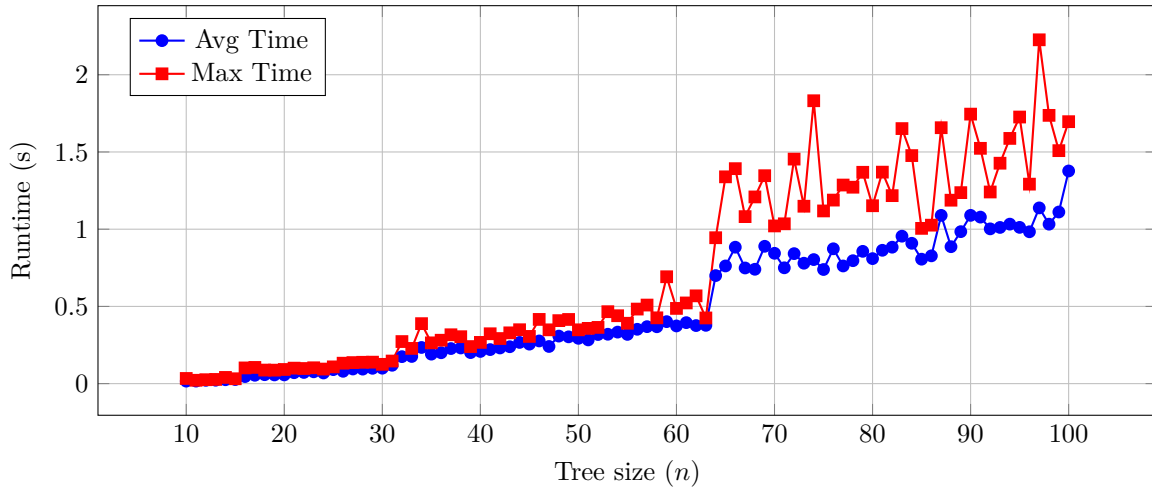


Figure 4.26: Random trees with exponential costs - Cicalese algorithm: runtime

Table 4.16: Random trees with exponential cost distribution - Cicalese algorithm (every 2nd instance)

n	Avg Time (s)	Max Time (s)	Avg Cost	Max Cost	Avg Ratio	Max Ratio
10	0.016237	0.033495	1.7121	2.1763	1.100966	1.373356
11	0.016435	0.021775	1.6764	2.3384	1.106673	1.221622
12	0.021375	0.025441	1.8340	2.4271	1.186964	1.433260
13	0.022067	0.027149	1.6666	2.2813	1.072361	1.183724
14	0.025236	0.040168	1.8195	3.3349	1.198633	1.390735
15	0.026477	0.031603	1.8378	2.3640	1.216127	1.555999
16	0.045667	0.101752	1.8317	2.5652	1.268623	1.446646
17	0.054146	0.104885	1.5905	1.9033	1.268534	1.497025
18	0.057846	0.087809	1.8084	2.3725	1.295600	1.688485
19	0.056449	0.087378	1.7525	2.5305	1.260531	1.664679
20	0.056650	0.091631	1.9674	2.9465	1.200211	1.423546
21	0.070721	0.100305	2.0354	2.7751	1.259849	1.498037
22	0.072349	0.097988	1.9210	3.1176	1.149977	1.289329
23	0.076577	0.102658	1.8545	2.5860	1.145669	1.356598
24	0.069105	0.094356	2.1877	3.0123	1.188728	1.571916
25	0.090613	0.108517	1.7902	2.5287	1.182837	1.608406
26	0.079870	0.132329	2.1941	3.0924	1.338275	1.703062
27	0.094799	0.136214	1.7815	2.5804	1.181060	1.535153
28	0.093992	0.138174	1.9527	2.7889	1.187824	1.453300
29	0.099203	0.139057	2.1008	3.0494	1.267033	1.452920
30	0.100028	0.124936	2.0204	2.5196	1.223907	1.493783
31	0.118946	0.146275	1.7871	2.6809	1.237473	1.594948
32	0.174762	0.272404	1.7303	2.6102	1.241535	1.464382
33	0.175950	0.228101	1.9243	2.3879	1.325479	1.325479
34	0.234167	0.388441	2.1530	2.7081	N/A	N/A
35	0.190970	0.264060	1.9725	2.9010	N/A	N/A
36	0.199970	0.281022	2.0900	2.7969	N/A	N/A
37	0.227082	0.316417	2.0376	3.0079	N/A	N/A
38	0.230939	0.303853	2.0335	3.0135	N/A	N/A
39	0.200918	0.240026	2.1867	2.9043	N/A	N/A
40	0.208437	0.267012	2.2071	3.0158	N/A	N/A
41	0.220411	0.322908	2.1467	2.6411	N/A	N/A
42	0.230934	0.292023	1.9165	2.6343	N/A	N/A
43	0.238974	0.328827	1.7774	2.4194	N/A	N/A
44	0.266803	0.348988	2.1116	2.8924	N/A	N/A
45	0.255801	0.305685	1.7865	2.3460	N/A	N/A
46	0.276369	0.415777	1.9606	3.1310	N/A	N/A
47	0.240607	0.348401	2.0294	2.8902	N/A	N/A
48	0.307847	0.407946	1.9049	3.0710	N/A	N/A
49	0.302979	0.415478	2.1192	2.6123	N/A	N/A
50	0.293423	0.348525	2.1187	3.5104	N/A	N/A
51	0.283124	0.357786	1.9496	2.7632	N/A	N/A
52	0.317979	0.364042	1.9771	2.6000	N/A	N/A
53	0.319823	0.465793	1.9892	3.1693	N/A	N/A
54	0.333592	0.439661	1.9241	2.6152	N/A	N/A
55	0.318824	0.390928	1.8618	2.5191	N/A	N/A
56	0.351867	0.482792	2.1865	3.2603	N/A	N/A
57	0.368880	0.508758	1.9598	3.3761	N/A	N/A
58	0.368294	0.426094	1.9520	2.6180	N/A	N/A
59	0.401352	0.691992	1.8215	2.4247	N/A	N/A
60	0.373618	0.487093	1.9104	2.4981	N/A	N/A
61	0.394903	0.522432	2.2282	3.4636	N/A	N/A
62	0.375968	0.568445	1.9815	2.3735	N/A	N/A
63	0.377460	0.425218	2.2127	2.8955	N/A	N/A
64	0.699574	0.944739	2.0672	3.3542	N/A	N/A
65	0.761989	1.339043	1.9128	2.4773	N/A	N/A
66	0.883377	1.391986	2.1432	2.9604	N/A	N/A
67	0.748847	1.081355	1.9384	2.7230	N/A	N/A
68	0.740471	1.208873	2.1372	2.7436	N/A	N/A
69	0.889268	1.346342	1.9011	2.9210	N/A	N/A
70	0.843911	1.021090	1.9027	2.4419	N/A	N/A
71	0.749992	1.034903	2.1806	2.4891	N/A	N/A
72	0.841749	1.453867	1.8915	2.4149	N/A	N/A
73	0.779666	1.148637	2.0378	2.9202	N/A	N/A
74	0.803237	1.831655	2.1075	2.8968	N/A	N/A
75	0.738934	1.118207	2.1036	3.1720	N/A	N/A
76	0.873154	1.188425	2.2788	2.9854	N/A	N/A
77	0.761878	1.285886	1.9191	2.7118	N/A	N/A
78	0.796022	1.271787	2.0897	3.0263	N/A	N/A
79	0.856574	1.368408	2.0692	3.5652	N/A	N/A
80	0.809499	1.151806	2.2019	2.7582	N/A	N/A
81	0.863389	1.369387	1.9043	2.4292	N/A	N/A
82	0.883448	1.217195	1.7939	2.3756	N/A	N/A
83	0.954642	1.651388	1.9296	2.6076	N/A	N/A
84	0.908694	1.476544	1.9166	2.7540	N/A	N/A
85	0.805750	1.005292	1.9630	2.9213	N/A	N/A
86	0.827420	1.026068	2.2813	3.3730	N/A	N/A
87	1.089388	1.657093	1.9245	2.3860	N/A	N/A
88	0.886662	1.187690	1.7277	2.7450	N/A	N/A
89	0.984090	1.236690	1.9441	2.5101	N/A	N/A
90	1.089562	1.745053	2.2210	2.8069	N/A	N/A
91	1.078212	1.523559	1.9545	2.5371	N/A	N/A
92	1.002155	1.240874	2.2054	3.1364	N/A	N/A
93	1.011360	1.427223	2.0733	2.7658	N/A	N/A
94	1.032717	1.587759	2.0612	3.0558	N/A	N/A
95	1.011799	1.726225	2.0039	2.8603	N/A	N/A
96	0.983281	1.291645	2.0126	2.5147	N/A	N/A
97	1.137877	2.226232	2.2059	2.9068	N/A	N/A
98	1.032485	1.737016	2.3638	3.4569	N/A	N/A
99	1.111267	1.508560	1.9973	2.5536	N/A	N/A
100	1.376897	1.696145	1.9522	2.5738	N/A	N/A

Dereniowski $O(\sqrt{\log n})$ -Approximation Algorithm

Table 4.17: Random trees with exponential cost distribution - Dereniowski algorithm (every 2nd instance)

n	Avg Time (s)	Max Time (s)	Avg Cost	Max Cost	Avg Ratio	Max Ratio
10	3.271125	17.785696	1.8367	2.4016	1.188598	1.623924
11	3.629943	20.215514	1.6837	2.3384	1.107890	1.239986
12	0.136612	0.703763	1.9165	2.7691	1.213336	1.595378
13	5.155277	26.768550	1.8225	2.2719	1.161811	1.518507
14	5.674060	27.903470	1.8356	3.3661	1.212446	1.397901
15	3.337283	15.047687	1.7821	2.1606	1.172650	1.433563
16	21.073328	85.748191	1.6668	2.8439	1.238863	1.529049
17	25.494491	165.556898	1.5193	1.9148	1.214376	1.582861
18	99.869352	406.402671	1.8736	2.6929	1.277778	1.553406
19	19.421447	69.304303	1.7725	3.0094	1.254327	1.843616
20	22.498086	106.415632	1.9989	2.9465	1.221528	1.402355
21	23.358083	72.395053	2.2114	3.0928	1.324501	1.669538
22	37.254589	192.448365	2.1058	3.3406	1.216357	1.623233
23	66.243801	206.180124	1.9605	3.2134	1.221255	1.773604
24	19.672028	104.269739	2.5429	3.0867	1.306232	1.571916
25	98.736290	355.952677	1.9154	2.5287	1.266486	1.749569
26	56.177485	202.758462	2.2349	3.4418	1.370396	1.985201
27	49.095161	221.547670	1.8147	2.5804	1.135261	1.346807
28	29.204365	141.400477	2.1097	3.0662	1.235612	1.754597
29	26.653767	131.919711	2.0997	3.3945	1.262877	1.494846
30	40.418576	102.952930	1.9985	2.4121	1.191372	1.459335
31	41.726362	116.834656	1.8344	2.0202	1.305434	1.583043
32	18.009437	57.953526	1.6247	2.0688	1.210094	1.366504
33	18.721550	65.473680	2.1445	3.3142	1.507570	1.507570
34	30.534315	60.208069	2.2975	2.4680	N/A	N/A
35	11.796997	24.369075	1.8985	2.8521	N/A	N/A
36	32.693106	127.268339	2.2920	3.2832	N/A	N/A
37	28.107762	69.320070	1.9708	2.6454	N/A	N/A
38	74.705023	297.670002	1.9312	2.7219	N/A	N/A
39	28.753502	107.382334	2.4597	3.5480	N/A	N/A
40	40.023030	113.296597	2.3885	3.4799	N/A	N/A
41	44.548128	84.873195	2.5159	2.9444	N/A	N/A
42	617.346981	1712.780073	1.8700	2.4136	N/A	N/A
43	3343.745612	25313.002871	1.7465	2.4194	N/A	N/A
44	526.433698	833.764386	2.4429	3.5024	N/A	N/A
45	1025.802671	2694.859194	2.0049	2.4161	N/A	N/A
46	376.982329	798.947795	2.1529	2.7224	N/A	N/A
47	182.393146	683.893516	2.2085	2.7435	N/A	N/A
48	7359.758340	35448.198180	1.9486	2.4580	N/A	N/A
49	2234.354507	8378.426965	2.5363	2.9495	N/A	N/A
50	3424.428952	13372.659564	2.2139	2.6620	N/A	N/A
51	603.334160	1764.191564	1.7830	2.3359	N/A	N/A
52	374.618495	682.347076	2.1870	2.7190	N/A	N/A
53	412.175755	1087.737078	2.1039	3.0948	N/A	N/A
54	11327.159077	44777.889020	1.8128	2.6372	N/A	N/A

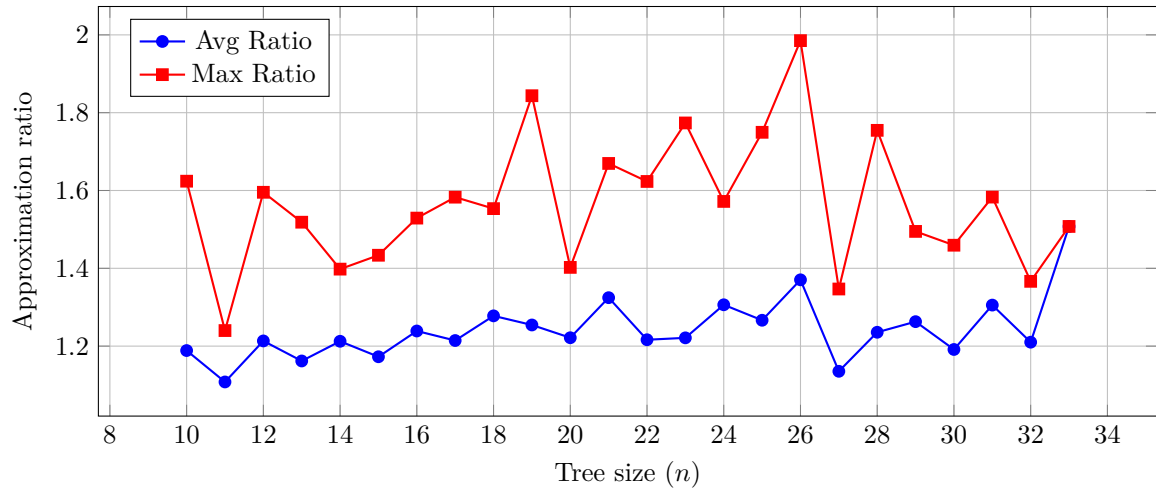


Figure 4.27: Random trees with exponential costs - Dereniowski algorithm: approximation quality

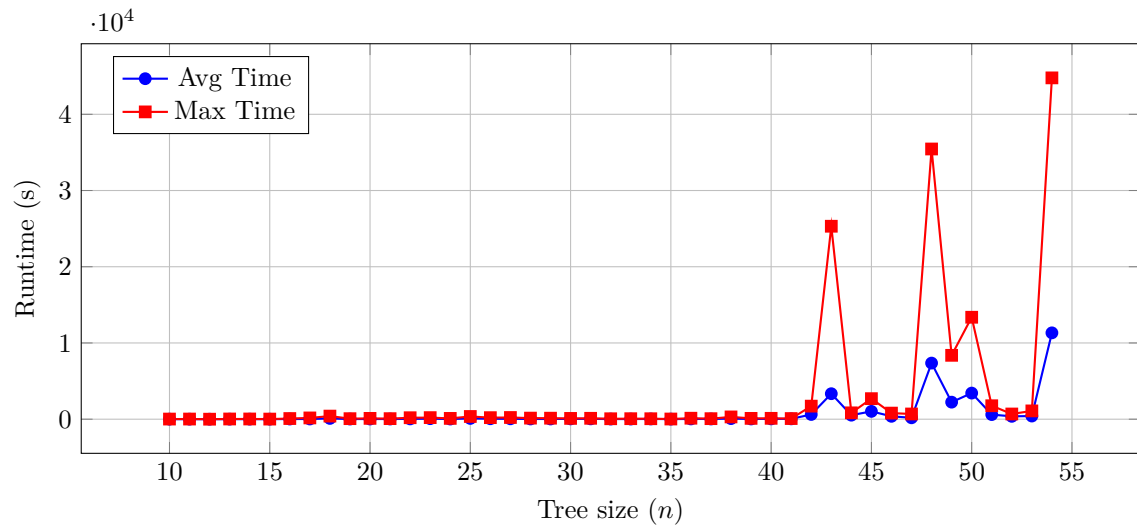


Figure 4.28: Random trees with exponential costs - Dereniowski algorithm: runtime

K-up $O(\log \log n)$ -Approximation Algorithm

Table 4.18: Random trees with exponential cost distribution - K-up algorithm (every 2nd instance)

n	Avg Time (s)	Max Time (s)	Avg Cost	Max Cost	Avg Ratio	Max Ratio
10	2.443456	22.215506	2.1098	2.8660	1.363845	1.932925
11	6.154900	57.546679	1.9326	2.8294	1.293482	1.512384
12	1151.116577	9780.632542	1.9381	2.9057	1.253727	1.674094
13	14.196982	85.125569	2.0658	3.0771	1.344387	1.728937
14	198.175667	1789.909311	2.1204	2.8122	1.482452	1.941569
15	5862.478971	46840.667775	1.8959	2.1669	1.235591	1.615380
16	318.129038	3075.309871	1.8765	3.1386	1.332374	1.729848
17	3.516439	21.537107	2.2755	2.7576	1.847921	2.528925
18	669.176482	5233.473255	2.4730	3.3256	1.829482	2.322135
19	6.937829	25.825382	2.4158	3.0508	1.852755	2.402326
20	167.503671	1599.696498	2.5691	3.3406	1.604565	2.177329
21	638.992803	5089.416766	2.5644	3.7624	1.616518	2.146588
22	152.358166	1272.209907	2.5881	3.0978	1.592249	2.216253
23	229.841830	1240.158336	2.5098	3.2451	1.621922	2.538157

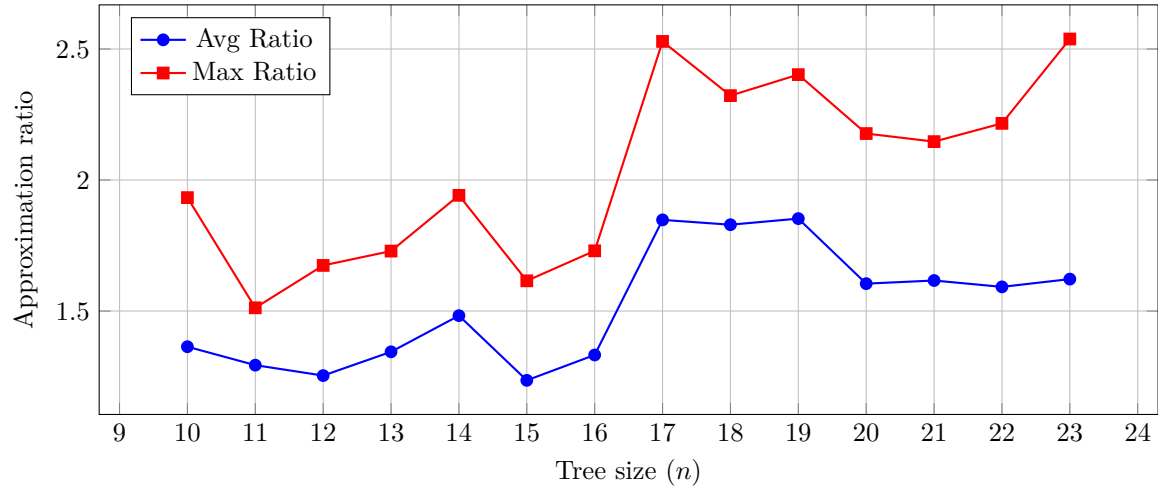


Figure 4.29: Random trees with exponential costs - K-up algorithm: approximation quality

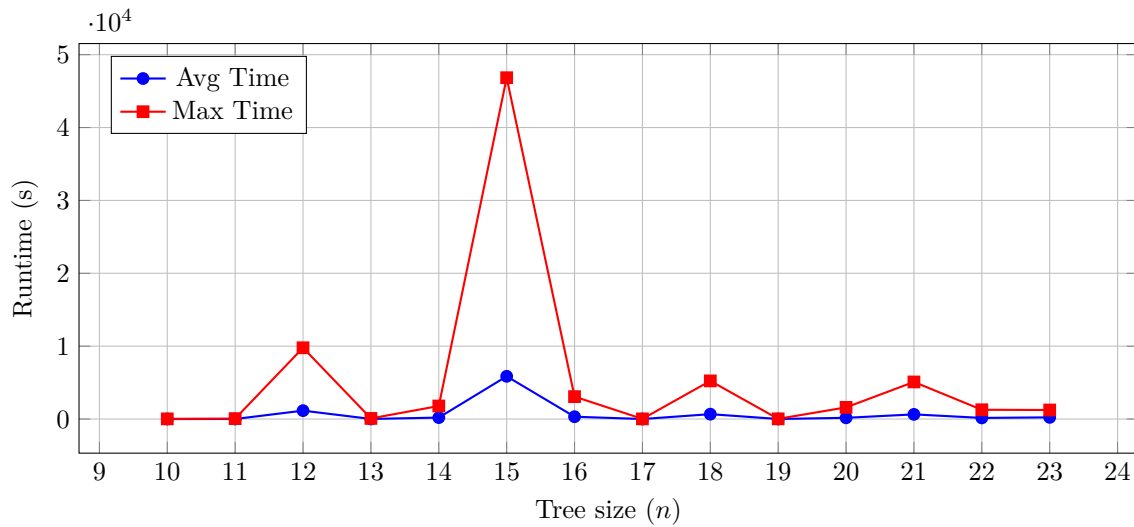


Figure 4.30: Random trees with exponential costs - K-up algorithm: runtime

Chapter 5

Conclusions

This thesis presented an experimental analysis of the generalized worst-case binary search problem in tree graphs. We explored various algorithms designed to tackle the Tree Search Problem with both uniform and non-uniform query costs.

Obtaining efficient search strategies for trees is important due to numerous practical applications, such as bug searching, network diagnostics and river pollution detection. Additionally, the graph search problem is also important from a theoretical point of view. In particular, for the case of uniform costs, the depth of the decision tree corresponds to the well-known graph parameter called the *tree-depth* [GHT12]. This parameter, tightly connected to other graph invariants like tree-width, is of wide interest in graph theory and parametrized complexity [Bod+95; NO06; BDO23]. The significance of the non-uniform cost model, on the other hand is because it is one of the few problems which are NP-hard already on trees. This is due to the fact that the decision trees have a non local structure, which in turn yields most of the classic dynamic programming techniques inapplicable. Approximation algorithm presented hereby, are therefore of particular interest, since they provide efficient ways of obtaining solutions of guaranteed quality for problems with such characteristics. It is interesting whether similar techniques can be applied to other, similar problems. Among them is for example the conflict-free connection coloring which differs only slightly since it requires that each path in the tree contains a vertex/edge with a unique color [Cha+21], a relaxed condition of the ranking coloring which is equivalent to the Tree Search Problem.

The above mentioned non-locality enforces the use of complex algorithmic techniques, and non-polynomial time subroutines. This provides a two-fold challenge: on the one hand, it makes the algorithms hard to implement and debug, on the other hand, it limits their practical usability to relatively small instances. Due to this, the experimental analysis conducted in this thesis required significant effort. This was particularly demanding while implementing the QPTAS for $T||V, c||C_{max}$ both in terms of the used data structures (boxed decision trees) with operations on them (efficient merging and rotating) as well as the algorithm by itself, consisting of few cases using complex subroutines.

It is an interesting question, whether it is possible to derive similar results, using simpler techniques. In particular, a further research direction would be to further simplify the QPTAS both in terms of the large constant in the exponent and the algorithm itself. Even a central question in this line of research is whether it is possible to derive constant factor approximation guarantee for the non-uniform costs or ideally, a PTAS. As of right now, no inapproximability results are known for this problem, besides a trivial non-existence of absolute approximation algorithms. This problem therefore is included among the specific list of problems for which a $O(\sqrt{\log n})$ -approximation exists, but there is no lower bound matching this result. A similar issue is prevalent among many separation problems such as treewidth, vertex separators and edge cuts [ARV09; KNS; FHL05] for which $O(\sqrt{\log n})$ -approximation algorithms are known, but no non-trivial lower bounds beyond constant factor inapproximability are known. Interestingly, these problems are connected to Tree Search Problem, since they can be used to derive relatively simple constant factor approximation algorithms for the average case version of the problem [Szy25; Høg+21; Das16].

The main conclusion drawn from the experimental analysis is that the state-of-the-art approximation algorithms are of practical use only for relatively small instances, up to several dozens of vertices. Beyond that, the $O(\log n / \log \log n)$ -approximation algorithm is the most efficient one. The $O(\log \log n)$ -approximation algorithm, while theoretically interesting, is impractical for real-world applications due to its high computational complexity.

Additionally, on the random instances used in our experiments, the costs of the solutions built by all algorithms were relatively low, and no particular algorithm consistently outperformed the

others. This suggests that on random instances, most heuristics usually perform well. Additionally, the approximation ratios observed in the experiments were significantly better than the theoretical worst-case guarantees, indicating that the algorithms are effective in practice despite their theoretical limitations. It should be remarked that the approximation ratios obtained during the theoretical analysis are worst-case guarantees and in general, the intrinsic difficulty of achieving better results for the Tree Search Problem is lack of mathematical techniques providing better lower bounds on the optimal solution. In particular the only lower bound used in the analysis of the QPTAS is the value of 1 (after costs normalization) which is rather weak. Moreover during its recursive usage in the $O(\sqrt{\log n})$ -approximation algorithm, each such usage induces the cost of $O(\text{OPT})$, which is a lot considering the fact, that the size of the instances processed at each level of the recursion is significantly smaller than the original instance. Transcending any of these obstacles is a possible further research direction.

Another interesting observation is the surprisingly good performance of the $O(2^n n)$ time exact algorithm. Despite its exponential time complexity, it was able to solve instances with up to 40 vertices within reasonable time limits. In fact the upper bound on the running time of this algorithm is the number of connected subtrees of T . Since we generated our random trees using Prüfer sequences, by [KP19] we know that the expected number of connected subtrees of a tree generated in this way is of order $O(1.4196^n)$. Although this is still exponential, it is significantly better than the worst-case bound of $O(2^n)$. This suggests that the exact algorithm may be practical for small to medium-sized instances, especially when the input trees have certain structural properties that limit the number of connected subtrees. In recent years many NP-hard problems, such as the Traveling Salesman Problem or the Steiner Tree Problem, have seen significant improvements in their exact exponential time algorithms [Gas10]. It is therefore an interesting research direction to investigate whether similar improvements can be obtained for the Tree Search Problem.

Interestingly, the naive $O(n \log n)$ implementation of the linear time algorithm for $T||V||C_{max}$ based on calculating the ranking coloring performed differently among the three considered classes of trees. On randomly generated and bounded degree instances the observed running times were almost linear while for the bounded diameter instances the observed running time matches the upper bound of $O(n \log n)$. On the other hand for the bounded diameter instances the depth of the decision tree was always at most 4. This is due to the fact that the depth of the decision tree is trivially upper bounded by $\left\lceil \frac{\text{diam}(T)+1}{2} \right\rceil$. For randomly generated and bounded degree instances the depth of the decision trees produced by the algorithm behaved logarithmically in n . For the latter it is not hard to see that $\text{rank}(T) = \Omega(\log_{\Delta} n)$ since each query results in at least one response of size at least $\frac{n-1}{\Delta}$. For random trees the observed logarithmic behavior of $\text{rank}(T)$ is more surprising. We conjecture that given a random tree T on n vertices, $\text{rank}(T) = \Theta(\log n)$. An interesting research direction would be to prove this claim.

The above bound $\text{rank}(T) \leq \left\lceil \frac{\text{diam}(T)+1}{2} \right\rceil$ also shows that for trees with bounded diameter, the running time of the QPTAS for $T||V, c||C_{max}$ is in fact polynomial in n and the algorithm becomes a PTAS running in $n^{O(\text{diam}(T)/\epsilon^2)}$. It would be interesting to verify whether a similar result can be obtained for trees with bounded degree as well. Note that for such trees the depth of the decision tree is of order $O(\log n)$, so a new technique would have to be designed. However, it is possible that other structural properties of such trees can be exploited to obtain better bounds on the ranking number.

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